

MAGNETOM

Replacement of Parts System

Magnet System

Harmony syngo MR
Symphony syngo MR
Sonata syngo MR
Trio
Upgrades to syngo MR:
from Impact/Expert
from Vision

Document Version

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1.1 Installing a new bursting disk



WARNING

Risk of asphyxiation

If not observed person working with cryogenics becomes dizzy or loses consciousness.

- » Helium gas must not be vented into the magnet room.
- » Ensure that the turret venting kit is connected to quench pipe.



CAUTION

Magnet field and pressurized helium vessel

If not observed, serious injury or property damage can occur

- » The magnet must be de-energized and depressurized before attempting this procedure



Protective earth connection test.

If a protective earth connection has to be disconnected for replacing a component, measure the protective earth connection resistance $\leq 100 \text{ m}\Omega$ with a ground test meter (Fluke 1503, or equivalent) between the protective earth connector on the "sub-component" (e.g. compressor) and the protective earth wire connected to the "main component" (e.g. SEP).

Document the successful measurement in the service report (measured value and measured equipment) as required by IEC 62353.

1.1.1 Required Tools

- Torque wrench capable of measuring 0.8 Nm
- Allen key with 4 mm; 3 mm sockets
- Socket wrench
- Vacuum grease
- Thread locking component
- Handy Detector (leakage meter)
- Tissue paper

1.1.2 Spare parts

- Bursting disk (20 psi)
- "O" ring

- Fiber gasket

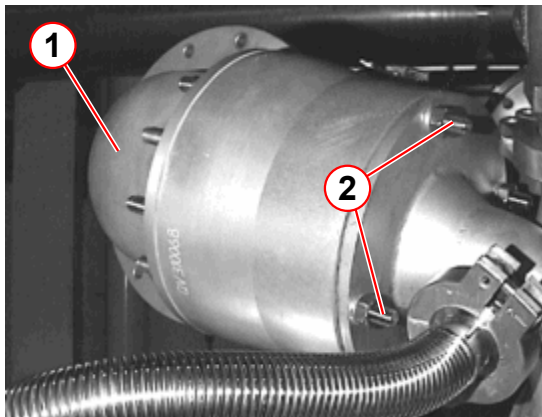
1.1.3 Exchange procedure

- Close the depressurization valve on the venting system
- Remove quench tube from 90° elbow
- Remove 90° elbow from quench valve
- Remove quench valve from turret
Record its orientation



Remove the 4 screws uniformly;
The valve is spring-loaded

Fig. 1: Quench valve



- (1) Elbow
- (2) Screw

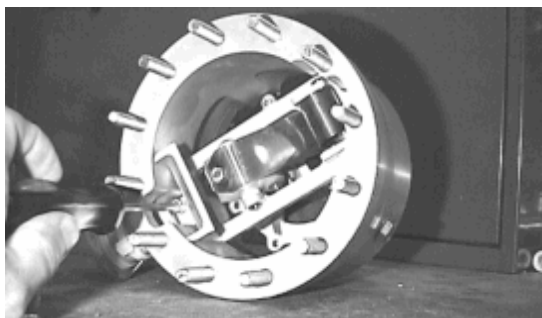
- Remove the quench valve
- Remove the gasket



While removing the valve, the turret flange must be temporarily blanked using tissue paper.

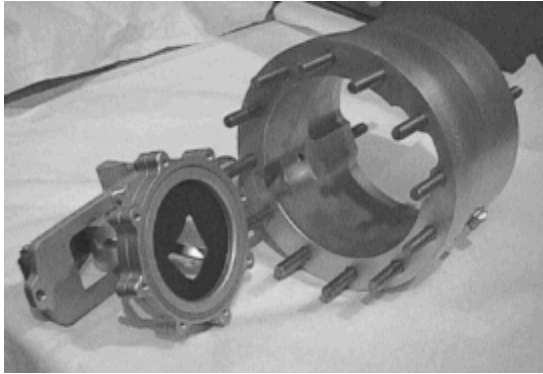
The following steps should be performed quickly to keep air from entering.

Fig. 2: Quench valve



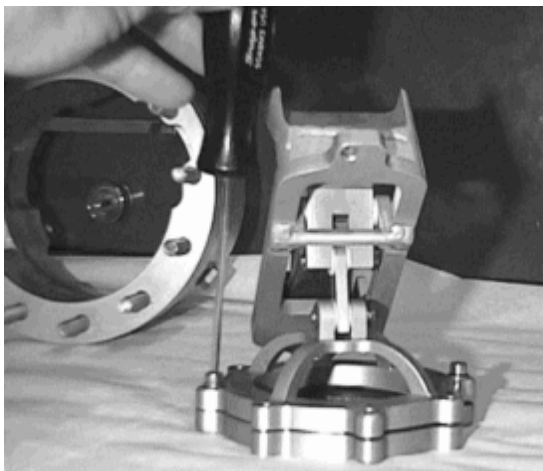
- Unscrew and remove the two M5 screws.

Fig. 3: Spring plate assembly



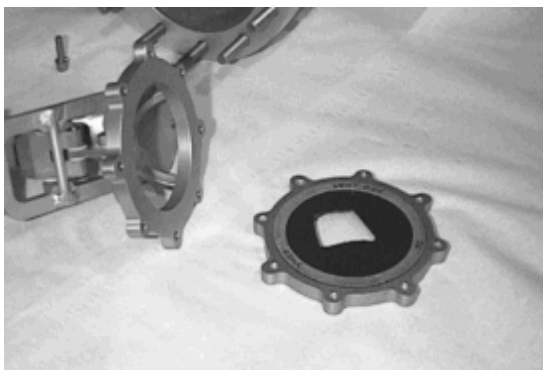
- Remove the spring plate assembly and the bursting disk assembly

Fig. 4: Burst disc assembly



- Inspect the bursting disk. Ruptured or cracked disks must be replaced.
- Unscrew and remove the eight M4 screws

Fig. 5: Broken bursting disk

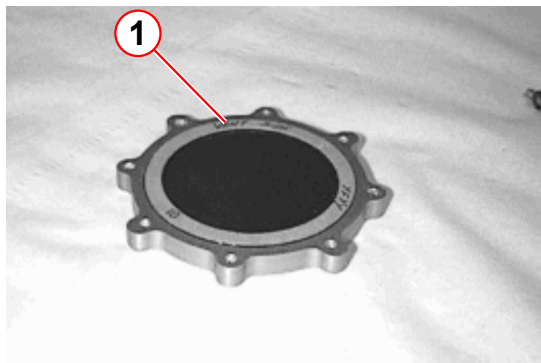


- Remove broken bursting disk

Fig. 6: Broken bursting disk

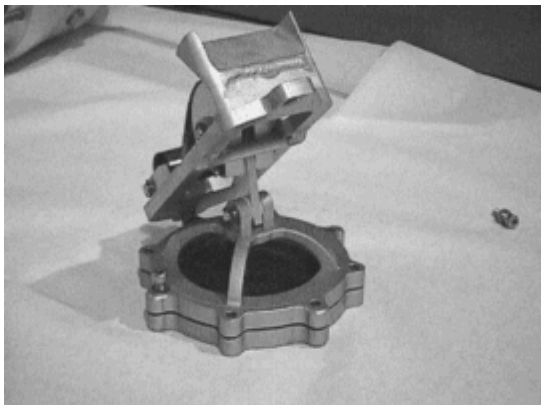


Fig. 7: Bursting disk assembly



(1) Vent side

Fig. 8: Bursting disk assembly



- Fit the new bursting disk into the bursting disk assembly as specified. The words "Vent side" (→ 1/ Fig. 7 Page 8) written on the bursting disk gasket should be on top.

- Re-assemble the bursting disk assembly.
The eight M4 screws and plain washers should be torqued to 0.8 Nm and locked with PermaBond

Fig. 9: Thread locking

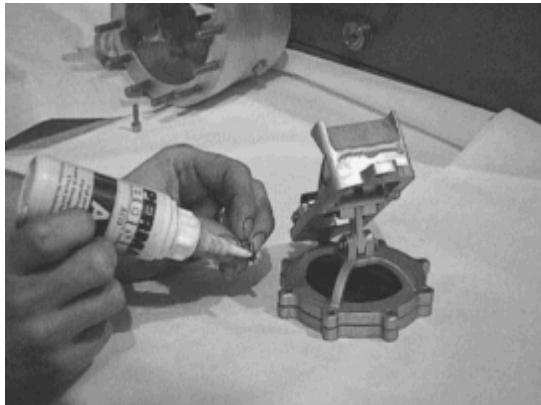


Fig. 10: Torque wrench

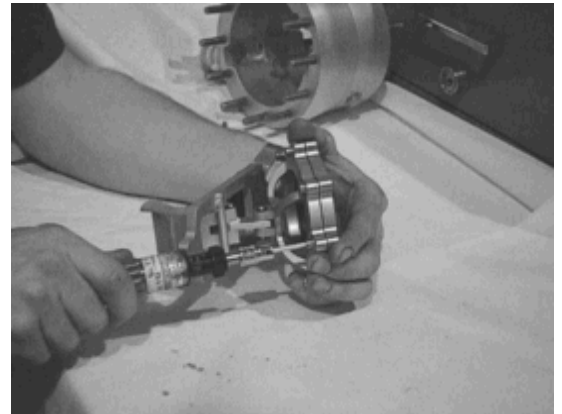
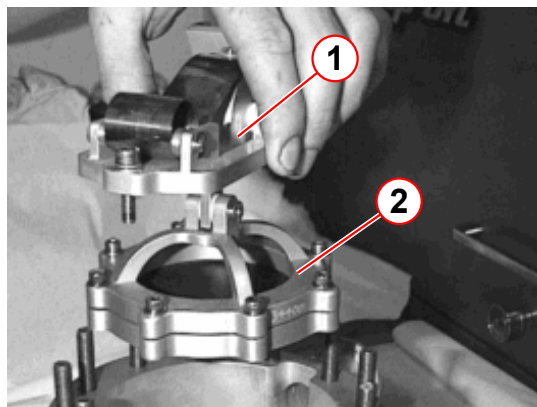


Fig. 11: Refitting the spring plate



- (1) spring plate
- (2) bursting disk assembly

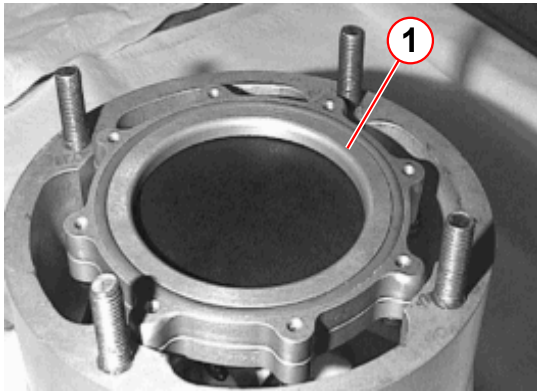
- Refit the spring plate assembly and bursting disk assembly into the quench valve body with the two M5 screws and washers. Torque evenly to 5 Nm.

- Before refitting the quench valve assembly to the turret flange, apply a **thin film** of vacuum grease to the turret flange, which fits into the fiber gasket.
Inspect the "O"ring seal in the bursting disk assembly.
Inspect the fiber gasket that fits to the turret flange.
If either seal appears damaged, it should be replaced.



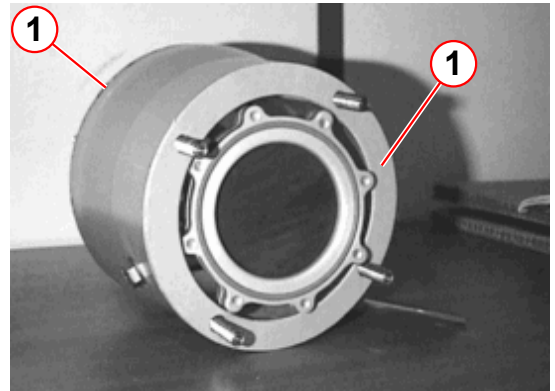
Don't use vacuum grease for the "O"ring and the mating turret flange.
The "O"ring and the fiber gasket are part of the magnet spare kit.

Fig. 12: Inspection of the "O" ring



(1) "O" ring

Fig. 13: Inspection of the fiber gasket



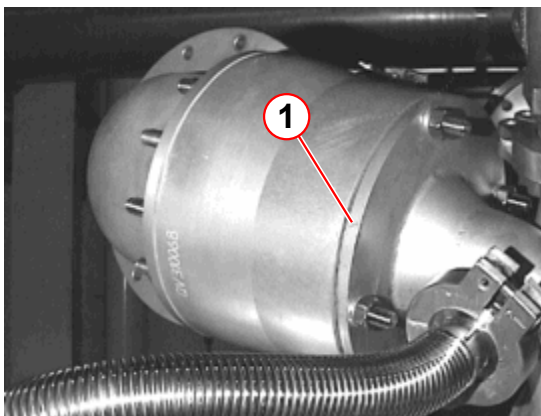
(1) Fiber gasket

- Remove the tissue paper used to temporarily seal the turret flange, and ensure that the "O"ring sealing surface is clean before refitting the quench valve assembly.
- Refit the quench valve assembly to the turret flange.

Ensure correct orientation.

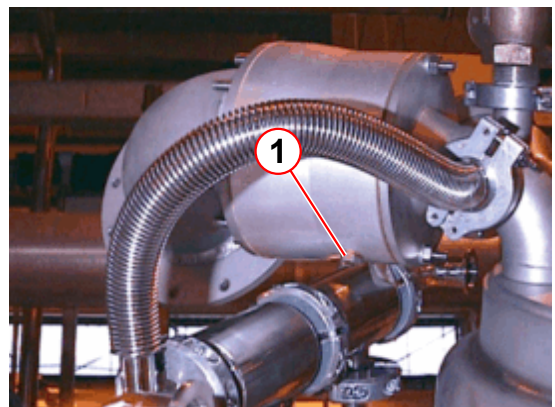
Torque to 0.8 Nm

Fig. 14: Quench valve assembly



(1) Turret flange

Fig. 15: Orientation of the quench valve



(1) Drain port

1.1.4 Final check

- Wait for the system to pressurize slightly, then check integrity of the valve, using a helium Handy Detector or check the pressure gauge. If the pointer stays in the yellow/green sector, the valve is tight.

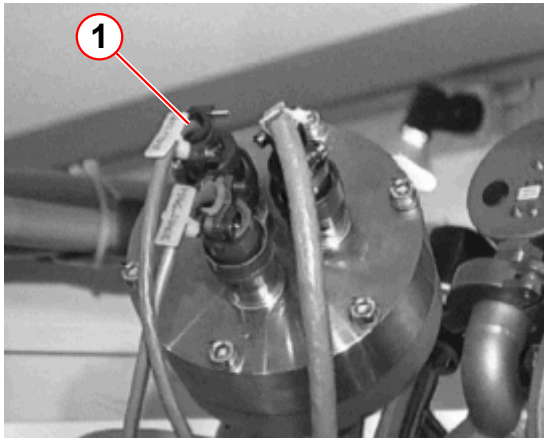
If a leak is present, the quench valve should be removed, the turret flange sealed, and the bursting disk assembly should be disassembled and reassembled again with particular care being given to sealing faces.

- Reconnect the venting elbow and the venting line to the quench valve body

- Replace the identification tag of the broken disk, which is fitted to the turreted flange, with the identification tag of the new bursting disk.

1.2 Connection of Helium Level Probe 2

Fig. 16: Terminal plate



(1) Connector "A"

- Disconnect the cable of helium level meter from Connector (→ 1/Fig. 16 Page 12) on the magnet.
- Connect the helium level adapter cable (service tool) to connector (→ 1/Fig. 16 Page 12).
- Connect the cable from helium level meter to adapter cable
- Calibrate helium level probe 2 via "Magnet & Cooling" in service software.

1.3 Baffle; Current lead; Syphon seal configuration

Fig. 17: Baffle plug seal configuration

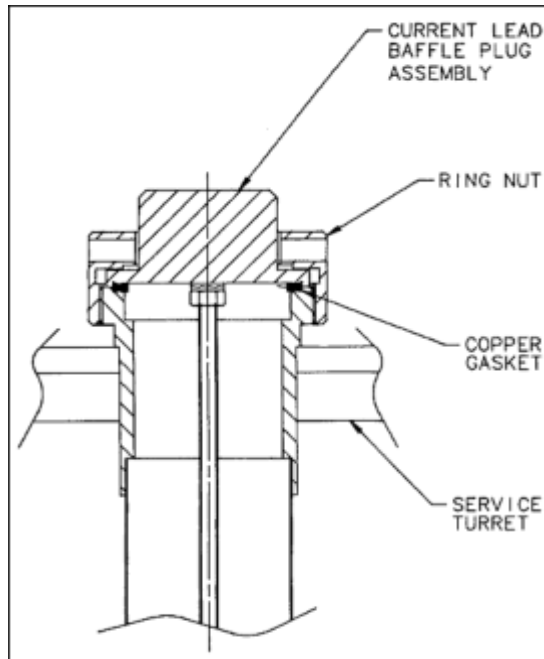


Fig. 18: Current lead seal configuration

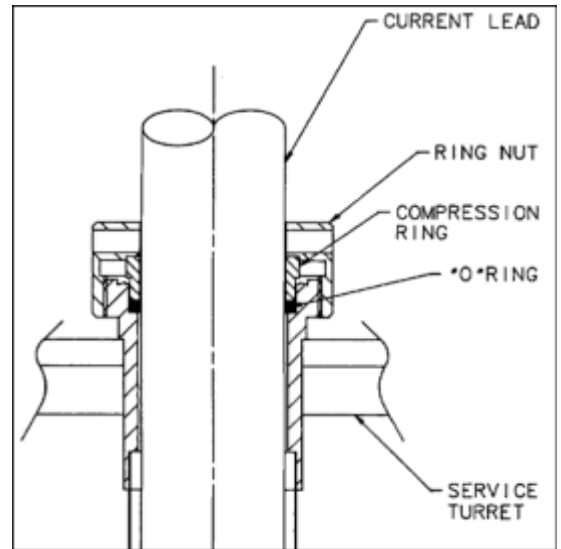
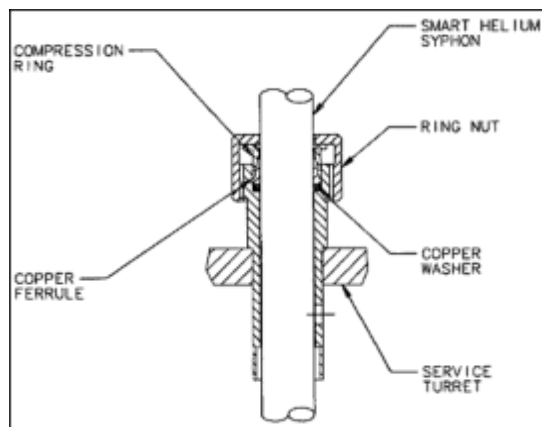


Fig. 19: Helium syphon seal



1.4 Replacing the pressure gauge



CAUTION

Pressurised helium vessel

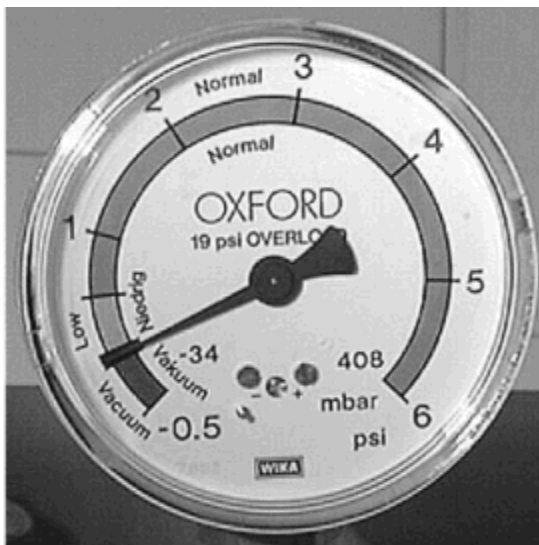
If not observed, serious injury can occur

- » Do not undo any of the fittings without first depressurising the helium vessel



Do not leave the system depressurised for more than the minimum necessary time, in order to avoid icing in the neck

Fig. 20: Pressure Gauge



- Depressurize helium tank via the bypass valve until the gauge reads 0 psi.
 - Remove clamp.
 - Remove pressure gauge.
 - Close bypass valve.
 - Attach new pressure gauge using the clamp.
- Ensure that seal is clean and dry.
- Check for leaks after the pressure has built up in the helium tank.
- (Handy Tector)

1.5 Replacing the smart syphon



CAUTION

Pressurised helium vessel

If not observed, serious injury can occur

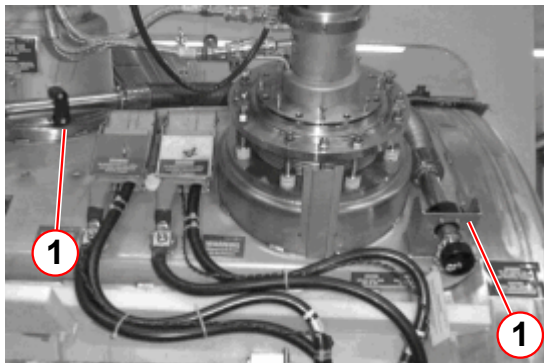
» **Do not undo any of the fittings without first depressurising the helium vessel**



Do not leave the system depressurised for more than the minimum necessary time, in order to avoid icing in the neck

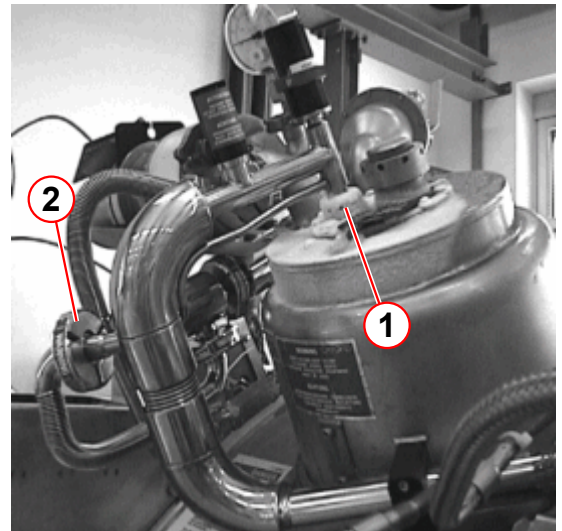
- Depressurize helium tank via the bypass valve until the gauge reads 0 psi .

Fig. 21: Smart syphon



(1) Mounting brackets

Fig. 22: Smart syphon



(1) Ring nut
(2) Clamp with high pressure seal

- Remove the screws from the mounting brackets.
- Remove the clamp and discard the old high pressure seal.
- Remove the ring nut, use the ring nut spanner.
- Pull the smart syphon out.
- Plug the syphon port with the baffle to prevent air entering the helium reservoir.



Make sure that the smart syphon is dry before inserting into the magnet. Purge the syphon with helium gas from a gas bottle.

- Insert the new syphon into the port. Removing the baffle before. Use the correct seals. Use a new copper washer.
- Tighten the ring nut using the ring nut spanner.
- Reconnect the venting line to the syphon, use a new high pressure seal.

- Close the bypass valve.



The system should start to re-pressurise.

- Check for leaks.

2.1 Replacing the Displacer - Leybold system



Servicing the APD coldhead and replacing the displacer is different from the Leybold coldhead. Training of service personnel on the APD coldhead is offered by CS TC.



This instruction is intended as a supplement to "Refrigerator" course



Only LICH II displacer unit may be installed on Harmony/Symphony systems with a Leybold coldhead.

The Double Damped Displacer as well as the single Damped Displacer are obsolete.

For magnets on Impact/Expert/Vision/SP systems the Single Damped Displacer must continue to be used for technical reasons

2.1.1 Auxiliary tools and measurement devices

- Dust Mask (e.g.(→ Fig. 52 Page 31))
- Displacer exchange kit - 10119990.
- Cold Head Service tool ((→ Fig. 26 Page 19), (→ Fig. 27 Page 19)); order number 4765 611 KE999.
- Displacer exchange tool kit; 8395753 necessary on Vision upgrade systems
- Helium gas 50 l / 200 bar, grade 5.0 (99.9990%).
- Pressure regulator 200/50 bar.
- Helium irrigation and filling configuration (→ Fig. 28 Page 20).

2.1.2 Spare parts

Tab. 1 Spare Part

	Displacer	LICH II displacer with control disc 74 61 333
Harmony / Symphony		x
Sonata		x
Harmony Impact/Expert	x	
Symphony Vision	x	
Sonata Vision	x	



The displacers are packed air tight in a container filled with gaseous helium.

- Control disc; order number 1562 516 KE030, not for LICH II
Control disc for LICH II displacer is part of the displacer order number
- Displ. kit (O-rings, rotating valve)
- The Displacer exchange kit includes all tooling and components required within the one kit, Displacer exchange kit - 10119990.

Fig. 23: Displacer exchange kit - Top tray

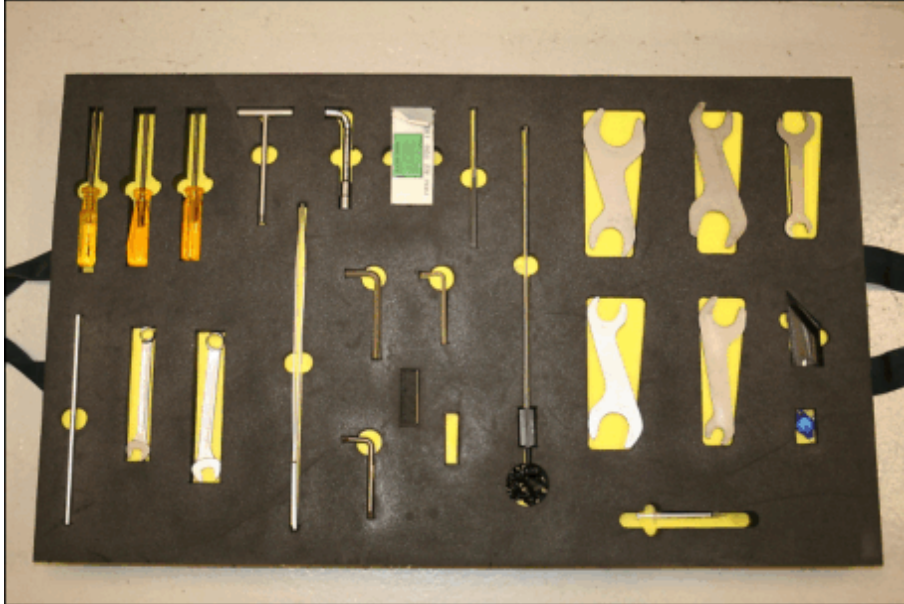


Fig. 24: Displacer exchange kit - Middle tray

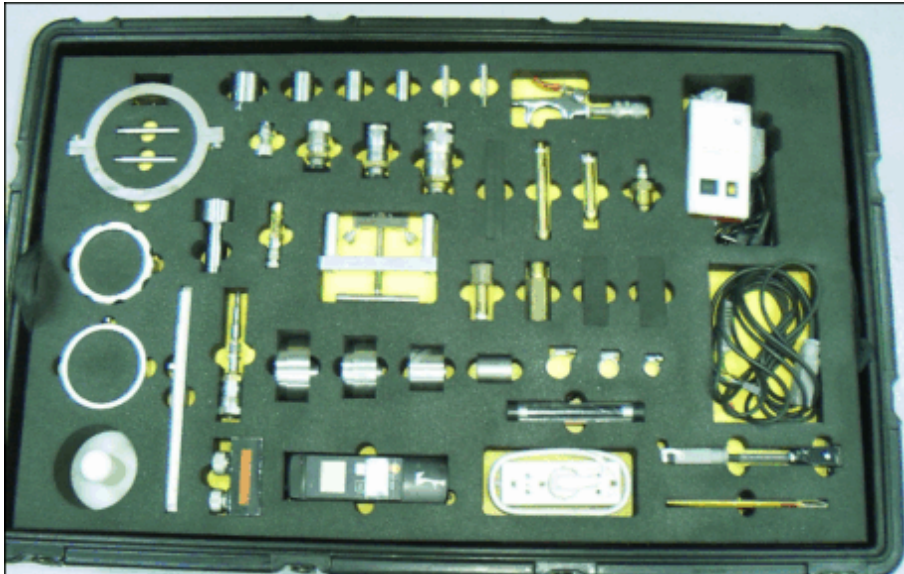


Fig. 25: Displacer exchange kit - Bottom tray

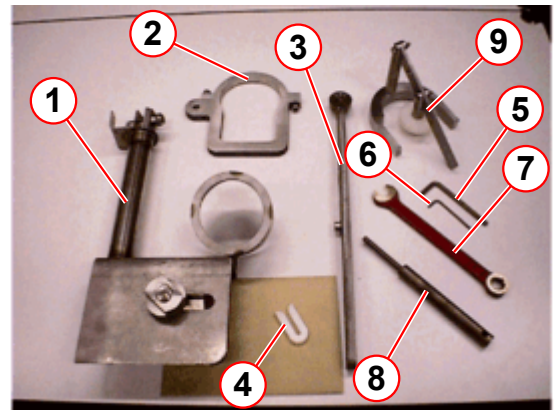


The cold head service tool allows displacer exchange with magnet at field.

Fig. 26: Cold Head Service Tool Kit



Fig. 27: Cold Head Service Tool Kit

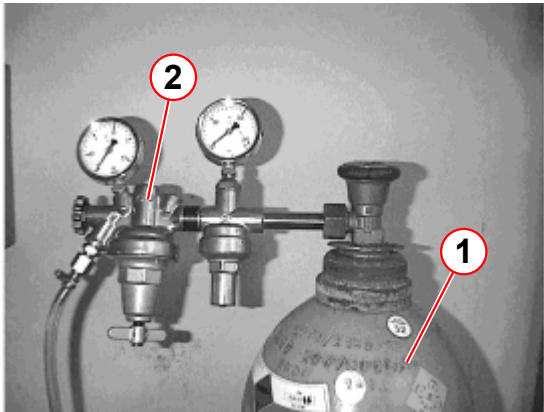


- (1) Cold head tool base
- (2) Motor clamp
- (3) Motor clamp arm
- (4) Iron shield clip
- (5) M6 Allen key
- (6) M3 Allen key
- (7) M17 combination spanner
- (8) Socket spanner for motor clamp
- (9) Control disc insertion tool



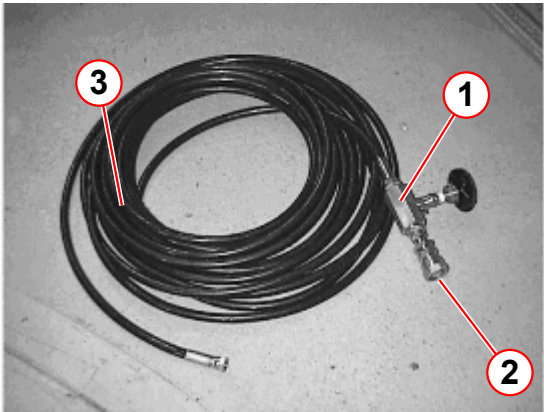
M6 and M3 Allen key are made from Beryllium Copper, use allen keys from the "Titanium Tool Ser MR".

Fig. 28: Irrigation configuration



- (1) Gas bottle filled with 5.0 (99.999%) helium
- (2) Pressure regulator

Fig. 29: Irrigation configuration



- (1) Shut off valve
- (2) Adapter Aeroquip 1/2" f
- (3) High pressure hose (Length 10 m)

2.1.2.1 Source of the supplier for the high pressure hose:

REGENT HOSE & HYDRAULICS	Tel: 0993 778278/9
UNIT 2 MINISTER INDUSTRIAL	Fax: 0993 779018
DOWNS ROAD MINSTER LOVELL	
WITNEY	
OXON OX8 5QS	
England	

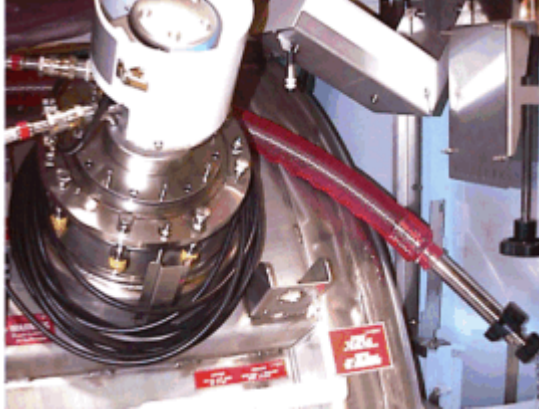
Order number	
Pressure hose:	RH/OMT/02515 SS
Adapter:	RH/OMT/S2-8/S5-4SP

2.1.3 Installation of the Cold Head Service Tool

- Remove the upper longitudinal support bar from the magnet cover frame to allow clear access to the cold head.
- Remove the 4 nuts securing the frequency converter to the magnet frame. Place the converter in a safe position.
- Remove the loud speaker.

- Remove the 4 M4 screws which secure the helium syphon. Remove the syphon end cap and pass the syphon back through the bracket. Refit the end cap (→ Fig. 30 Page 21).

Fig. 30: Cold head

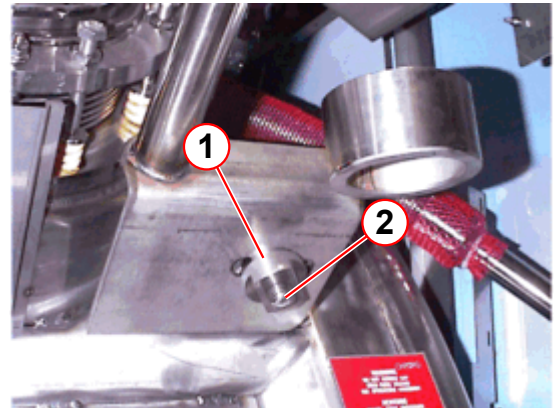


- Pass the bolt "A" through the syphon bracket and fix the tool base to the bracket hand tight with nut "B" (→ Fig. 31 Page 21), (→ Fig. 32 Page 21).

Fig. 31: Cold head tool base



Fig. 32: Tool base



- (1) Nut "B"
(2) Bolt "A"

2.1.3.1 Symphony systems only



CAUTION

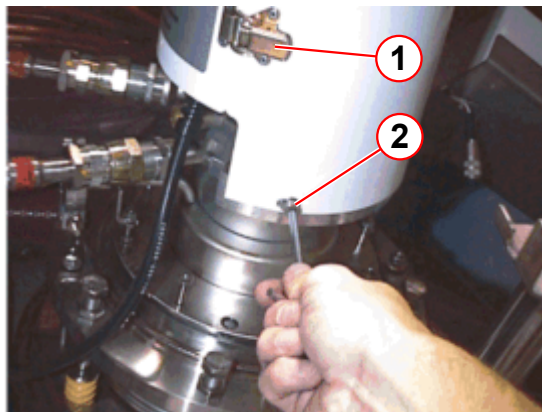
Cold head shield is magnetic

If not observed, serious injury or property damage can occur.

- » Do not attempt to remove the cold head iron shield with the magnet at field.

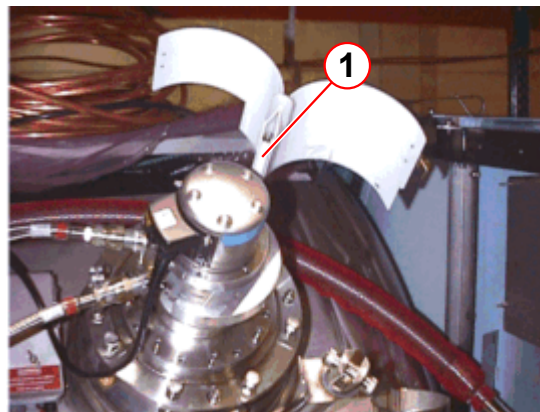
- Remove the screw "C" of the iron shield, using the non-magnetic M6 Allen key (→ 2/ Fig. 33 Page 22).
- Release the clasp "D" and open the wings of the iron shield (→ 1/ Fig. 33 Page 22). Hold the wings together using the plastic clip (→ Fig. 34 Page 22).
- Keep the iron shield in position.

Fig. 33: Iron shield



- (1) Clasp "D"
(2) Screw "C"

Fig. 34: Iron shield wings

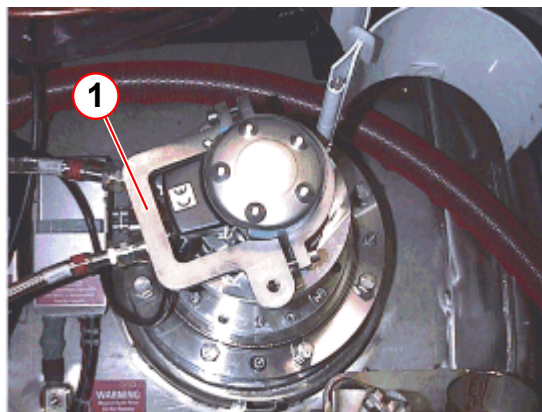


- (1) Plastic clip

2.1.3.2 Harmony and Symphony

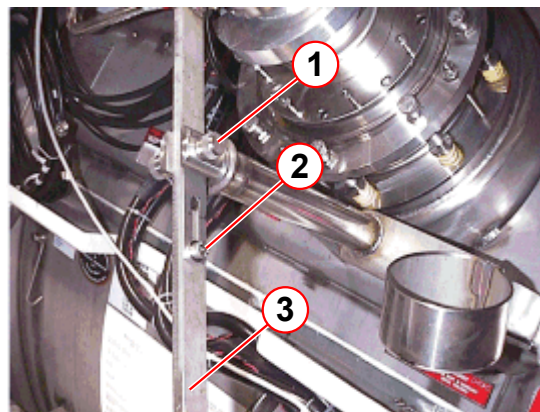
- Fit the motor clamp around the middle of the coldhead motor and secure it hand tight. Using the M6 Allen key (→ Fig. 35 Page 22).

Fig. 35: Motor clamp



- (1) Motor clamp

Fig. 36: Motor clamp arm

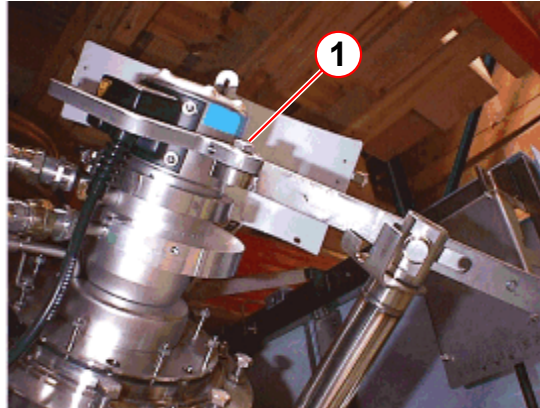


- (1) Pivot bolt
(2) Bolt "E"
(3) Motor clamp arm

- Place the motor clamp arm in the slot of the tool base and loosen the locking plate bolt "E" (→ 2/ Fig. 36 Page 22).
- Fix the motor clamp arm with pivot bolt (→ 1/ Fig. 36 Page 22).

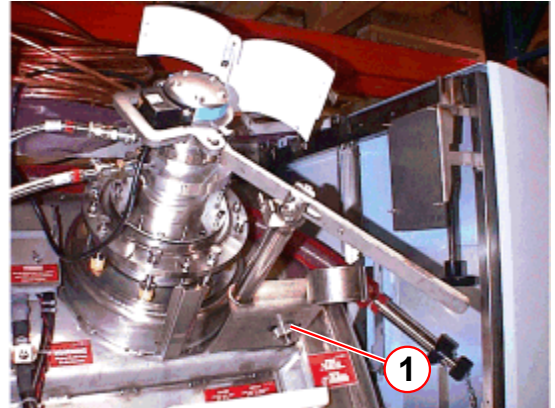
- Slide the motor clamp arm forward and attach hand tight to the motor clamp with bolt "F" (→ 1/Fig. 37 Page 23). It will be necessary to loosen the locking plate bolt "E" (→ 2/Fig. 36 Page 22) to align all parts.

Fig. 37: Cold head tool



(1) Bolt "F"

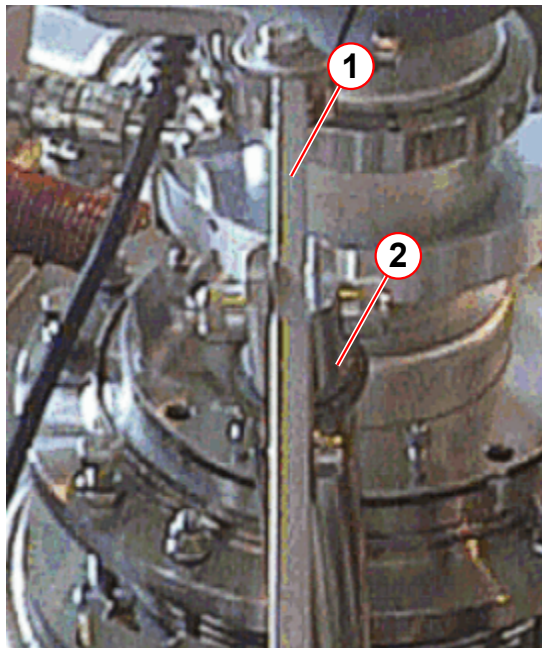
Fig. 38: Cold head tool



(1) Nut "B"

- When all parts have been aligned, secure bolt "F" (→ 1/Fig. 37 Page 23).

Fig. 39:



(1) Motor clamp arm
(2) Tool base



The motor clamp arm must be aligned with the tool base.
Ensure that the clamp arm is not tilted.

- Secure the tool base with nut "B" (→ 1/Fig. 38 Page 23), use the 1 3/16" non-magnetic spanner.
- Secure locking plate with bolt "E" (→ 2/Fig. 36 Page 22).

- Secure motor clamp (→ Fig. 35 Page 22).

2.1.4 Replacing the Displacer

2.1.4.1 Safety recommendations



The cold head, compressor, elbows, adapters, and pressurized hoses are filled with helium under high pressure.

The static pressure is approximately 12 - 16 bar, Operating pressure will be approximately 19 - 22 bar.



CAUTION

Check if all service tools are non-magnetic.

If not observed, serious injury or property damage can occur

» Use only nonmagnetic tools

2.1.5 Order of changing the displacer

1. (→ Preliminary steps / Page 24)
2. (→ Disconnecting the helium pressure lines from the cold head / Page 25)
3. (→ Depressurizing the cold head / Page 26)
4. Removing the cold head motor and the spacer
5. (→ Removing the control disc / Page 27)
6. (→ Removing the displacer / Page 28)
7. Replacing the rotating valve
8. Preparing the control disc
9. Heating and cleaning the cold head
10. (→ Replacing the Displacer / Page 24)

2.1.5.1 Preliminary steps

Establishing the irrigation configuration.

Establishing the configuration for irrigating and refilling the cold head with helium.

The configuration is shown in (→ Fig. 28 Page 20).

Purge the pressure hose.

**CAUTION**

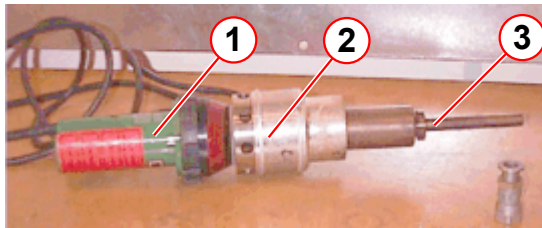
Helium gas bottle is magnetic.

If not observed, serious injury or property damage can occur.

» Place it outside of the magnet room.

Assembling the heat gun

Fig. 40: Heat gun



- (1) Heat gun
- (2) Adapter ring
- (3) Adjustment screw



The heat gun contains magnetic parts.



The flow of the heated air to the two stages can be regulated with the adjustment screw on the hot air cylinder.

2.1.5.2 Disconnecting the helium pressure lines from the cold head

Tools

1 3/16" - 1 1/8" wrench

- Switch off the compressor.
Switch off F3 in the RCA
- Disconnect both pressure lines from the cold head.

	RGD 5100
1/2" = high pressure connection	Red = high pressure connection
1/2" = low pressure connection	Green = low pressure connection

Working with self-sealing couplings

Fig. 41: Attaching and detaching self sealing couplings

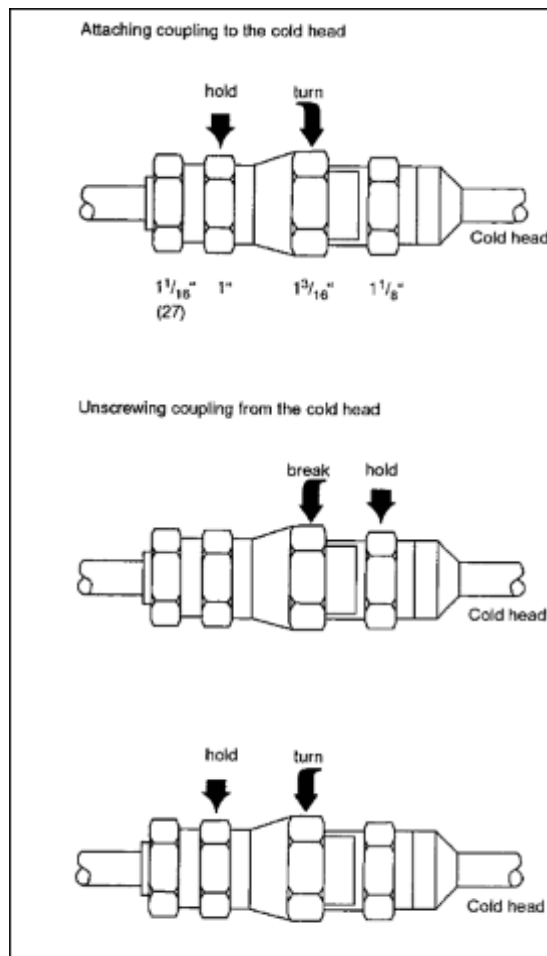
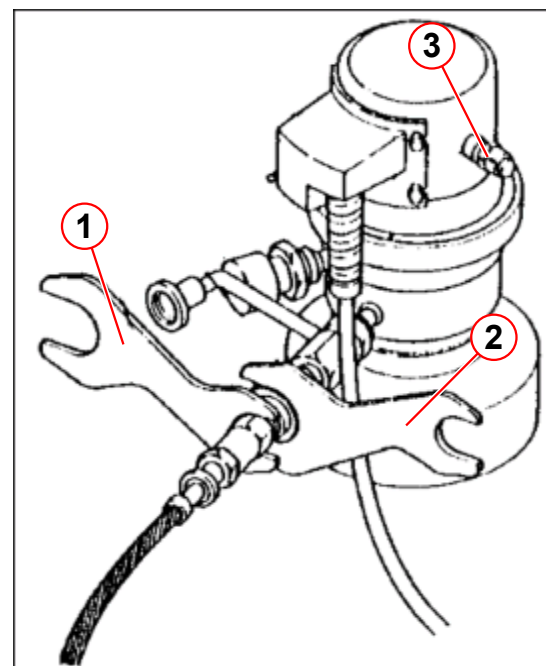


Fig. 42: Disconnecting the helium pressure hoses



Do not twist the pressure hose. Use a second wrench when removing the Aeroquip couplings

2.1.5.3 Depressurizing the cold head

Tools

- Adapter, Aeroquip 1/2"f/DN 16 small flange
- Wrench 1 3/16" - 1"



The adapter Aeroquip contains magnetic parts.

Fig. 43: Depressurizing the cold head



- Connect the Aeroquip adapter 1/2"f/DN 16 small flange to the high pressure connection of the cold head. It should open up slightly allowing the helium to escape.
- Connect the Aeroquip adapter 1/2"f/DN 16 small flange to the low pressure connection of the cold head.
- Connect the hose for the helium irrigation and filling configuration to the high pressure connection. Ensure that the self-sealing mechanism of the Aeroquip does not open up.

2.1.5.4 Removing the motor unit

Tools from the Cold Head Service Tool Kit

- 10 mm socket wrench (→ 8/ Fig. 27 Page 19)

Fig. 44: Removing the cold head motor



- Remove clamp with socket wrench.
- Remove the motor from the cold head by gently pushing down on the motor clamp arm.
Park the clamp arm in the cradle of the tool base.
- Remove the spacer.

2.1.5.5 Removing the control disc

Tools

- Extractor for control disc.

Fig. 45: Removing the control disc



- Position the extractor on the housing of the cold head.
- Insert the two knurled screws up to the end stop into the threaded holes of the control disc (→ Fig. 45 Page 28).
- Turn the threaded spindle in the center of the extractor to unseat and pull out the control disc.

2.1.5.6 Removing the displacer

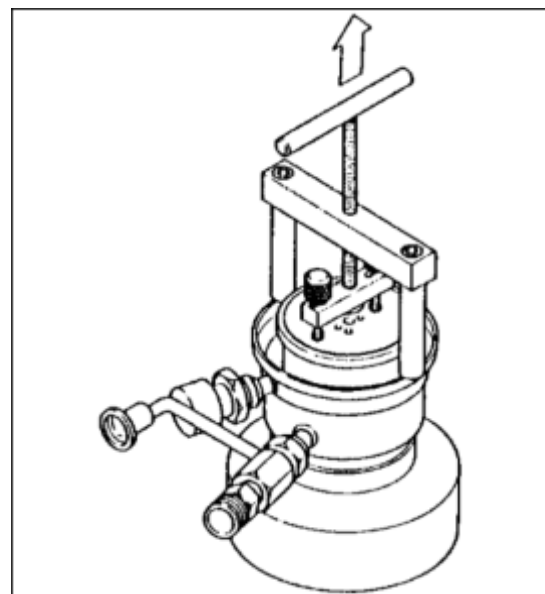
Tools

- Extractor for displacer or
- Extractor for guide sleeve

Fig. 46: Removing the displacer with extractor



Fig. 47: Removing the displacer with extractor for control disc



- Install the extractor.
- Pull out the displacer.

2.1.5.7 Replacing the control valve

Tools

- File
- Power supply for motor

Material:

- Paper towels, cleaning fluid
- New control valve

Fig. 48: Removing the control valve



- Remove the old control valve and don't lose the spiral spring.
- Ensure that the new control valve is leveled using a hairline ruler. Clean the valve and reattach it to the motor shaft.
- Check whether the valve moves smoothly in its support in both the lateral and the vertical direction when pressure is applied to the spiral spring. If necessary, widen the slot in the valve with a file.
- Check the direction of rotation of the valve. Disconnect the motor cable from the filter plate and connect it to the power supply. Switch on the power supply. The valve should turn in clockwise direction.

2.1.5.8 Prepare the new control disc

Material

- Paper towels and cleaning fluid
- O-rings
- Clean the control disc
- Provide the new control disc with new O-rings

Fig. 49: Control disc

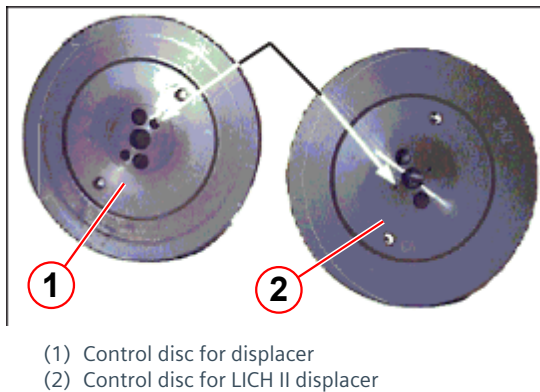
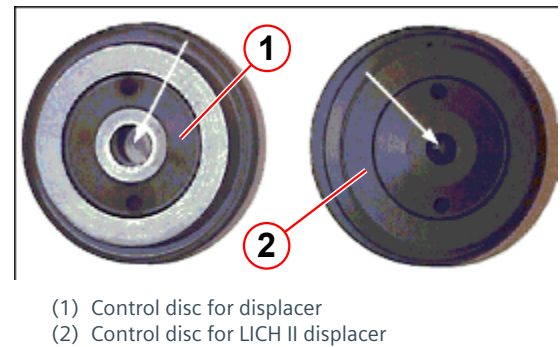


Fig. 50: Control disc



Diameter of the drill holes are different between both control disc

2.1.5.9 9) Heating and cleaning the housing of the cold head



WARNING

Danger of lead contamination during the heating process.

Dispersed lead dust may be inhaled and could result in personal injury.

- » **Wear a respirator mask to a minimum filter standard of P2 (refer to (→ Fig. 51 Page 31) or (→ Fig. 52 Page 31)) during the cold head heating process.**

The following Safety precautions **must** be followed.

- Wear a protective work apron and protective gloves (especially if there are open wounds on the hands).
- It is prohibited to eat, drink and smoke while working with lead/ displacers.
- Never touch the eyes or the mouth with your hands while working with lead/ displacers.
- Thoroughly wash your hands with soap and water after completion.
- Hazardous materials containing lead may not be placed or stored in containers which, due to their shape or labeling, may be confused with food containers..
- Do not dispose of lead scrap (lead remnants and production resources contaminated with lead) into waste receptacles. Collect the lead in a separate container and have it collected up as special refuse by the responsible department according the local legal requirements.

Our refrigeration supplier has recommended the use of personal inhalation protection during cold head displacer exchanges. The reason is that lead dust may come out of the cold head during the heating process. The 2nd stage of the displacer is filled with lead balls.

The European standards that are applicable are listed below. Any equipment procured in the United States must adhere to OSHA standards and be NIOSH / MSHA certified as suitable for the task specified.

Tab. 2 European Standards

Description	European Standard
3M - Model 6000 Half Mask	EN 140
3M – 2135 (P2) Filter	EN 143

Tools

- Heat gun
- Thermo sensor
- Flash light

Tools

- Pair of long tweezers

Material

- Dust Mask (e.g. (→ Fig. 52 Page 31))

Fig. 51: Example for an appropriate mask



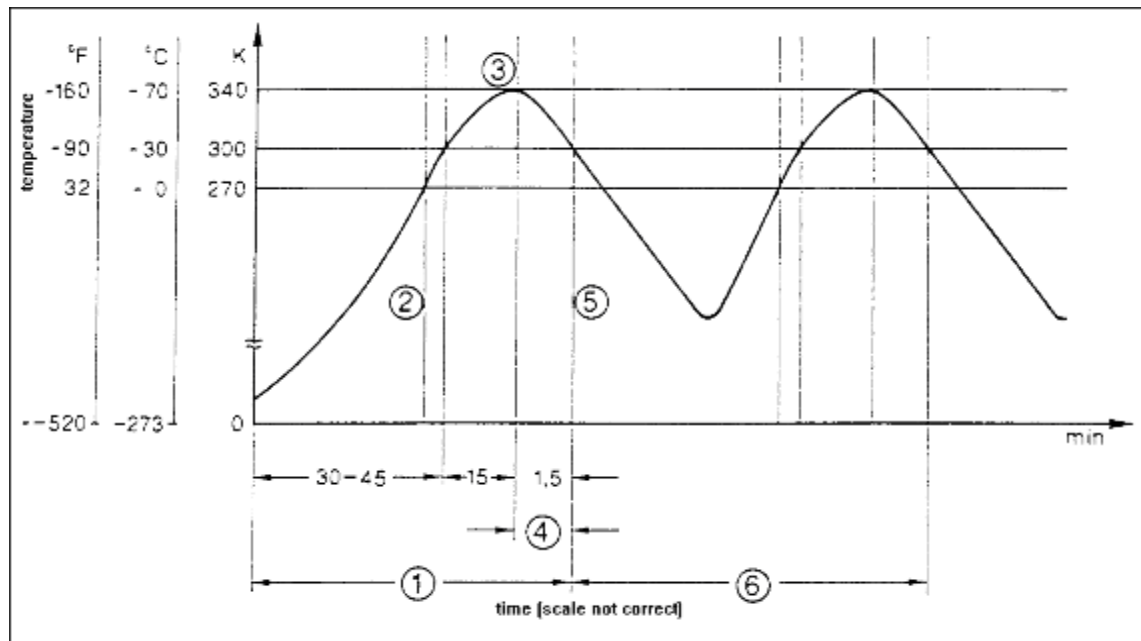
Fig. 52: Mask with part number 70 14 868



- Paper towels and cleaning fluid

Working with the heat gun

Fig. 53: Diagram for working with the cylinder heater



2.2 Explanations for the diagram

1. Time interval required to reach the operating temperature of the cold head
T < 100 Kelvin (= -170° Celsius) and the cooling period after the critical temperature
T = 300 Kelvin (= 30° Celsius), (one cycle)!
2. T = 270 Kelvin (= 0° Celsius). Melting point for ice. Wipe off the water droplets using a cloth or let them evaporate. You can now clean the cylinder.

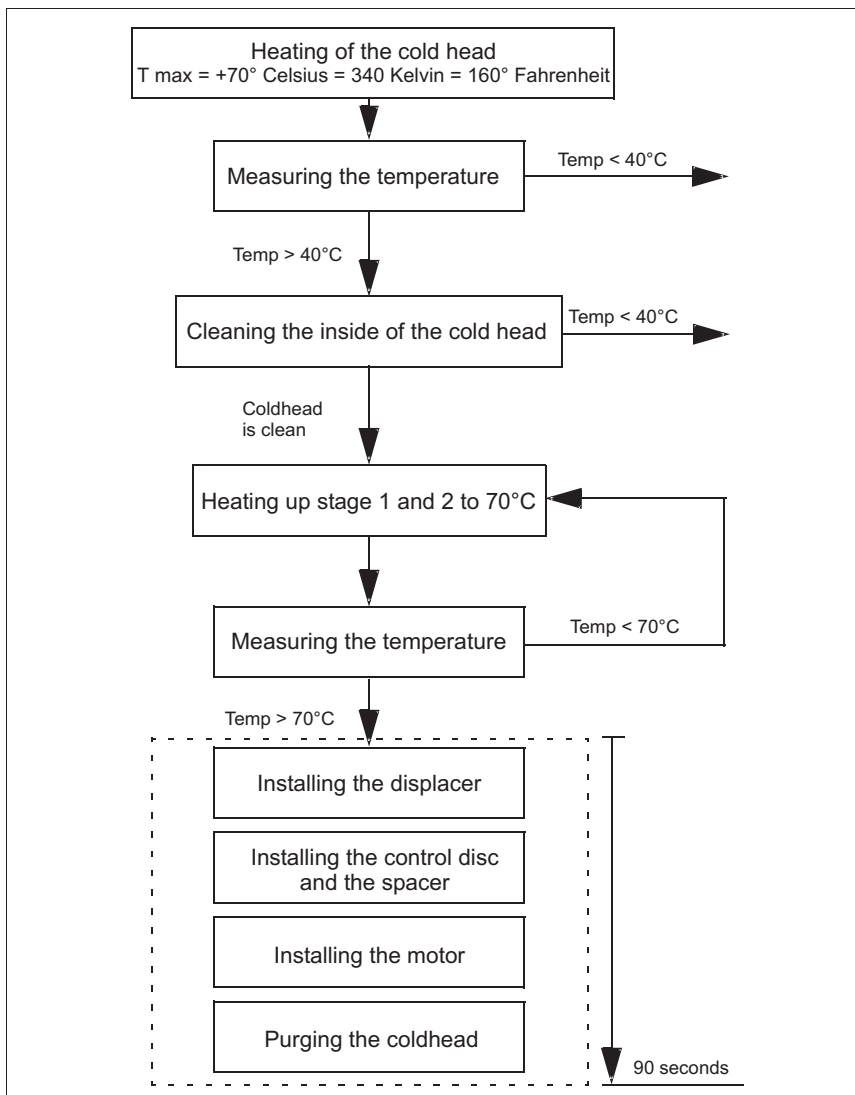
When the temperature falls below 270 Kelvin (= 0° Celsius), the drying and cleaning process must be repeated because the temperature inside the cylinder will reach the dew point for humidity.
3. T = 340 Kelvin (+70° Celsius). Maximum temperature threshold which should not be exceeded to protect the copper band and the cold head.
4. **Important step:**

After cleaning the cylinder, you have approximately 90 seconds to reinsert the displacer, reattach the control disc, spacer and the motor, close the cold head, and irrigate it with 5.0 helium.

Perform these steps quickly and carefully. In case of delays, impurities will freeze to the walls of the cylinder.
5. T = 300 Kelvin (+ 30° Celsius). Critical temperature level. If the temperature falls below this level, water will begin to condense at the cylinder walls. The cylinder will need to be dried again.
6. Item one (time interval ...) can be repeated as required.

2.3 Order of installation of the new the displacer

Fig. 54:



2.3.1 Heating of the Cold head

The heat can be set with the control knob at the heat gun.

- Set the heat to power stage "seven" and switch on the gun. Subsequently, reduce the setting to power stage "four" or lower. Both stages should show the same warm-up temperature. Measure the temperature with the thermo sensor at least once every minute.

$T_{\max} = +70^{\circ} \text{ Celsius} = 340 \text{ Kelvin} = 160^{\circ} \text{ Fahrenheit}.$

Fig. 55: Cleaning the displacer cylinder



Overheating will destroy the housing of the cold head. It will also damage the cryostat which in turn will cause helium boil-off.

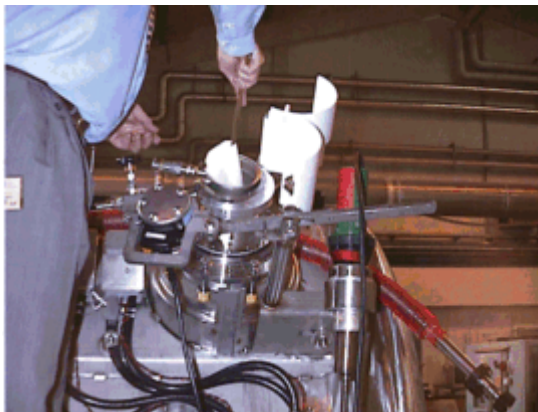
2.3.2 Cleaning the inside of the cold head

- Use paper towels to clean the inside wall of the housing of the cold head. Subsequently, clean the entire inside wall by wrapping a cloth soaked in cleaning fluid around a pair of long tweezers.



Do not use any towels containing plastic material. It will melt due to the high temperature (+40 - +70 °C) causing severe contamination of the surface.

Fig. 56: Cleaning the displacer cylinder



- Let the residual film of cleaning fluid evaporate.
- To remove the lint, purge the inside wall with helium gas.



Do not scratch the contact surface of the cold head. The honed inside wall of the housing of the cold head is highly sensitive to scratches.

2.3.3 Installing the new displacer

2.3.3.1 Tools

- Ring compressor for control piston
- Ring compressor for first stage
- Ring compressor for second stage

2.3.3.2 Material

- Container with new displacer
- Cotton gloves
- Irrigation and fill configuration
- Helium gas 5.0

Establish the purging configuration before heating the cold head.

- Connect the Aeroquip coupling to the high pressure connection.

Fig. 57: New displacer



Fig. 58:



2.3.4 Preparing the displacer

- Unpack the new displacer. The transport container protects the new displacer against humidity. The displacer is packed in helium gas under atmospheric pressure.
- For this reason, the container should remain closed when not in use. Wear cotton gloves when removing the displacer from the transport container or when replacing the old displacer. Expose the container only briefly to the ambient air to avoid contamination.
- Check the seals (control piston, first and second stage) at the new displacer. Press the seals down by using the respective ring compressors.



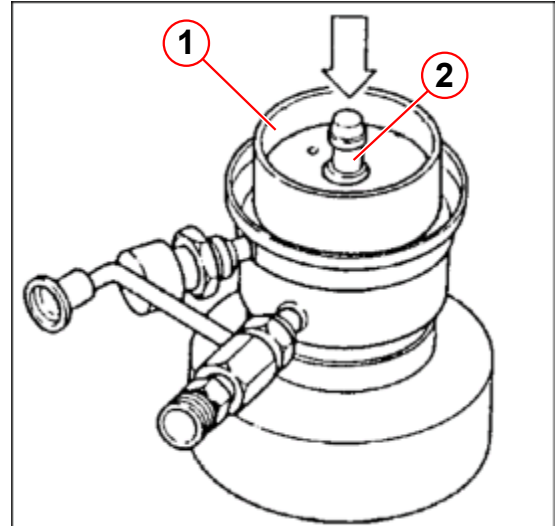
Do not use force when sliding the ring compressor over the seals.

- Hold the ring compressor with the cone-shaped side pointing away from you. Turn the ring compressor slightly and slide it over the first and second stage as well as the control piston so that the seals are pressed down. Subsequently, remove the ring compressor from the second stage. Turn the displacer slightly while inserting it quickly and carefully into the housing of the cold head (→ Fig. 59 Page 37).

Fig. 59: Inserting the displacer (RGD 5100)



Fig. 60: Removing the ring compressor



- (1) Ring compressor
- (2) Control piston

- Ensure that the seals of the second stage do not move out from the displacer. The ring compressors of the first stage and the control piston should not be removed until the displacer can be pressed up to the end stop into the housing (→ Fig. 59 Page 37).

2.3.5 Installing the control disc

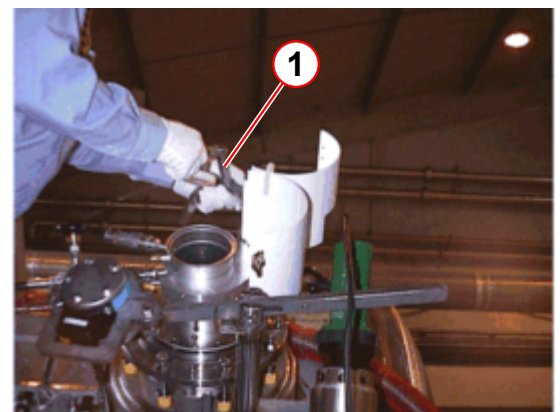
2.3.5.1 Tool from the Cold Head Service Tool Kit

- Control disc insertion tool.

Fig. 61: Inserting and pressing down the control disc (RGD 5100)



Fig. 62:



- (1) Insertion tool

- Press down on the clean control disc with both thumbs and carefully insert it into the housing of the cold head (up to the end stop). Use the control disc insertion tool.
- Check that the gas bores are clear.

2.3.6 Installing the motor

2.3.6.1 Tools from the Cold Head Service Tool Kit

- 10 mm socket wrench



The flange of the motor and of the coldhead will be hot from the hot air of the heat gun during heating the cold head.

Fig. 63: Motor clamp



- Insert the rubber spacer with the O-ring. Make sure that one of the four holes of the spacer is placed between the two helium gas connections (high and low pressure).
- Swing the motor toward the cold head.
- Slide the clamp over it and tighten it.



Attach the motor so that the connection cable is routed between the high pressure and low pressure connection.

2.3.7 Purging and filling the cold head with helium

2.3.7.1 Tools

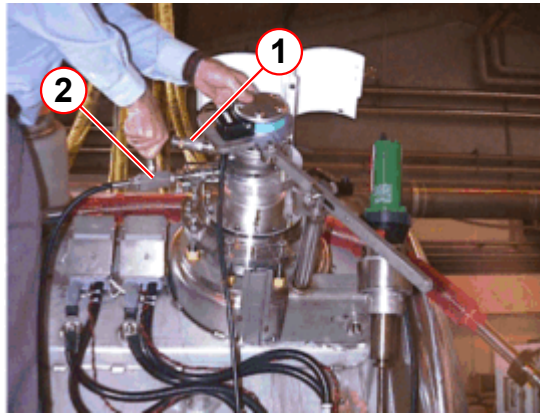
- Wrench
- Adapter, Aeroquip 1/2"/DN 16 small flange.

2.3.7.2 Material

- Irrigation and fill configuration.
- Helium gas 5.0.

Establish the purging configuration before heating the cold head.

Fig. 64: Purging the cold head



- (1) Adapter
(2) Shut-off valve

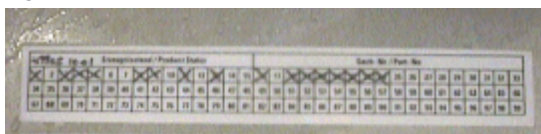
- Attach the Aeroquip coupling 1/2" (→ Fig. 63 Page 38) to the high pressure hose of the fill configuration to the red high pressure connection of the cold head. Slowly open the bottle valve, the pressure regulator and the shut-off valve. Slowly fill the cold head up to 5 bar with helium.
- Attach Aeroquip adapter 1/2"f/DN 16 small flange to the green low pressure connection to allow helium to flow from the cold head.
- Activate the switch at the power supply for the motor.
- The helium escapes in pulses from the cold head via the green low pressure connection. After approximately 5 pulses, close the shut-off valve (→ Fig. 64 Page 39) while the motor is running. After the helium of the last pulse has escaped, open the shut-off valve again. Keep it open for another 5 pulses, then close the valve.
- Open and close the shut-off valve approximately 5 times.
- Disconnect the Aeroquip adapter from the low pressure connector of the cold head, while the motor is running and the gas escapes.
- Switch off the motor power supply.
- Fill the cold head to 14 - 15 bar, close the shut-off valve. Disconnect the high pressure hose from the cold head.
- The replacement procedure for the displacer is now completed.

2.3.8 Start-up of compressor

- Reconnect both pressure helium lines to the cold head, high pressure first.
- Reconnect the motor cable.
- Switch on F3.
- Switch on the compressor.
- Check the helium fill pressure.

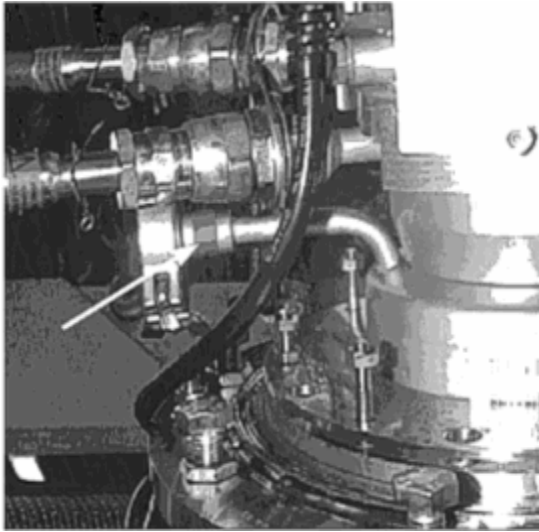
2.3.9 Final check if LICH II displacer was installed

Fig. 65: Revision level



- Mark the revision level if necessary
ES 22 for OR70 (1.5 Tesla)
ES 21 for OR24 (1.0 Tesla)

Fig. 66:

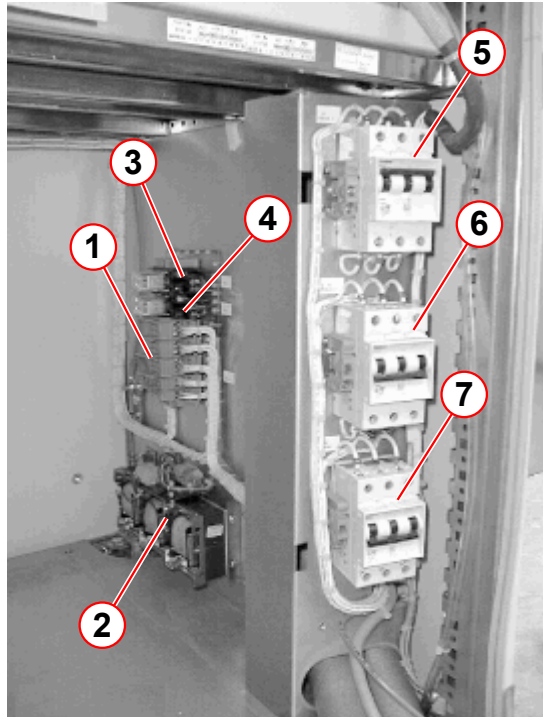


- Mark the cold head with red tape at the vacuum pumping port.

2.4 Electrical components and their location in the RCA

The electrical components (transformer, relays) are located to the right of the helium compressor. You can access them only after the compressor has been removed.

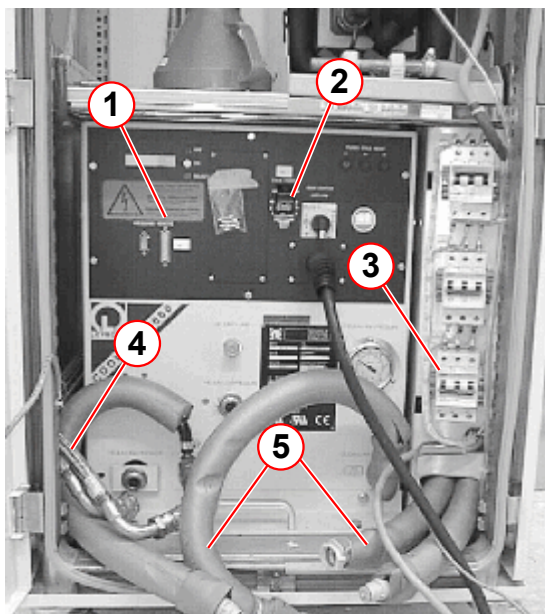
Fig. 67: Electrical components



- (1) X102
- (2) T1
- (3) K1
- (4) K2
- (5) F1
- (6) F2
- (7) F3

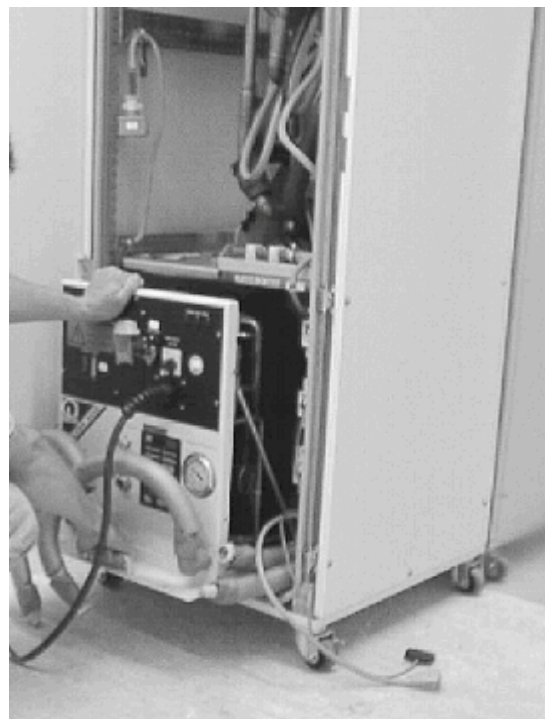
2.5 Replacement of the Compressor (COOLPAK)

Fig. 68: Coolpak



- (1) X7
- (2) X5
- (3) F3
- (4) Flexlines
- (5) Water lines

Fig. 69: Coolpak



2.5.1 Removing the COOLPAK

Step	Instructions	Required Tools
1	Switch off compressor Switch off F3 Switch off F4 on top of CCA	
2	Disconnect the water lines	
3	Disconnect the high and low pressure helium flexlines	Wrench 1 3/16" and 1"
4	Remove cover from circuit breaker Only necessary if the power cable is not pluggable	
5	Unplug the power cable or Disconnect power cable from F3 Notice: Input voltage on top Output voltage at the bottom	Screw driver

Step	Instructions	Required Tools
6	Disconnect cable X7 and X5	
7	Unscrew compressor from the base plate	Spanner
8	Pull out the compressor (weight approx. 100kg) The front of the compressor should be placed on top of a sturdy support.	

2.5.2 Installing the COOLPAK

Step	Instructions	Required Tools
1	Slide compressor into the cabinet	
2	Tighten the compressor on the base plate	Wrench
3	Connect the line voltage cable Verify that the direction of rotation is correct	
4	Install cover for circuit breaker	
5	Connect the high and low helium flexlines 1. High-pressure (red) 2. Low-pressure (green)	Wrench 1 3/16" and 1"
6	Connect the water lines	
7	Connect cable X7 and X5	
8	Check the helium pressure on the pressure gauge (compressor off) Operation at 50 Hz: Nominal value: 15 bar (218 psig) Minimum value: 14 bar (203 psig) Operation at 60 Hz Nominal value: 14 bar (203 psig) Minimum value: 13 bar (189 psig)	
9	Switch on F4 Switch on F3 Switch on the compressor Typical operating pressure (compressor on) Operation at 50 Hz: 20 - 21 bar (290-305psig) Operation at 60 Hz: 19 - 20 bar (276-290psig)	

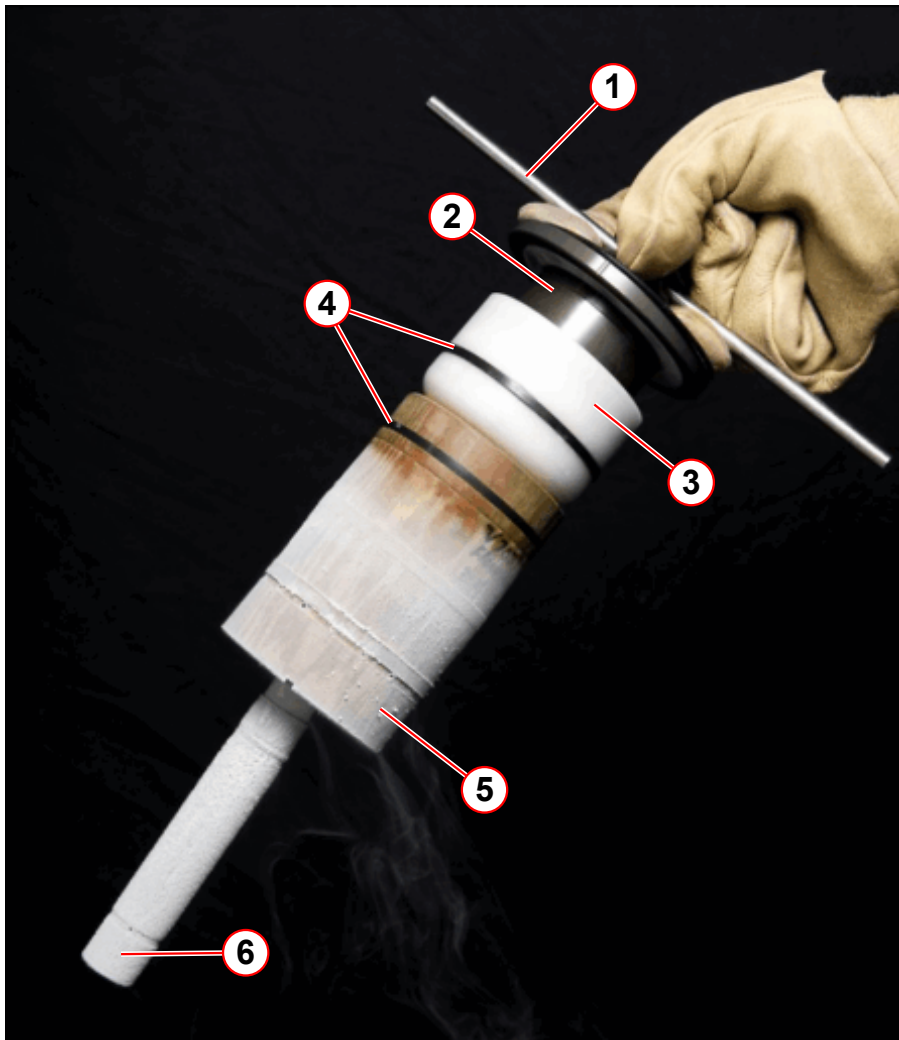


Drain the water from the pipes, before returning the replaced unit

2.6 Replacing the Displacer - APD system

2.6.1 Replacing the Displacer - APD systems

Fig. 70: Displacer APD



- (1) Extraction bar 7.5 mm
- (2) Valve stem
- (3) Slack cap
- (4) Gas seals
- (5) 80K stage
- (6) 20K stage

2.6.1.1 General

This procedure describes the displacer exchange for the APD (Whisper Plus) cold head fitted to the 10K magnet types Harmony, Symphony, Sonata, Trio (upgrade to syngo) and Impact/Expert from Vision, manufactured by Siemens Magnet Technology.



Read the procedure and check serviceability of replacement parts before carrying out the replacement process.



This displacer replacement procedure must only be performed by Certified Engineers, who have completed the relevant training course(s).

2.6.1.2 Safety recommendations



The cold head, compressor, elbows, adapters, and pressurized hoses are filled with helium under high pressure. The static pressure is approximately 12 - 16 bar. The operating pressure will be approximately 19 - 22 bar.



WARNING

Pressurized Helium Vessel.

Failure to observe the following safety information may result in physical injury.

- » Do not disconnect any of the turret fittings without first depressurizing the helium vessel.
- » Only trained and accredited engineers must perform this procedure.



WARNING

Risk of Asphyxiation.

Failure to observe the following may result in dizziness and loss of consciousness.

- » Do not vent helium gas directly into the an enclosed place, with no ventilation.
- » Restrict unauthorised access to the Imaging room.
- » Personal Oxygen monitors must be worn if a fixed room monitor has not been installed.



CAUTION

Risk of Cold Burns.

Failure to observe the following safety information may result in physical injury.

- » Do not touch cold surfaces with bare hands.
- » Personnel safety equipment must be worn.

**CAUTION**

Risk of Hot Burns.

Failure to observe the following safety information may result in physical injury.

- » Do not touch hot surfaces with bare hands.
- » Personnel safety equipment must be worn.

**WARNING**

Strong Magnetic Fields.

Failure to observe the following safety warning may result in physical injury.

- » Only non-magnetic equipment must be taken into the magnet room, when a magnetic field is present.

**CAUTION**

Flammable Cleaning Fluids.

Failure to observe the following safety caution may result in burns.

- » Protective clothing, gloves and eye protection must be worn.
- » Follow the manufactures instructions when handling flammable cleaning fluids.

**WARNING**

Danger of LEAD CONTAMINATION during the heating process.

Dispersed lead dust may be inhaled and could result in personal injury.

- » Wear a respirator mask to a minimum filter standard of P2 during the cold head heating process.

The following Safety precautions **must** be followed.

- Wear a protective work apron and protective gloves (especially if there are open wounds on the hands).
- It is prohibited to eat, drink and smoke while working with lead/ displacers.
- Never touch the eyes or the mouth with your hands while working with lead/ displacers.
- Thoroughly wash your hands with soap and water after completion.
- Hazardous materials containing lead may not be placed or stored in containers which, due to their shape or labeling, may be confused with food containers..
- Do not dispose of lead scrap (lead remnants and production resources contaminated with lead) into waste receptacles. Collect the lead in a separate container and have it

collected up as special refuse by the responsible department according the local legal requirements.

Personal inhalation protection **MUST** be worn during this procedure.

- **Lead dust** can escape during the heating process.
- The **2nd stage** of the displacer is filled with **lead balls**.

The **European** standards applicable are listed below. Any equipment procured in the **United States** must adhere to **OSHA** standards and be **NIOSH / MSHA** certified as suitable for the task specified.

Fig. 71: Example of appropriate mask



Fig. 72: Mask (part number 70 14 868)



Tab. 3 European Standards

Description	European Standard
3M - Model 6000 Half Mask	EN 140
3M – 2135 (P2) Filter	EN 140

2.6.1.3 Spare parts

The replacement **APD displacer kit** contains the displacer and valve stem assembly in a sealed bag. The kit also contains all the replacement O-rings, rotary valve and seals required.

- APD replacement displacer kit order number **08395738**.

2.6.1.4 Tools and equipment required

- Check that all service parts, tools, and equipment are available and serviceable **before proceeding**.

The following **service tools** are required when replacing a displacer at field:

- **Displacer exchange kit – 10119990.**
- **Displacer exchange tool kit (Trio system) order number 8395753**

Further tooling requirements (to be supplied by the local engineer):

- 1 x 50 liter (200 bar) cylinder of helium gas 99.999% purity (also known as C5 or CP grade).
- Helium leak detector, capable of detecting leaks to 1×10^{-4} mbar l/s.
- Cotton gloves.
- Dust mask order number **07014868**
- **Lint free** paper towelling.
- Petroleum ether or alcohol based cleaning fluid, which leaves no oily residue.
- 5 mm **non magnetic** allen key.

2.6.1.5 Displacer exchange kit

The Displacer exchange kit includes **all tooling and components required** within the one kit, **Displacer exchange kit - 10119990**.

Fig. 73: Displacer exchange kit - Top tray

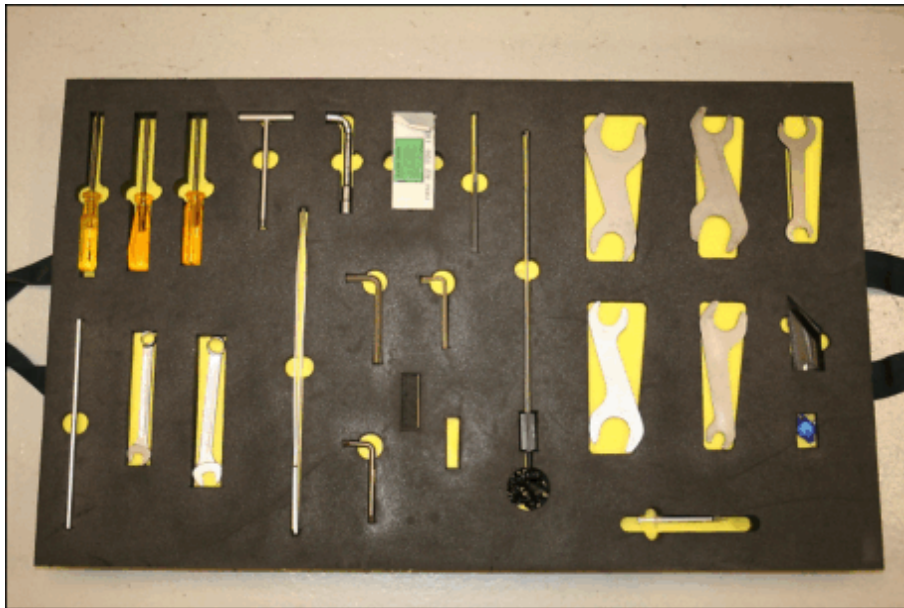


Fig. 74: Displacer exchange kit - Middle tray

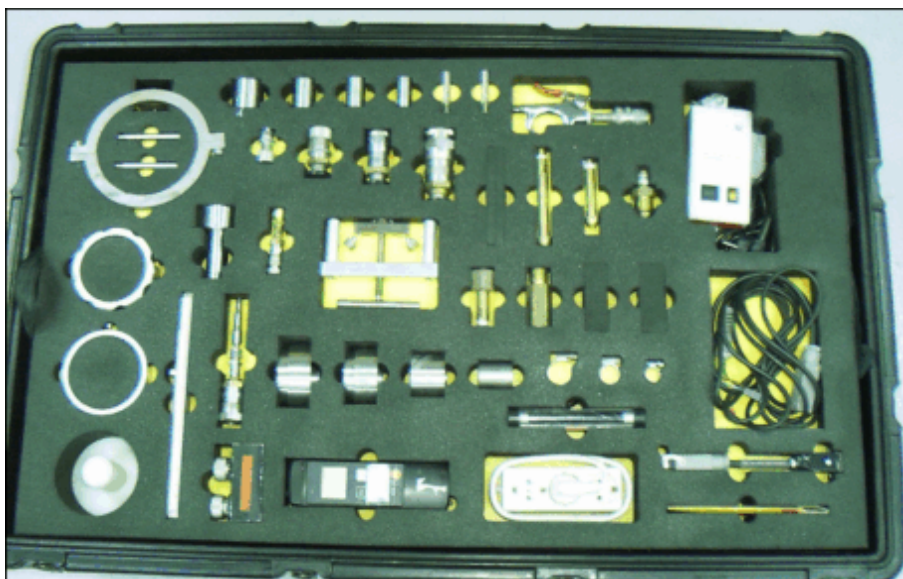


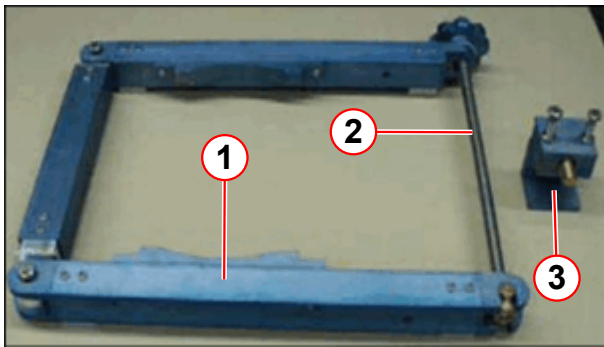
Fig. 75: Displacer exchange kit - Bottom tray



2.6.1.6 Displacer exchange tool kit (Trio magnets)

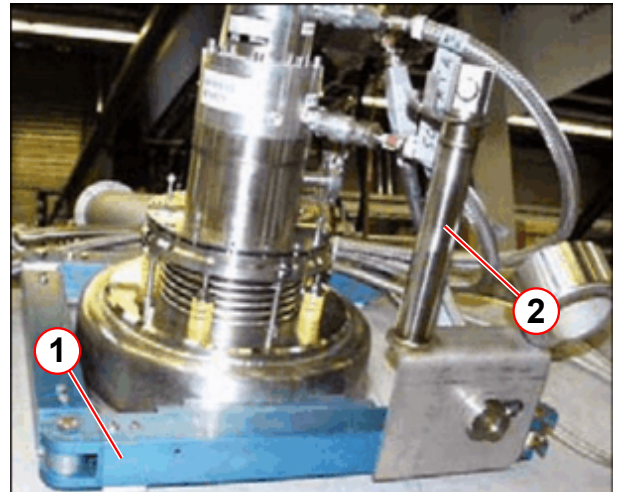
An additional tool kit is necessary if exchanging the displacer **at field**. Order part number **8395753**.

Fig. 76: Displacer exchange tool kit - Trio system



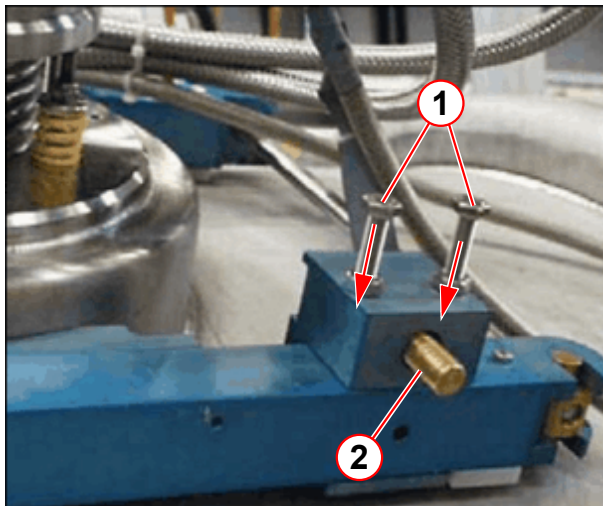
- (1) Frame
- (2) Locking bar
- (3) Clamp

Fig. 77: Displacer tooling - Trio systems



- (1) Trio exchange tooling
- (2) Motor removal tooling

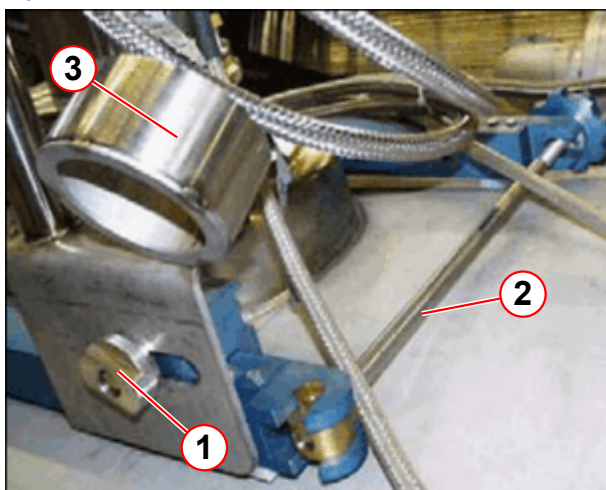
Fig. 78: Fit the clamp to the frame



- (1) Screws and nuts (2 off)
- (2) Mounting stud

- Fit the clamp to the frame using the 2 off screws.

Fig. 79: Fit the tool base to the frame

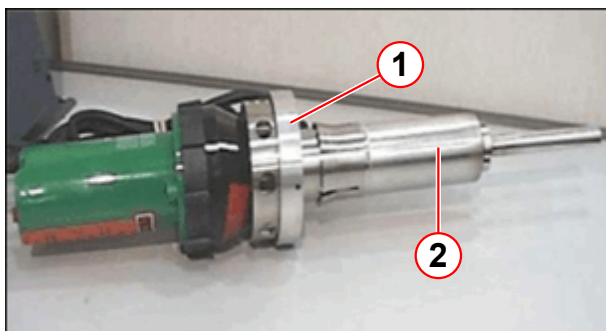


- (1) Tool base locking nut
- (2) Locking bar
- (3) Tool base and heat gun holder

- Align the mounting **stud** with the slot in the tool tray.
- Secure with the tool base **locking nut** (pos1).
- Tighten the **locking bar** (pos 2).

2.6.1.7 Heat gun assembly

Fig. 80: Heat gun tooling



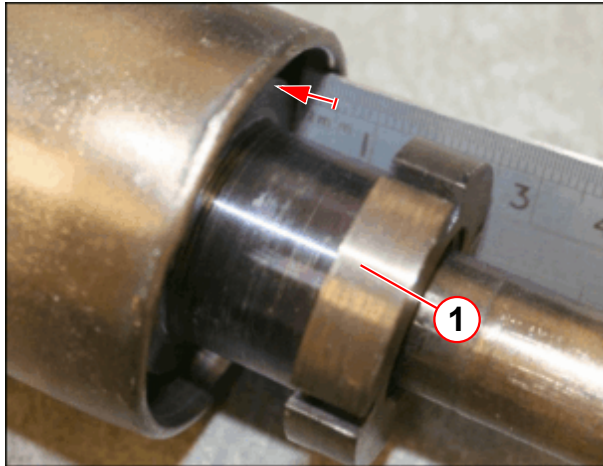
- (1) Adaptor ring
- (2) Air cylinder

- **Assemble** the adaptor **ring** and **air cylinder** onto the heat gun.



The heat gun contains magnetic parts.

Fig. 81: Set the gap to 5 mm



(1) Air flow adjustment screw

- Set the initial measurement to by turning the **air flow adjustment screw**.
 - » set the gap to **5 mm**.



The flow of the heated air to the two stages can be regulated with the adjustment screw on the hot air cylinder.

2.6.1.8 Magnetic fields



WARNING

Strong Magnetic Fields.

Failure to observe the following safety warning may result in personal injury.

- » Only non-magnetic equipment must be taken into the magnet room.



Low ceiling heights restrict the ability to work safely.

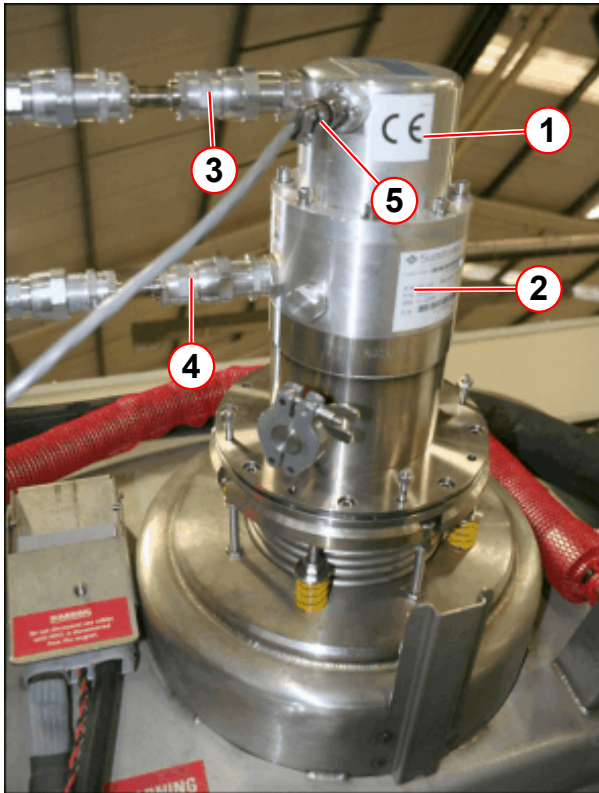
2.6.1.9 Displacer exchange preparation

- **Remove** the required **magnet covers** and **equipment** necessary to allow **clear access** to the **cold head**.
- **Position** the helium **gas cylinder**, in a **safe position** as required.

OR70 only

- To allow **clear access** to the cold head:
 - **Ensure** that the **power to the frequency converter is OFF** otherwise the magnetic field can **destroy** the frequency converter.
 - **Remove** the **4 off nuts** securing the **frequency converter** and place it in a safe location.
 - **Remove** the loud speaker.

Fig. 82: APD Cold head - OR70



- (1) Motor housing
- (2) Valve stem housing
- (3) Supply helium gas line
- (4) Return helium gas line
- (5) Cold head motor power cable

- Switch **off** and electrically **isolate** the **compressor**.
- **Disconnect** the motor **cab**le from the **cold head** (pos 5).
- **Disconnect** the **Supply** and **Return** gas lines, using three **Aeroquip** wrenches (pos 3,4).



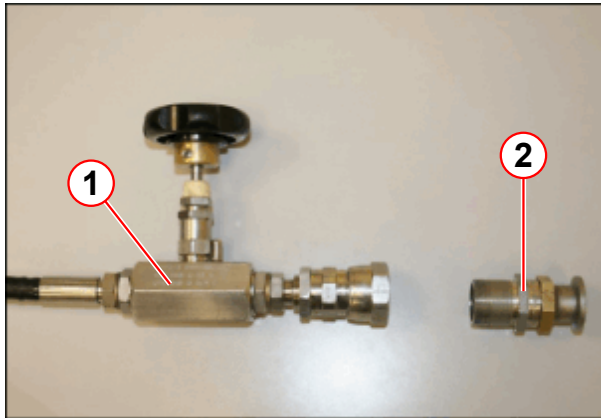
To prevent movement of the fixed Aeroquip couplings, always use three wrenches when removing the connections.

- **Connect** the **high pressure regulator** to the **gas cylinder**.
- **Connect** the **gas supply hose** to the **regulator**.



Do not connect to the cold head at this time.

Fig. 83: Gas line purge tool



- (1) Shut-off valve
(2) Male purge adaptor

- **Purge** the gas line using a **male purge adaptor** (pos 2).
- It is recommended that the **gas line** is **purged** for a minimum of **30 seconds** with a **low flow** of helium gas.

Fig. 84: Mark the orientation of the motor



- (1) Orientation mark

- Mark the **orientation** of the motor to the **cold head body** using a marker pen.

2.6.1.10 Fitment of the cold head motor tooling (at field)

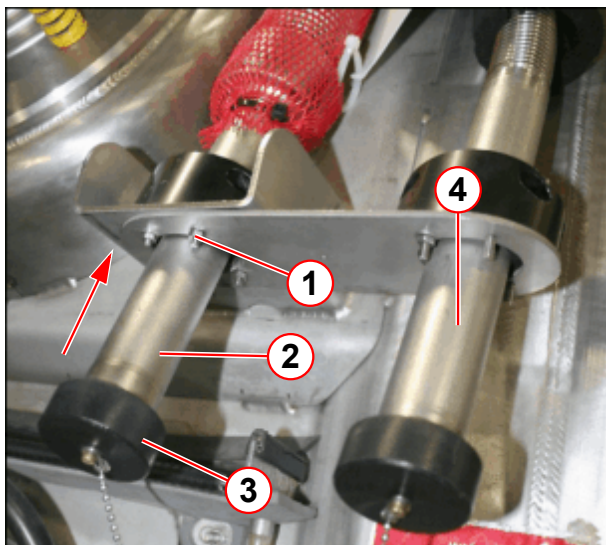


CAUTION

Check if all service tools are non-magnetic.
If not observed, serious injury or property damage can occur.
» Use only non-magnetic tools.

If the displacer exchange is being performed with the **magnet at field**, the following tooling **must be fitted** to allow **safe removal of the motor**.

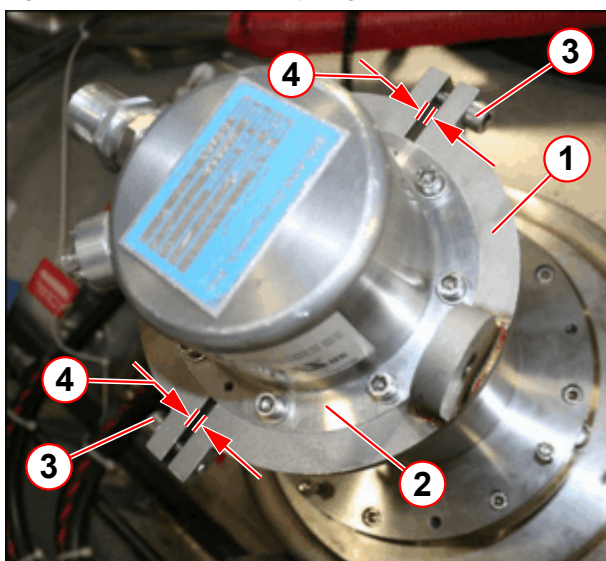
Fig. 85: Removing the helium syphon - OR70



- (1) Securing nuts
- (2) Helium syphon
- (3) Syphon end cap
- (4) Vent line

- Remove the **securing nuts** securing the helium **syphon** (pos 1).
- Remove the **syphon end cap** and pass the syphon back through the **bracket**.
- **Immediately refit the end cap** and move the syphon aside.

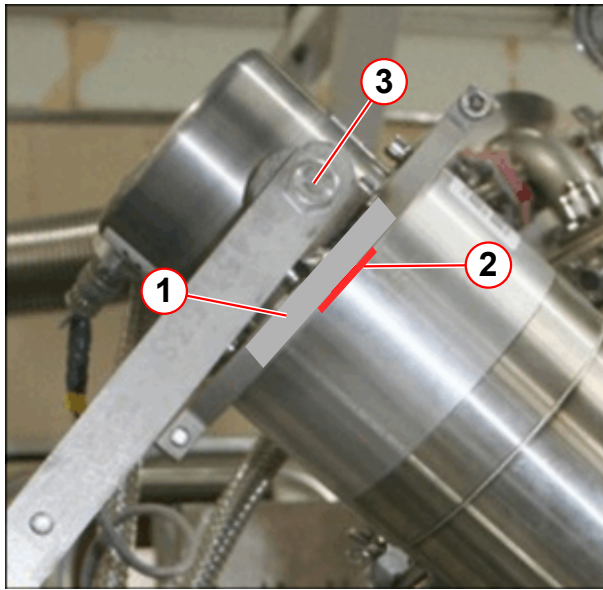
Fig. 86: Fit the motor clamp ring



- (1) Motor clamp ring
- (2) Motor flange
- (3) Screws (2 off)
- (4) Maintain equal spacing

- Place the **motor clamp ring** around the housing.
- **Assemble the screws by hand.** (pos 3)
 - » **do not tighten** at this stage.
- Maintain **equal spacing** when assembling the **clamp ring** (pos 3).

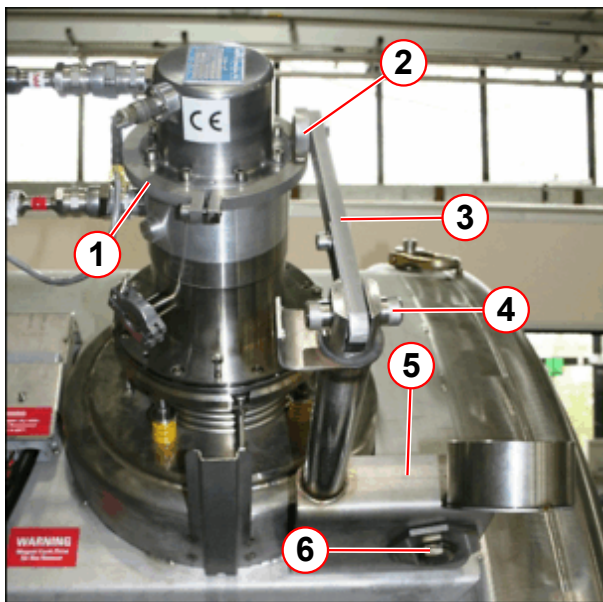
Fig. 87: Align the motor clamp ring



- (1) Motor clamp ring gap
- (2) Valve stem housing
- (3) Pivot arm

- Connect the **pivot arm** (pos 3).
- Ensure the **motor clamp** does not touch onto the **valve stem housing** (pos 2).

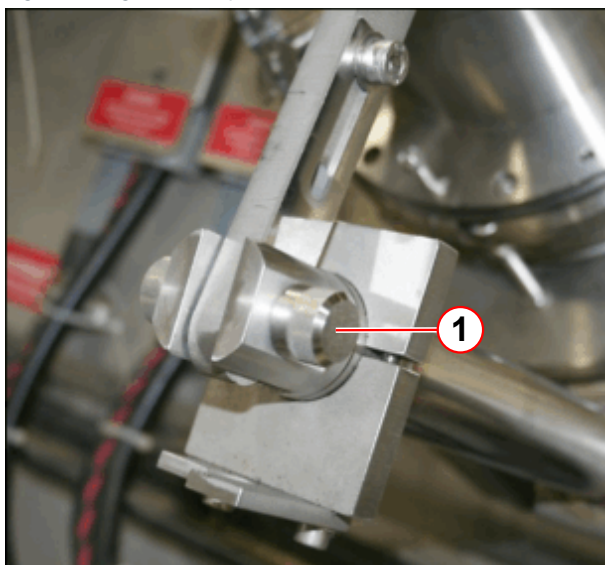
Fig. 88: Motor removal tooling (at field)



- (1) Motor clamp ring
- (2) Pivot arm connection to motor clamp ring
- (3) Pivot arm
- (4) Pivot bolt
- (5) Tool base and heat gun holder
- (6) Locking nut

- Attach the **tool base** to the **syphon bracket**.
- **Hand tighten** the **locking nut** (pos 6).

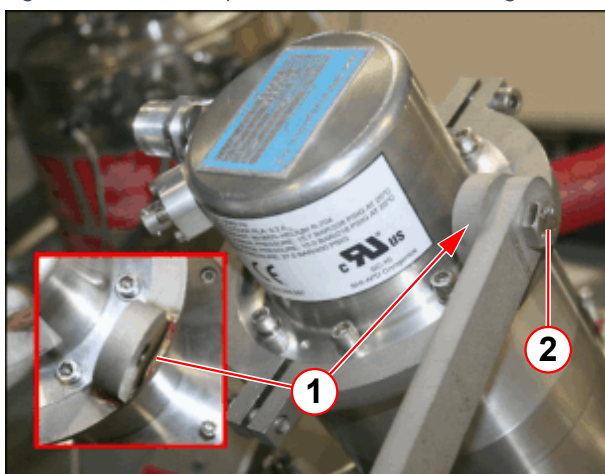
Fig. 89: Tighten the pivot bolt



(1) Pivot bolt

- Place the **pivot arm** in the **slot** of the tool base pillar.
- **Secure** the arm by **tightening** the **pivot bolt**.

Fig. 90: Fit the clamp arm to the motor housing



(1) Motor clamp ring
(2) Set screw

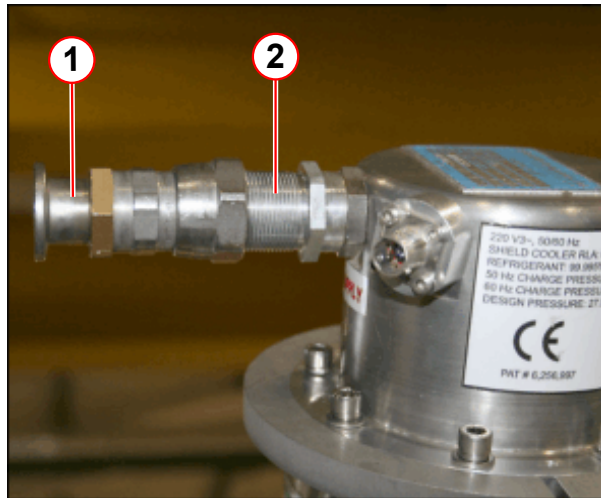
- Slide the arm forward and **attach** it to the **motor clamp**. **Tighten** the set screw **by hand**.
 - To allow for movement do not fully tighten the set screw.
- When all parts are **aligned**, tighten the:
 - set screw
 - pivot bolt
 - locking nut

- **Connect** the **heat gun** to mains power.
- Place the heat gun in the **cradle**.

2.6.1.11 Displacer exchange

Depressurise the cold head

Fig. 91: Depressurize the cold head



- (1) Female Aeroquip purge adaptor
- (2) SUPPLY coupling

- Screw the **female Aeroquip purge adaptor** securely onto the **supply coupling**, until **gas is heard** to escape.
- When **no more gas** is heard escaping, **remove the purge adaptor**.
- Screw the **female Aeroquip purge adaptor** securely onto the **return coupling**, until **gas is heard** to escape.
- When **no more gas** is heard escaping, **remove the purge adaptor**.



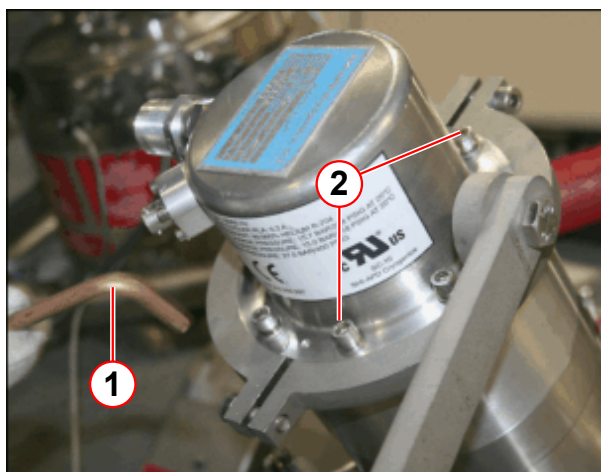
CAUTION

Risk of Air Ingress and Ice forming.

Failure to observe the following caution may result in damage to the cold head and displacer before it is disassembled.

- » Once the gas pressure within the cold head has subsided immediately remove the female Aeroquip adaptor to prevent air ingress into the cold head.

Fig. 92: Remove the M5 bolts



- (1) Allen key (non magnetic)
- (2) M5 x 90 Allen bolts (8 off)

- Remove the **8 off M5 x 90 allen bolts**, securing the **motor and valve stem housing** to the **cold head**.

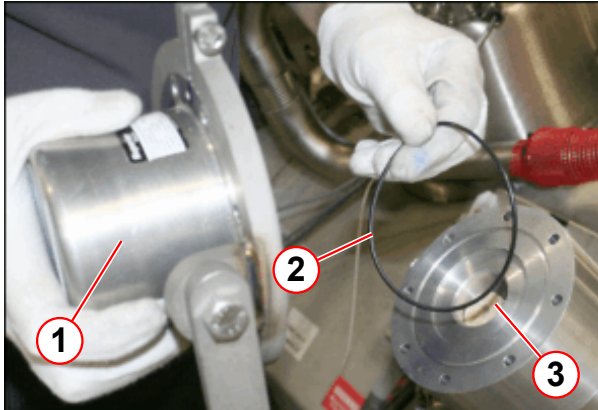
! CAUTION

Risk of Air Ingress and Icing.

Failure to observe the following caution may result in damage to the new displacer.

- » Once the 8 off M5 x 90 Allen screws have been removed complete the rest of the removal procedure as quickly as possible in a single operation.

Fig. 93: Remove the rotary valve



- (1) Motor housing
- (2) O-ring
- (3) Rotary valve

- Lift away the **motor housing** and remove the **valve stem housing O-ring**.
 - » In case of accidental damage, retain the **old** O-ring as a backup.
- Remove the **rotary valve** (pos 3).
 - the rotary valve can sometimes **remain in the valve stem housing** when the motor housing is removed.

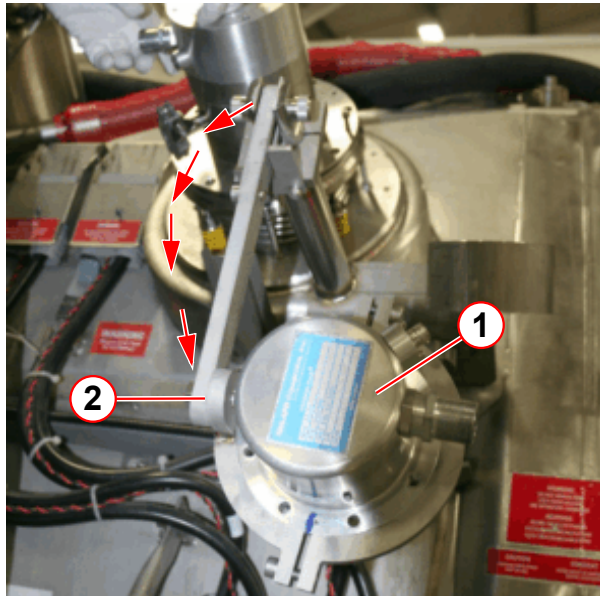
! WARNING

Risk of Cold Burns.

Failure to observe the following safety information may result in physical injury.

- » Do not touch the cold displacer surfaces with bare hands.

Fig. 94: Magnet at field - Support the motor housing in the service tool cradle



- (1) Motor housing
- (2) Pivot arm

- Swing the **cold head motor** away, **using the cradle** of the cold head service tool.



Whisper Plus cold heads prior to serial number 30, have 2 off M3 x 60 screws that secure the motor and valve stem housing together.

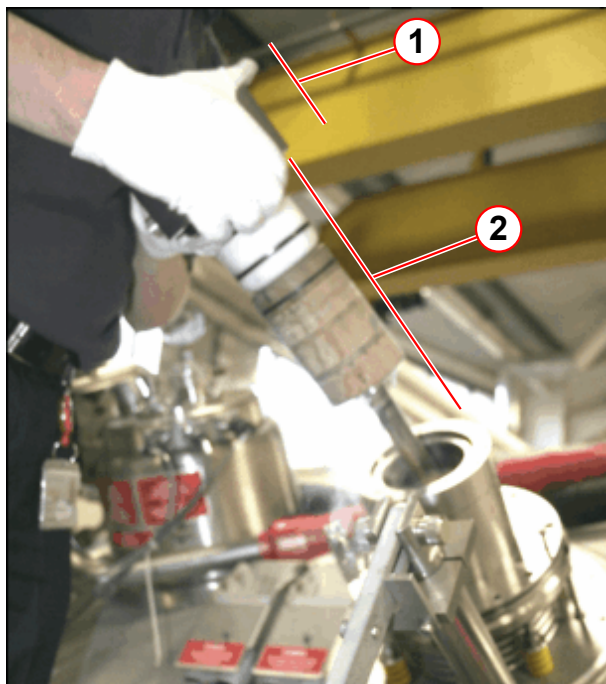
For these units remove the motor and valve stem housing as a complete assembly. After the assembly is removed, the 2 off screws can be discarded.

If the cold head is prior to serial number 30:

- **Remove the 2 off M3 x 60 screws**, securing the **motor** and **valve stem housing** together.
- **Remove the O-ring** from the **cold head**.

Remove the displacer

Fig. 95: Removing the valve stem and displacer



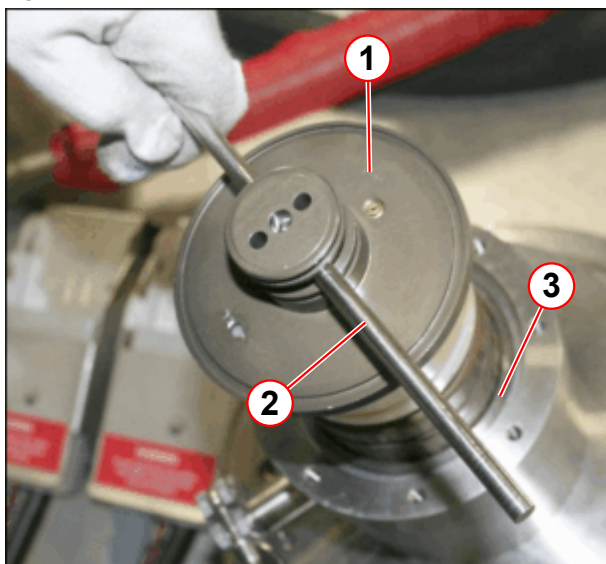
- (1) Valve stem housing
- (2) Displacer

Valve stem separation

It is possible for the **valve stem** to **stick in the valve stem housing** when being removed from the cold head.

In this case, the **valve stem** and **displacer** will be extracted as **one unit**, when the valve stem housing is removed.

Fig. 96: Valve stem extractor



- (1) Valve stem
- (2) 7.5 mm extraction bar
- (3) Cold head O-ring

If the **valve stem** and **displacer assembly** have **remained** in the **cold head**:

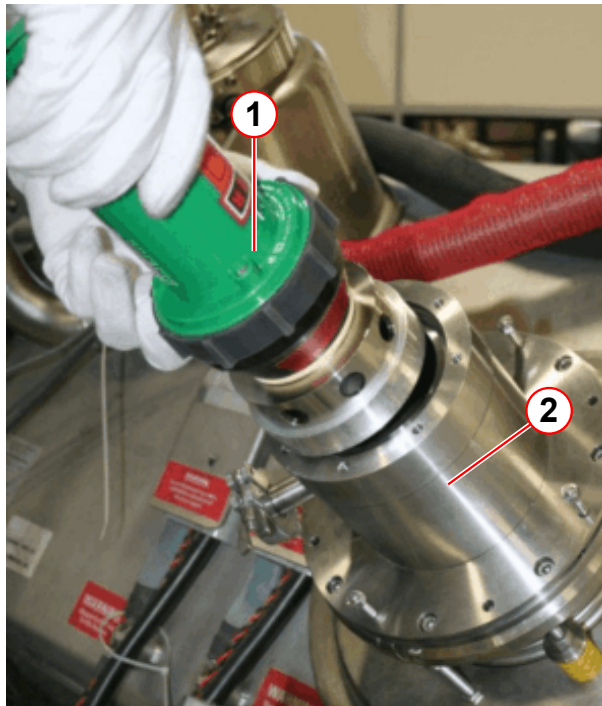
- Insert the **7.5 mm extraction bar** through the hole in the valve stem.
- Pull **upwards**, and position safely as these components will be **very cold**.
 - » Wear **protective gloves** during exchange procedure.

Fig. 97: Clean the cold head cylinder



- Remove the O-ring from the **cold head cylinder**.
- **Clean** the upper part of the **cold head cylinder**, (where the O-ring is located).
- **Do not clean inside the cold head.**
 - » Due to the very cold temperatures **any cleaning materials will freeze.**

Fig. 98: Insert the heat gun into the cold head cylinder



- (1) Heat gun
(2) Cold head cylinder

- **Insert** the **heat gun** into the cold head. **Select control setting '3'.**
 - The heat gun can be **left on** (at this low setting), during the next preparation stages.

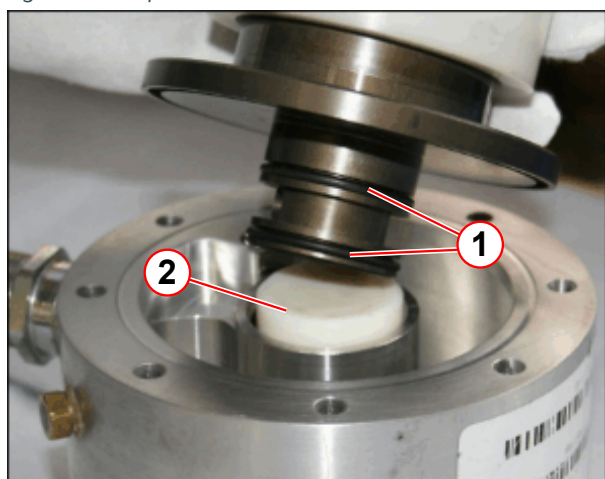
Separation of the valve stem

Fig. 99: Separate the valve stem housing and displacer



(1) Stem extraction tool (large diameter shown)

Fig. 100: Displacer and valve stem



(1) O-rings

(2) Extraction tool (small diameter)

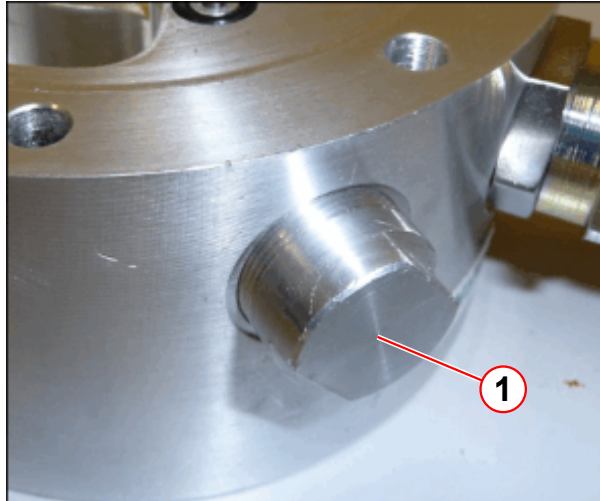
- The **displacer** can **remain** in the **valve stem housing** during removal.
- Use the **valve stem extraction tool** to separate the displacer.
 - Place the **large diameter** of the **extraction tool** on a smooth flat surface.

- Fit the **smaller diameter** into the **valve stem housing** and gently push down on the housing as shown.

2.6.1.12 Preparing the new displacer

Variable orifice cleaning and preparation

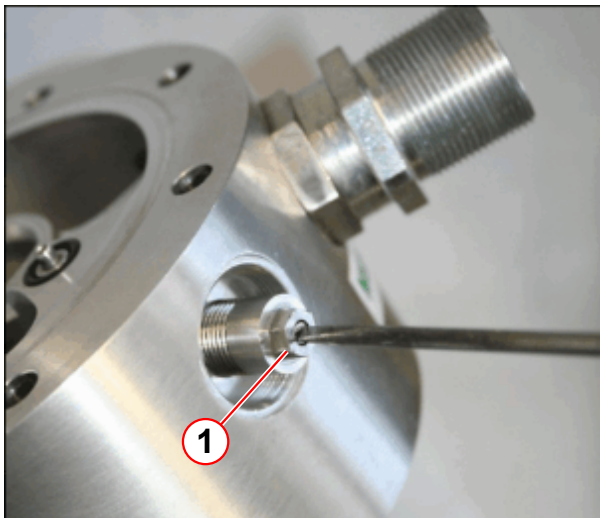
Fig. 101: Variable orifice cover



(1) Orifice cover

- **Remove** the variable **orifice cover**, using a **17 mm** wrench.

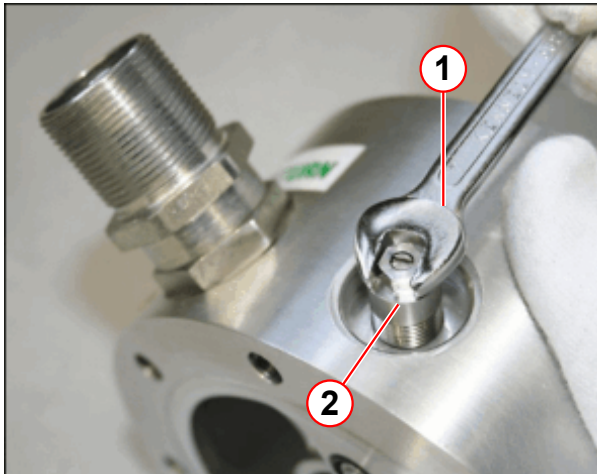
Fig. 102: Lock the variable orifice screw



(1) Turn the screw until level with the nut

- Use a **3 mm** flat bladed screwdriver to turn the variable orifice screw counter-clockwise, until it locks in the variable orifice nut:
 - » **count** the number of **turns** until the screw locks tight and is **level** with the **end of the nut**.
- **Record** the number of **turns**.

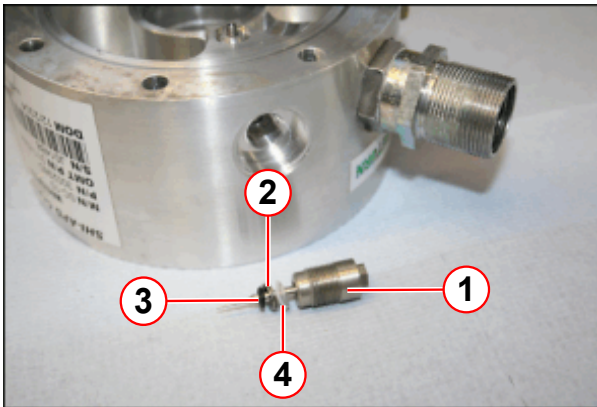
Fig. 103: Remove the variable orifice nut



(1) 7 mm combination wrench

- Use a **7mm** wrench to **slacken** the **variable orifice**.

Fig. 104: Remove the variable orifice needle valve



(1) Needle valve assembly
(2) St. Steel washer (2 off)
(3) O-ring (1 off)
(4) PTFE washer (1 off)

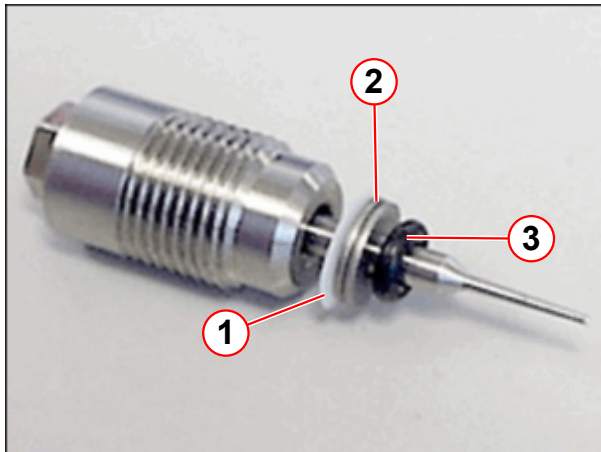
- **Unscrew** and **remove** the variable orifice **needle valve** (pos 1).
- **Remove:**
 - 1 off **O-ring** (pos 3),
 - 2 off **St Steel washer** (pos 2),
 - 1 off **PTFE washer** (pos 4),
- **Discard** the removed O-ring and washers.

- **Clean** the variable orifice **needle valve**.
- Blow **dry** with **helium gas**.



DO NOT apply GREASE to the O-ring.

Fig. 105: Fit new washers and O-ring

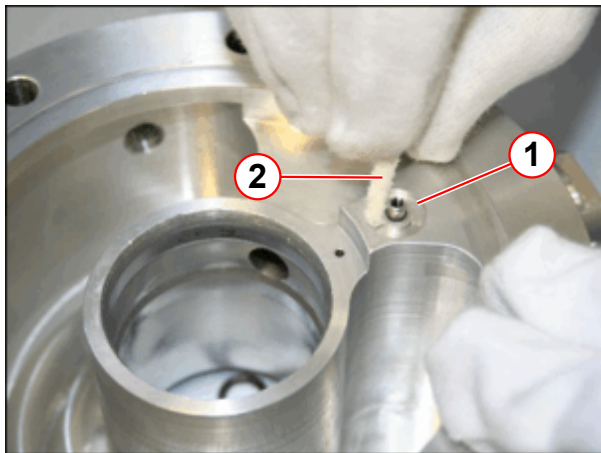


- (1) PTFE washer (1 off)
- (2) St. steel washer (2 off)
- (3) O-ring (1 off)

- **Fit the new O-ring and washers** (from the new displacer kit), to the **shaft of the needle valve**.

- PTFE washer (1 off)
- St Steel washer (2 off)
- O-ring (1 off)

Fig. 106: Clean the valve seat

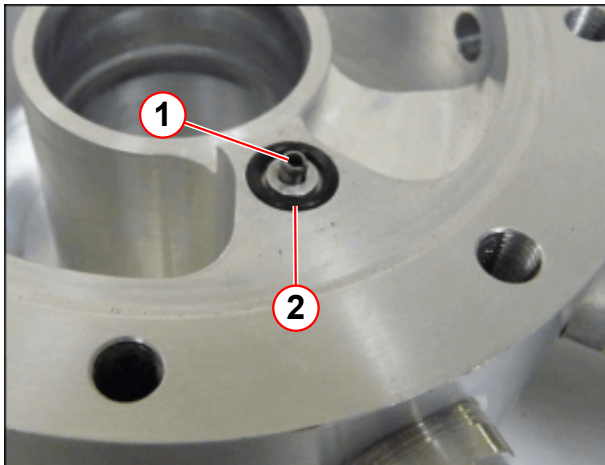


- (1) Pipe cleaner and cleaning fluid

- Using **pipe cleaners** and the **cleaning fluid**, clean the **valve seat** in the valve stem housing.

- **Insert the needle valve assembly** into the valve stem **housing** and screw down **by hand**.
- **Tighten** using the **7mm wrench**.
- Use the **3mm** flat bladed screwdriver to **screw the needle valve** into the valve stem **housing**. **Apply the number of turns** recorded during removal.
- **Recheck the tightness** of the **needle valve** with the **7 mm** wrench.
- **Refit the variable orifice cover**, using the **17 mm** wrench.

Fig. 107: Roll pin with the O-ring fitted

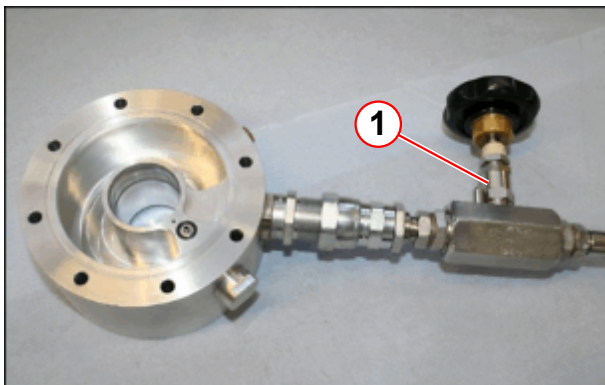


- (1) Roll pin
- (2) O-ring

- **Clean all surfaces** of the valve stem housing to **remove** any **contamination**.
- Fit the **new valve seat O-ring**, at the roll pin (pos 2).

Prepare the valve stem housing

Fig. 108: Connect the He gas supply to the housing



- (1) Shut off valve

- **Prepare** the valve stem **housing** for assembly of the new displacer.
- **Connect** the He **gas supply**.
 - **do not open** the gas shut off valve at this time.

- **Clean** the motor of all **contamination** from the old displacer.
- Check the **motor shaft roll pins** are **centered** on the **shaft** and that the pins and valve spring are undamaged.
- Screw the female **Aeroquip purge adapter** to the supply coupling of the **cold head motor**.
 - Be sure **gas** is able to **flow** through the **purge adapter**.
 - This will allow gas to **escape** through the **motor** when it is **installed** onto the valve stem assembly.

Prepare the new rotary valve



The rotary valve is a loose fit to the motor shaft and may fall off during assembly.

- Rub away the **rotary valve surface** with the **sharpening stone**.
- This will reduce the possibility of a high pitched sound coming from the new rotary valve once installed and operating.

Checking and preparing the motor for purging using the Leybold simulator

Fig. 109: Leybold simulator



- Connect the **Leybold simulator** to a suitable 220 volt AC supply.
- Connect the **cold head power cable** removed from the **filter panel** to the **simulator**.
- Switch on the **Leybold Simulator** to run the **cold head motor**.



Using the Leybold simulator tool with an APD motor, the motor rotation will be counter clockwise due to a change in the phases from Leybold to APD motors.

This is adequate for the testing and purging procedure.

Displacer preparation

Fig. 110: Displacer kit

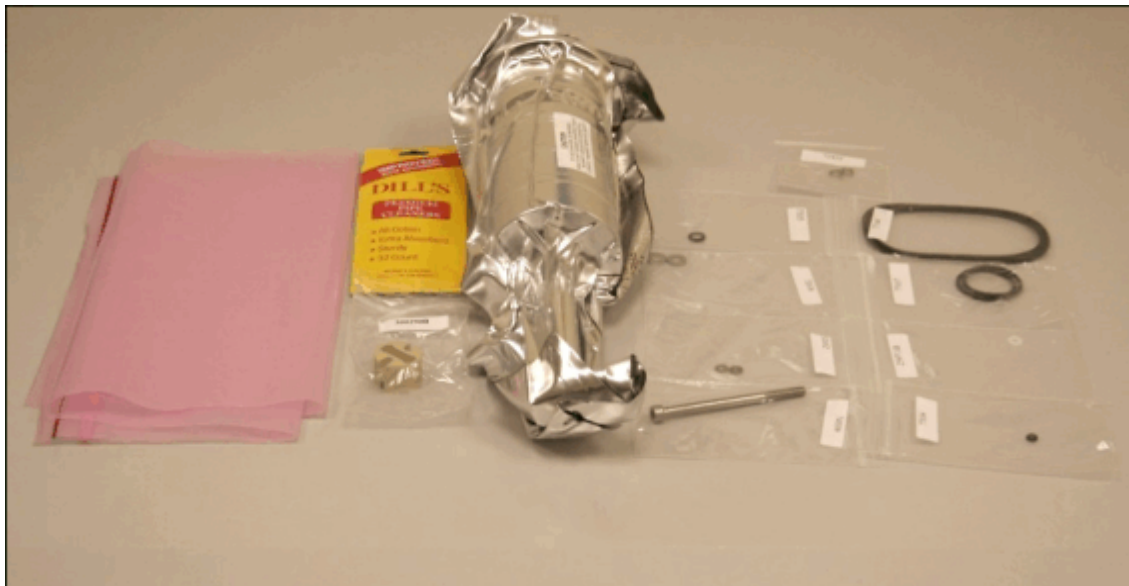


Fig. 111: Seal compression tools



- **Clean the inner surfaces** of the displacer **seal compression tools**.
- Use **lint free** paper towelling, lightly soaked in cleaning fluid



CAUTION

Contamination of the Displacer.

The displacer body is made of Hygroscopic material and performance will be compromised if contaminated.

- » Avoid prolonged atmospheric exposure.
- » Avoid grease.
- » Wear cotton gloves during exchange procedure.
- » Place the displacer in a helium background whenever possible throughout this procedure.



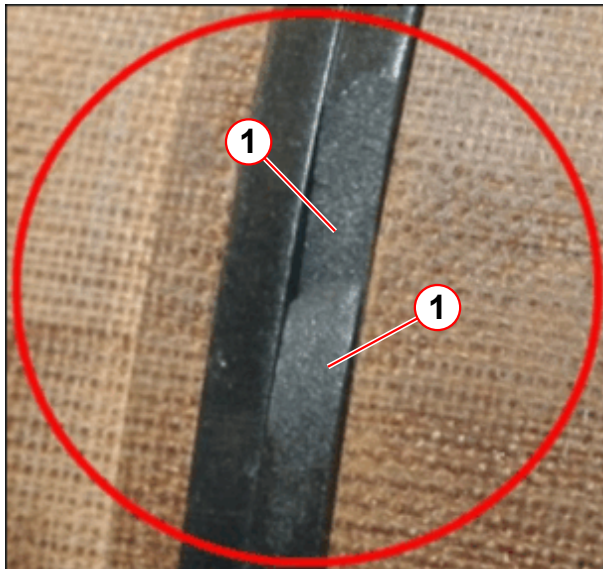
Wear cotton gloves when handling the displacer.

- Remove the **displacer** from the sealed packaging.

Fig. 112: New Displacer and packaging



Fig. 113: Displacer seals - tapered ends at 180 deg to each other



(1) Tapered ends

- Inspect the **displacer seals** to ensure the tapered seal **ends** are at **180° to each other**.

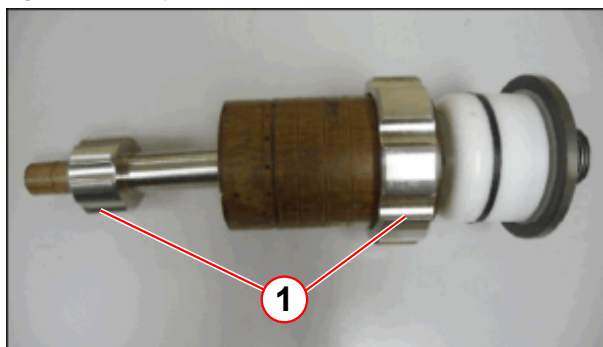
» This will **prevent** gas flow past **the seals** when installed.



The seal compression tools have an internal taper. The large end of the taper must be passed over the seal first, continue to slide the tool onto the seal until the seal is fully compressed into its seat.

Repeat for all seals. **DO NOT FORCE** the tools. Excessive force will result in damage to the seal.

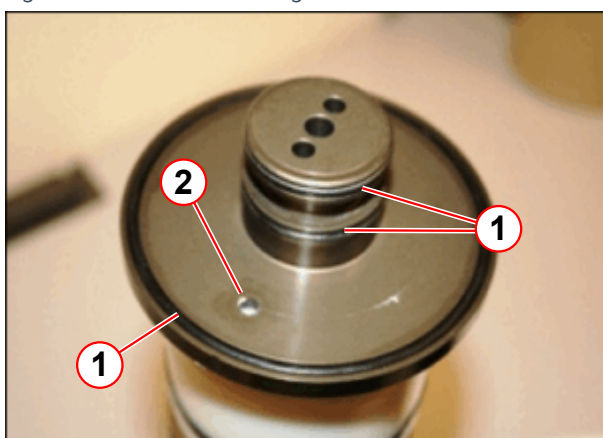
Fig. 114: Compression tools fitted



(1) Compression tools

- **Carefully** press the **seals** into their **seats** using the displacer seal **compression tools**

Fig. 115: Valve stem O-rings



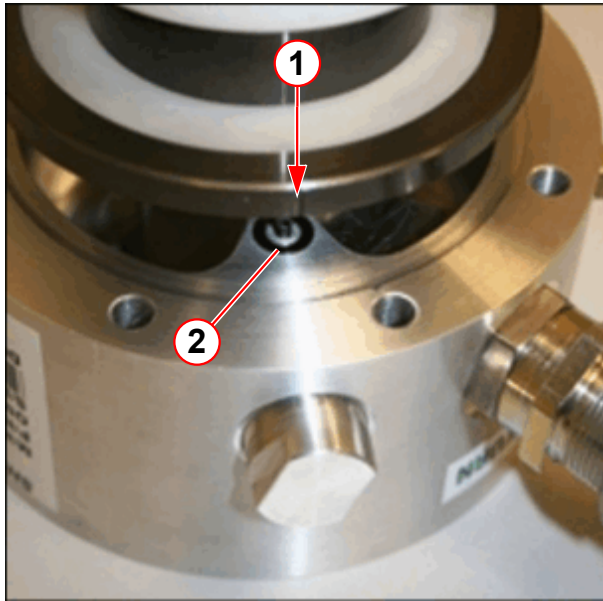
(1) O-rings
(2) Valve stem mating hole

- Very **lightly grease** the valve stem **O-rings**, and fit them to the **valve stem**.
- Do **not** allow **grease** to come into contact with the **displacer**.



Exercise extreme care when applying Apiezon 'N' grease to the thin O-rings. Due to the thin size of these O-rings any stretching will prevent them seating correctly.

Fig. 116: Valve stem alignment



- (1) Roll pin
- (2) Alignment slot

- **Align** the **roll pin** (in the valve stem housing) with the **mating hole** in the valve stem, by aligning **the slot** (pos 1), on the **outer** diameter of the **valve stem**.

Fig. 117: Displacer in sealed bag



- Place the assembled displacer and valve stem assembly into the **resealable bag** provided.
- Open the **gas regulator** valve and fill the bag with helium gas.
- **Maintain** a gas flow of **4 psi** into the bag.
 - **regulate the flow** at the pressure regulator to avoid any accidental **pressure increase**.
- The assembly is now **prepared for installation**.

Heating and cleaning

Fig. 118: Heat gun - set to position 8



For cleaning, **both cold head stages** inside the cylinder must be **heated to 25-30°C**.

- Set the **heat gun** to **position 8**.
 - **Begin to heat** the cold head cylinder.



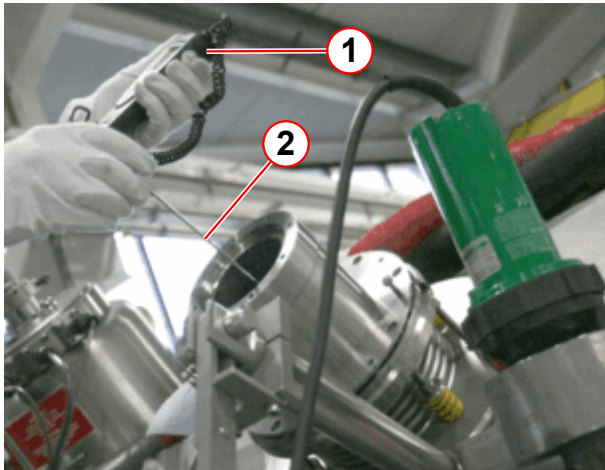
WARNING

Damage to internal magnet components.

Overheating of the cold head cylinder can damage the internal shield links and insulation.

- » Do not leave the heat gun in the cold head cylinder un-attended.
- » Continuously monitor the cold head cylinder temperature.

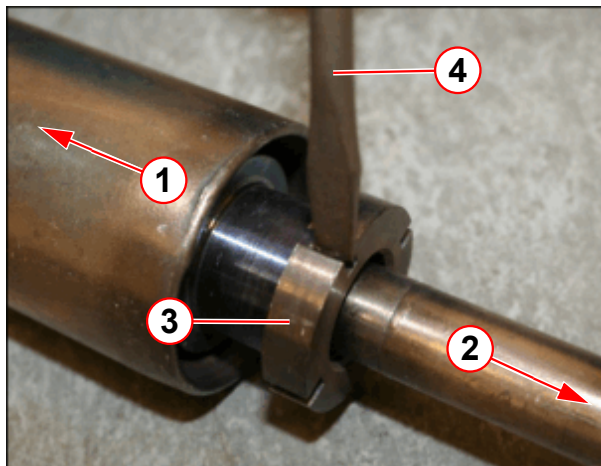
Fig. 119: Check the cylinder temperature with the probe



- (1) Electronic temperature sensor
- (2) Surface probe

- Check the **cylinder temperature** frequently using the **temperature measurement** instrument and **surface probe**.
 - **remove the heat gun**.
 - touch the **surface probe** on each of the stages.

Fig. 120: Adjusting the air flow temperature



- (1) Heat to 20k stage
- (2) Heat to 80k stage
- (3) Air flow adjustment screw
- (4) Screwdriver

- **Maintain an equal temperature** at each stage.
 - **regulate** the hot **air flow** from one stage to the other as required.
 - adjust the **air flow adjustment screw** to adjust the **temperature** from one stage to another (as required).
 - use a **screwdriver** to turn the air flow **adjustment screw**.

**WARNING**

Risk of Hot Burns.

Failure to observe the following safety information may result in physical injury.

- » **Do not touch hot surfaces with bare hands.**
- » **Personnel safety equipment must be worn.**

- It will take approximately **5 min** to raise the cylinder temperature **above a sub zero level**. After this, the temperature should be **checked** every **60-90 seconds**.
- When the **cylinder temperature** has reached **25-30°C**,
 - **wrap** cleaning cloth/lint free paper around the **tweezers**.
 - **moisten** in **cleaning fluid**.
 - **clean** around the upper and lower **cylinder bores** and **copper stages**.



Take great care not to scratch the cylinder and copper stages!!



This cleaning must be done QUICKLY. The cold head cylinder temperature will return to a sub zero value and cause the cleaning material to become frozen and stick to the cylinder.

Approximately 10 seconds cleaning time is recommended.

- If cleaning materials **freeze to the cold head displacer bore**:
 - » **Withdraw** the tweezers and **remove** as much of the cloth/lint free paper as possible.
 - » Use the **heat gun** and continue heating until the cloth **defrosts** and can be removed, or has burned away.
- Continue to **clean the cylinder** (as many times as required), until it is as clean as possible, keeping the **temperature within** the **25-30°C** range.

Heating the cold head

- When the cylinder is clean, **continue to apply heat**. Monitor until the **critical temperature** of **55°C** is reached.
- During the heating process, **prepare necessary tooling/spare parts** required for the next steps.
 - ensure they are **positioned safely** and **ready for use**.

Installation of the new displacer



Wear cotton gloves when handling the displacer.

From **55°C to 60°C** you will have **approximately 2 minutes** to prepare the new displacer **for installation**.

During this time the next steps can be carried out. **Continued monitoring** of the cold head temperatures **will not be required** if this process is **followed without problems**.

- When the final **55°C measurement** has been made, reinstall the **heat gun**.
- **Remove** the prepared displacer from the **helium filled bag**.
- **Check** all **seals are compressed** using the compression tools and **then remove them**.
- Move the displacer to the magnet.



The following steps must be completed within 90 SECONDS.

The 90 seconds will start when the heat gun is removed.



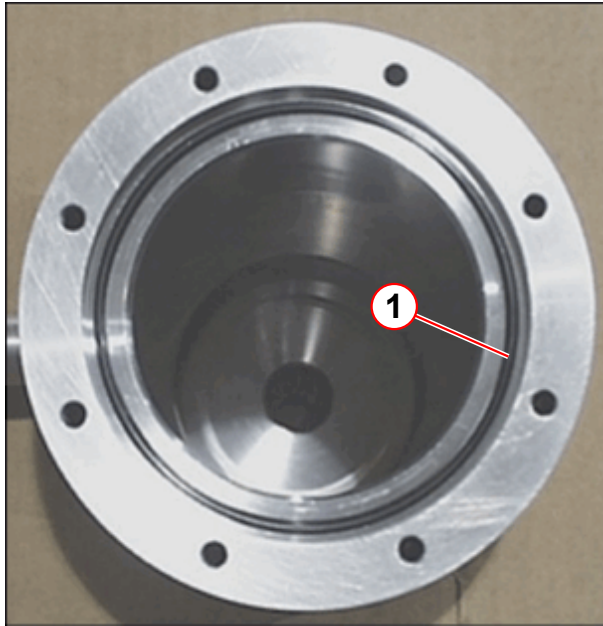
If there is any reason to abort the installation of the displacer, abort within the first 40 seconds to prevent moisture contamination destroying the new displacer.

Reinsert the displacer into the plastic bag and refill with helium gas.



As long as no moisture contamination has occurred on the new displacer, the process can be recovered and another attempt made when the cold head cylinder and displacer is prepared once again.

Fig. 121: Cold head cylinder O-ring



(1) O-ring

- Fit a **new O-ring** to the **cold head cylinder**.
- Check that the **O-ring** is **seated correctly**.



The O-ring can expand due to the temperature of the cold head body. Fit the second new O-ring if required.



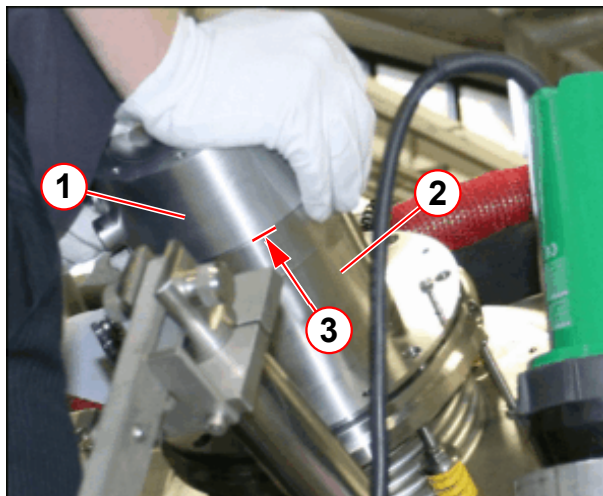
Take great care when inserting the displacer. This operation must be performed quickly.

Fig. 122: Insert the displacer



- **Carefully insert** the displacer into the **cold head**.
 - use a **twisting motion** to ensure the second stage displacer **locates** into its **cylinder**.
 - Ensure the **seals** do not become **unseated** or dislodged during installation.
 - There will be a continuous low **flow of helium gas** throughout this procedure.

Fig. 123: Push down until the valve stem housing is seated on the O-ring



- (1) Valve stem housing
- (2) Cold head cylinder
- (3) O-ring position

- Continue to **push the displacer down** into the cold head, until the **valve stem housing** seats onto the cold head **O-ring seal** (pos 3).



If a slight resistance is felt while inserting the displacer, stop and check if the displacer seals have become unseated.

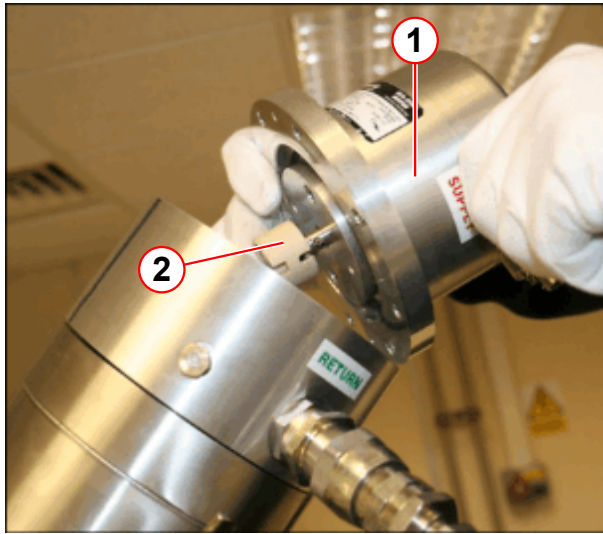
Fig. 124: Fit a new O-ring to the valve stem housing



- (1) O-ring

- Fit a **new O-ring** to the top of the **valve stem housing**.

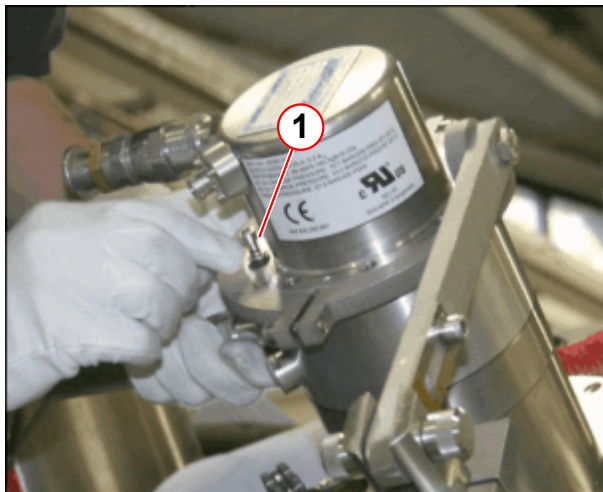
Fig. 125: Fitment of the motor housing



- (1) Motor
- (2) Rotary valve

- Fit the **motor**, (complete with rotary valve fitted), into the **valve shaft**.
 - Hold the rotary valve **in position** with the end of your finger, while the motor is placed onto the valve stem housing.

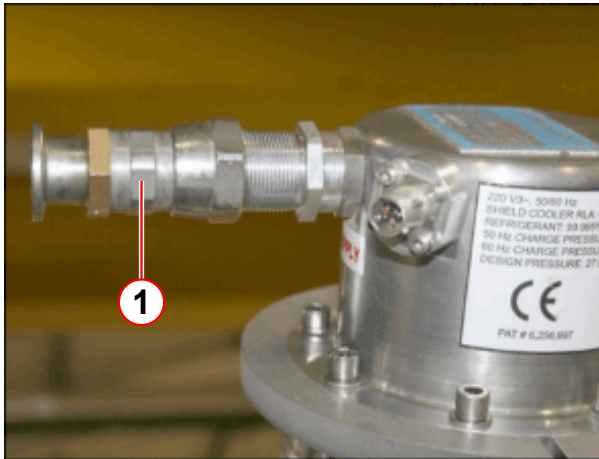
Fig. 126: Secure the cold head motor



- (1) M5 x 90 Allen bolts (8 off)

- Press and **hold the motor down** on to the **valve stem housing**.
 - ensure the **motor** is kept **in place** to **avoid** the **O-rings** moving out of position.

Fig. 127: Purging adaptor



(1) Female aeroquip purging adaptor

- A gentle stream of **gas** should now be flowing from the **female** Aeroquip **purging adaptor**.
- Allow the gas flow to continue whilst **securing** the **motor** to the **cold head cylinder**

- Ensure the motor is **aligned** with the **orientation marks**.
- **Secure** the cold head **motor** using a minimum of 3 off **bolts**.
 - **tighten** the bolts in a **diametric pattern** to spread the load on the O-ring.
 - **at this stage, low pressure gas is flowing**.
- When the **motor** is secured, **remove** the **purging adaptor** from the **Supply coupling**.



When the flow of gas through the purge adapter has stopped the critical 90 seconds procedure is complete.

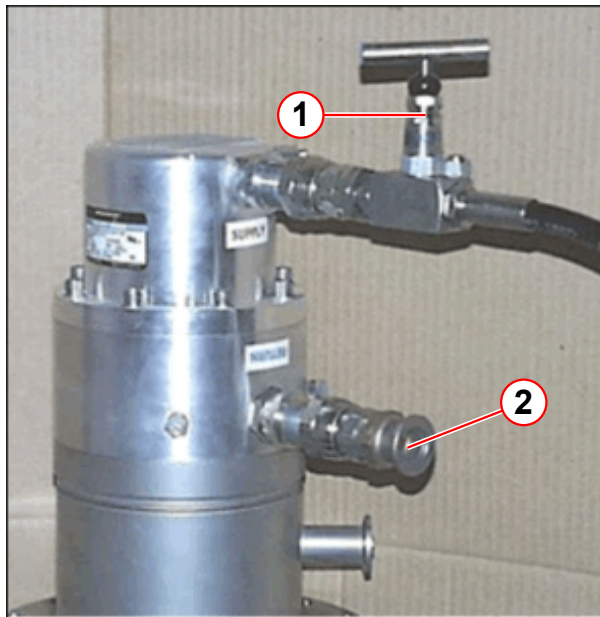
- **Fit** the remaining **motor bolts** and **torque all bolts to 15 Nm**.



Ensure all motor bolts are installed and are secure before commencing the high pressure purging of the cold head.

Purging the Cold head with Helium Gas

Fig. 128: Purge adaptor fitted



- (1) He gas supply valve
(2) Purge adaptor

- **Disconnect** the **gas supply** and purge adaptor from the **Return Aeroquip** couplings.
- **Connect** the gas supply to the **cold head Supply Aeroquip** coupling.
- Loosely screw the **purging adapter** into the **Return Aeroquip** coupling (by hand).
- To prevent air **ingress** into the cold head, **do not** screw the **purge adaptor** fully onto the **return** coupling.

- **Reconnect** the **power cable** to the cold head **motor**.
- **Open** the gas supply valve and **set** the regulator to **approximately 5 bar**.
- **Switch on** the motor **power supply**.
- **Screw** the **purging adapter** fully onto the gas **Return Aeroquip** coupling.
 - » The helium **gas** will **discharge** in **surges** from the **Return side** of the cold head.
- After **5 surges**, **close** the gas **supply shut off valve**.
- With the motor **still running**, allow the **pressure to discharge** from the cold head, then quickly **OPEN** the **gas supply valve**.
 - **Do not** allow **air ingress** via the purge adapter
- **Repeat** this sequence **5 times**.
- Continue to purge **continuously** (for at least 15 minutes) or until approximately **50 bar** is left in the gas cylinder.
- **This action:**
 - **purges** the cold head
 - **reduces heat input** from the heat gun, to the **cylinder**



The cold head motor tooling may be removed during purging.

Final steps of purging

- **Increase** the pressure regulator to **16 bar**.
- **Remove** the **purge adaptor** from the **Return** Aeroquip coupling.
 - Ensure the **gas pressure** is **trapped** within the **cold head**.
- **Switch off** the cold head motor **power supply**.
- **Close** the gas supply **shut-off valve**.
- **Disconnect** the **gas supply** from the **cold head Supply** Aeroquip coupling.

2.6.1.13 Reconnection of compressor and components

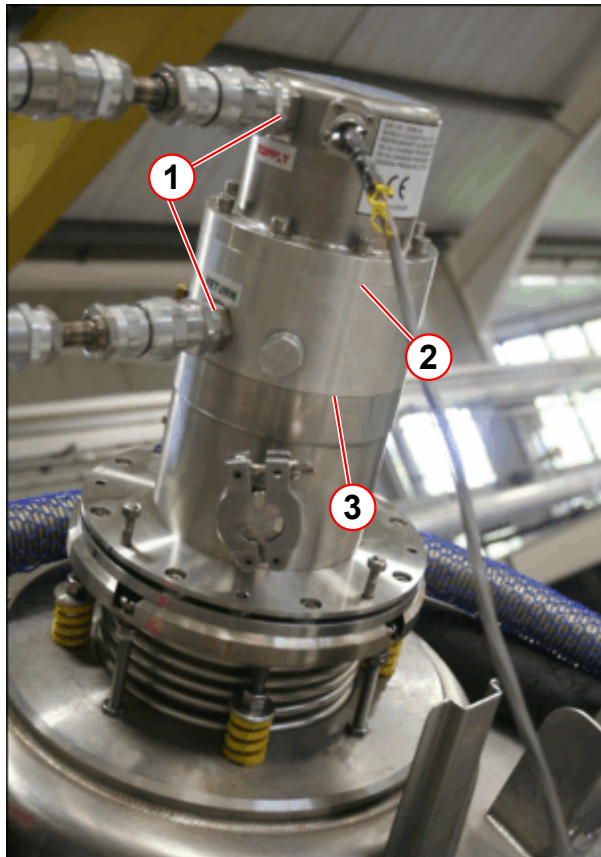
- **Connect** the **cold head/compressor** electrical connections.
- **Connect** the **Supply** and **Return** Aeroquip couplings.
- **Check** and record the compressor **static pressure**.
- **Check** all cold head **electrical connections**.
- **Turn on** the **compressor**.
- **Refit** the **he syphon** onto the bracket. Ensure the **syphon end cap** is **fitted**.
- **Refit covers** and any system **equipment** removed **during disassembly**.



Ensure no tooling or equipment has been left on or in the vicinity of the magnet.

2.6.1.14 Final steps

Fig. 129: Leak check points



- (1) Aeroquip fittings
- (2) O-ring seals
- (3) Orifice seal

Use a suitable **helium leak detector** capable of detecting leaks **down to 1×10^{-4} mbar l/s**.

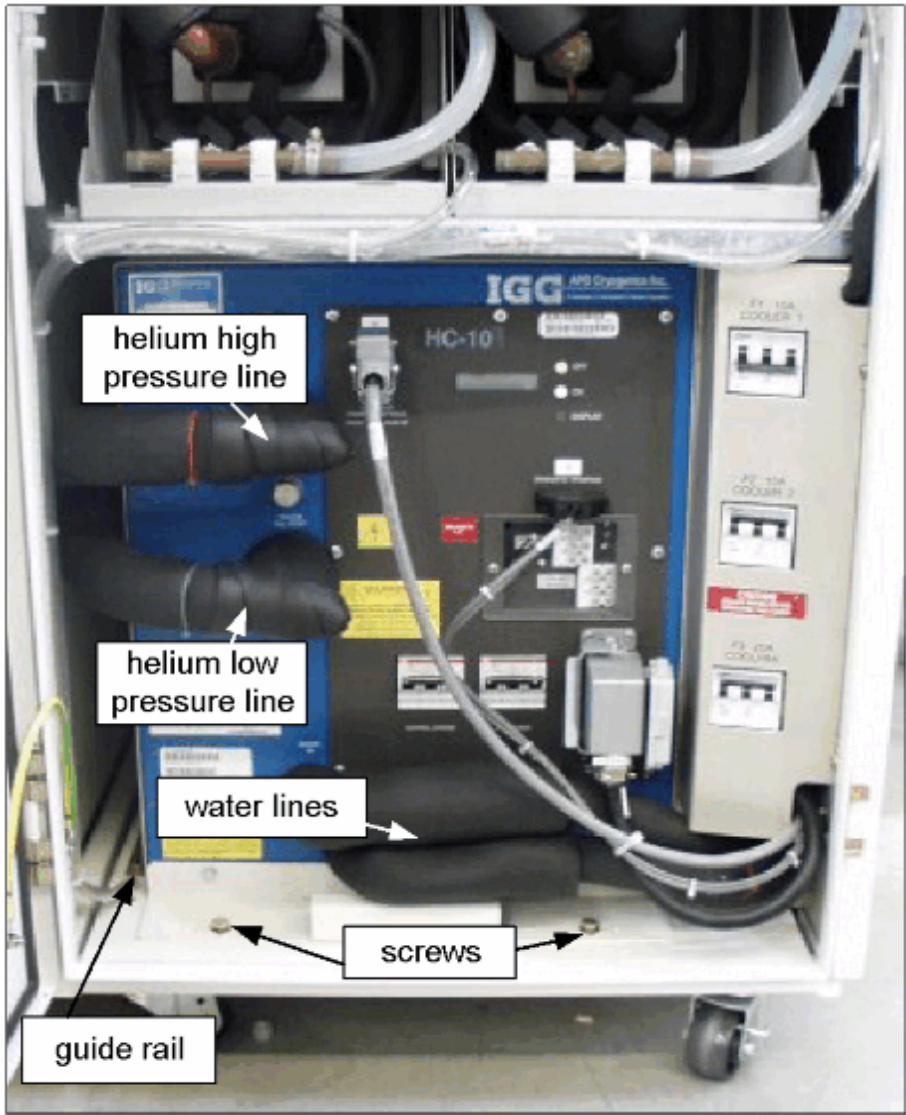
- Record the **shield temperatures**.
- After 2 hours:
 - » Check and record the **compressor pressure**.
 - » Check and record the **shield temperatures**.

2.6.1.15 Magnet ramping

If the magnet was **ramped down** prior to changing the displacer, ramp the magnet back to field. Follow the standard ramping procedures.

2.7 Replacement of the Compressor (APD)

Fig. 130: APD compressor



2.7.1 Removing the APD compressor

Step	Instructions	Required Tools
1	Switch off compressor Switch off F3	
2	Unplug the power cable Unplug the shield collar cable Unplug the Diagnostic cable	

Step	Instructions	Required Tools
3	Disconnect the high and low pressure helium flexlines Using two wrenches	Wrench 1 3/16" and 1"
4	Unscrew compressor from the base plate Removing the compressor can be facilitated by removing the guide rail screw first	
5	Carefully pull out the compressor. The compressor should be pulled onto a sturdy support (e.g., wood, pallet) of approx. 8-10 cm.	

Fig. 131: APD compressor



Fig. 132: APD compressor



Step	Instructions	Required Tools
6	Pull the compressor on a pallet	
7	Close the ball valve of primary water, supply and return	Spanner
8	Disconnect the water lines from the compressor	



The weight of the compressor is approximately 95 kg

Fig. 133: Water lines



Fig. 134: Water lines





Drain the water from the pipes, before returning the replaced unit

2.7.2 Installing the new compressor

Step	Instructions	Required Tools
1	Connect the water lines	
2	Slide compressor into the cabinet	Spanner
3	Tighten the compressor on the base plate	
4	Connect the high and low helium flexlines 1. High-pressure (red) 2. Low-pressure (green)	
5	Connect the electrical cable	
6	Check the helium pressure on the pressure gauge Operation (compressor off) at: 50 Hz: 15.6 - 15.9 bar (226 - 230 psig) 60 Hz: 14.9 - 15.2 bar (216 - 220 psig)	
7	Open the ball valve of the water supply	



In order that the compressor can run with the auto restart module, the compressor must be started from the remote switch.

DO NOT switch the compressor ON using the on/off button on the front panel of the compressor.

Step	Instructions	Required Tools
8	Check if remote switch is in OFF position Switch fuse F3 ON Close MAIN POWER AND CONTROL SYSTEM circuit breaker on the compressor CHECK that the display is ON but DO NOT switch the compressor ON using the on/off button on the front panel of the compressor	
9	Switch the compressor ON using the Remote on/off switch Typical operating pressure (compressor on): 50/60 Hz: 20.0 - 22.8 bar (290 - 330 psig)	

2.8 Replacing the F70 Compressor with Tooling kit

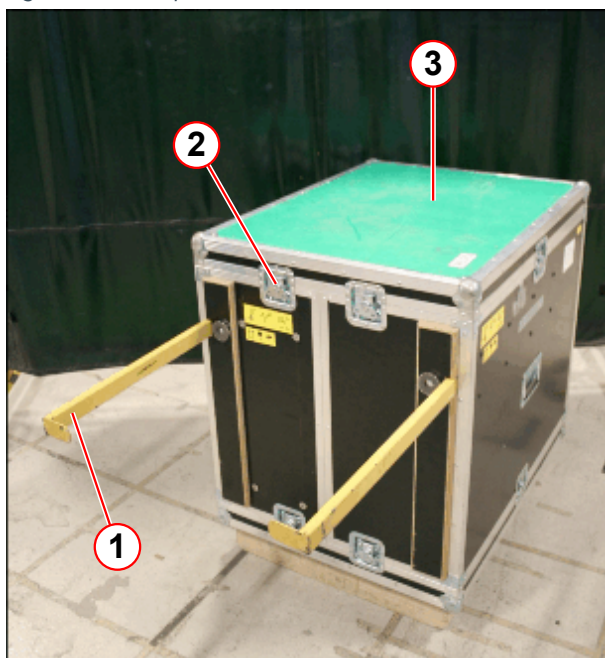
2.8.1 General

This procedure describes the replacement of the compressor unit when using the replacement tooling and transport kit (material number **11060700**)

The replacement compressor unit is shipped in a specially designed wooden transport box. The transport box contains the tooling and accessories required for a safe replacement of the compressor unit:

- It allows for the safe transport and replacement of the compressor unit
- The transport box allows 1 person to pack a damaged compressor unit.
- The shipping crate is suitable for worldwide transport and complies to shipping regulations IEC 60721-3-2.

Fig. 135: Transport box



- (1) Lifting handles (x 2)
- (2) top lid latch (x 6)
- (3) Lid with PTFE sheet covering

The transport box is fitted with 2 x lifting handles to allow movement. Built-in wheels allow movement over a short distance without requiring auxiliary lifting tools.



If necessary, the transport box can also be moved using a pallet truck.

2.8.1.1 Prerequisites

The document refers to compressor units identified by material no:

- 10121435 - Sumitomo F70
- 07461119 - compressor unit M/N HC-10
- 07758191 - SHI Compact compressor unit CSW71TW

- 07014827 - Robust Coolpack
- 04296518 - RCA - compressor unit Coolpak 6000 (MKI)
- 07120913 - Helium compressor unit. Coolpack. MKII

2.8.2 Safety Information

Carefully read the Operator Manual supplied in the transport box prior to using the tooling for the first time!

- The Operator Manual contains additional Safety Information. As well as general safety, transport, installation, operation, and maintenance.
- Also observe the special Safety Information and Notes included in the chapters!

NOTICE

Tripping hazard

Tripping hazard caused by edges, thresholds, cables and lines when moving the compressor unit on the casters.

- » Check for hazards before moving the compressor unit.



CAUTION

Unsafe lifting and movement of the tooling

Personal injury or equipment damage if safety procedures are not followed.

- » It is not permitted to lift or lower loads on inclines!
- » Do not park the shipping crate or tooling on inclines.
- » Only lift compressor units that the tooling is designed for.



CAUTION

Personal injury

When performing the replacement procedure personal injury can occur.

- » Safety shoes must be worn.
- » Wear protective gloves and suitable clothing when handling the equipment.
- » The operator must ensure that non-authorized personnel do not use the tooling.

**CAUTION**

High-pressure helium gas.

Follow the Safety Information provided to avoid injury.

- » Protective clothing, gloves, and eye protection must be worn.

NOTICE

Risk of injury and damage to equipment

The weight of the compressor unit is 100 kg. (220 lbs)

- » Use the supplied lifting equipment provided in the transport box.
- » Assemble the lifting equipment correctly to the instructions provided.

2.8.3 Completion Time and Number of Persons

Tab. 4 Completion time

Work steps	Time	Persons
Preparation	n.a.	1
Replacement	90 min	1
Startup/Tune-up/QA	n.a.	1
Final steps	30 min	1

2.8.4 Tools

- Transportation lifting kit (material number **11060700**, F70 compressor kit).
- 14-mm and 17-mm open-ended wrench.
- 17-mm socket.
- Torque wrench (0–20 Nm).
- 8-mm Allen (Hex) key.

- 7-mm socket and ratchet.

Fig. 136: Edwards Leak detector



- Use the Edwards hand-held leak detector (material no **10614850**).

The Edwards is the **recommended equipment** for leaks **greater than 1×10^{-4} mbar l/s**.

NOTE: The hand-held device is calibrated in **ml/sec**:-

1×10^{-4} mbar l/s = 1×10^{-4} ml/sec at normal pressure and temperature.

2.8.5 Disconnecting and Removing the Compressor Unit



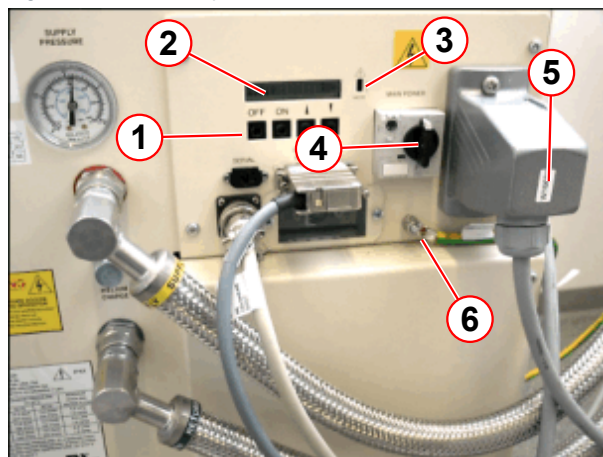
CAUTION

High-pressure helium gas.

Failure to observe the following Safety Information may result in injury.

- » Protective clothing, gloves, and eye protection must be worn.

Fig. 137: F70 Compressor control buttons



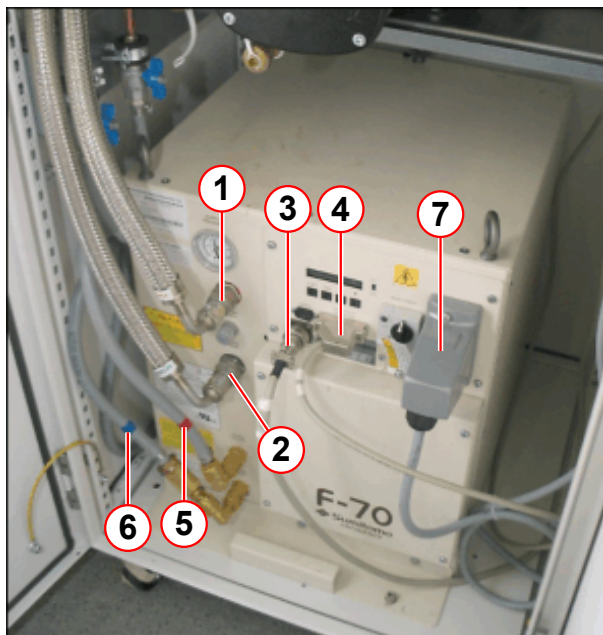
- (1) Control buttons
- (2) LCD display
- (3) Configuration selector switch
- (4) Main power switch
- (5) Mains supply cable
- (6) Ground wire

- Turn **off** the compressor unit at the **main power switch** (pos 4).
 - The **ON/OFF** buttons are disabled in **mode 2**.
- Switch off F3
- Switch off F4 on top of the CCA

- The **alarm box** displays a warning LED, and an audible alarm.

- Press the **AUDIO ALARM OFF** button to cancel the audible alarm.

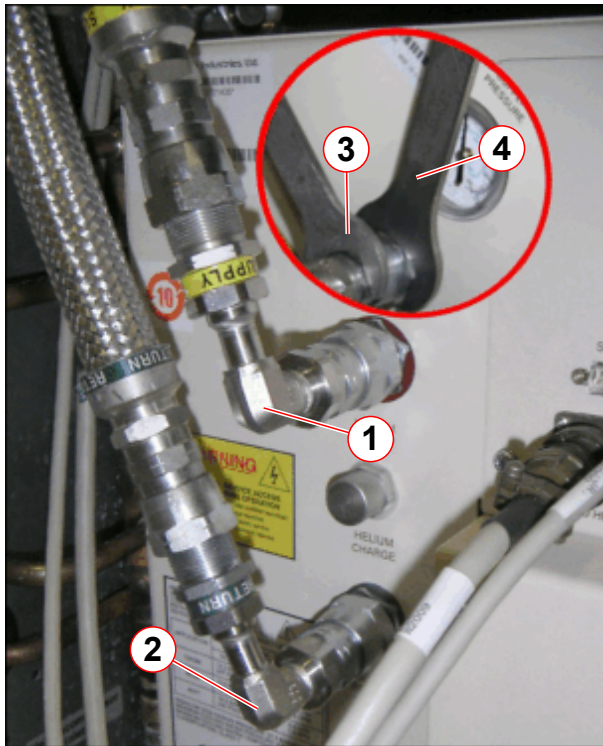
Fig. 138: F70 installation



- The base of the compressor unit is fitted with 2 nylon slide rails (pre-fitted) to ease movement.

- (1) Helium supply (high pressure)
- (2) Helium return (low pressure)
- (3) Cold head cable (to filter plate)
- (4) Diagnostic interface
- (5) Water line (inlet)
- (6) Water line (outlet)
- (7) Mains power connections

Fig. 139: F70 He compressor connections

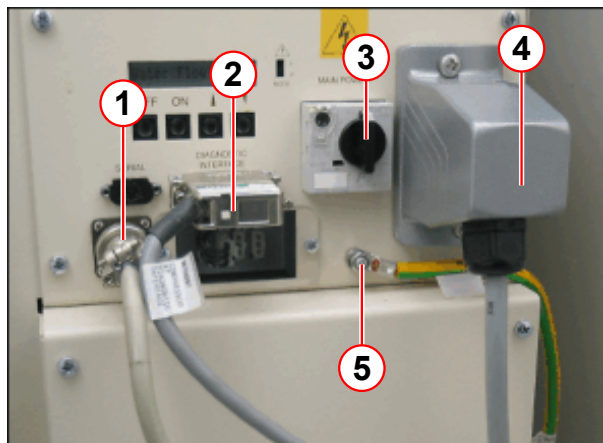


- (1) SUPPLY (high pressure) 90-degree elbow
- (2) RETURN (low pressure) 90-degree elbow
- (3) Hold nut (1" spanner/wrench)
- (4) Turn nut (1 3/16" open end crowfoot spanner/wrench)

Each helium pressure line is connected to the compressor unit via a **90 degree elbow**.

- **Remove** the self-sealing helium pressure lines in the following order:
 - **Return** (low pressure).
 - **Supply** (high pressure).
- **Hold** nut (3) and **turn** nut (4).

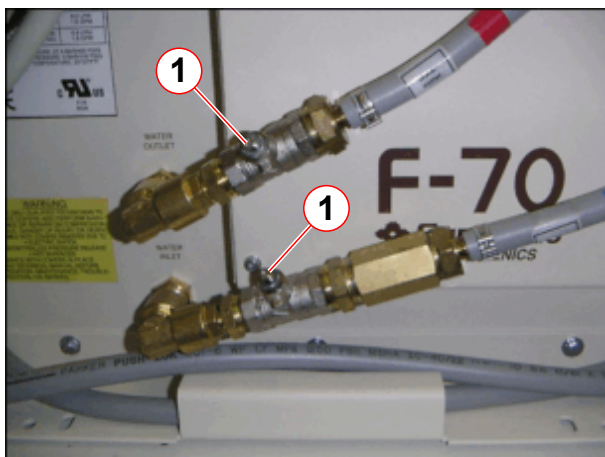
Fig. 140: F70 Electrical cable connections



- (1) Cold head cable
- (2) MSUP DB25 Diagnostic interface cable
- (3) Main power switch
- (4) Mains supply cable
- (5) Ground wire

- Disconnect the **mains voltage**, **cold head motor**, and **MSUP** (DB25 diagnostic interface) cable.
- Disconnect the **ground wire** (5).

Fig. 141: F70 Compressor water inlet/outlet shutoff valves



(1) Shutoff valve

- Close the **shut-off valves** on the **inlet** and **outlet** water lines.
- Disconnect the **water lines** from the compressor.



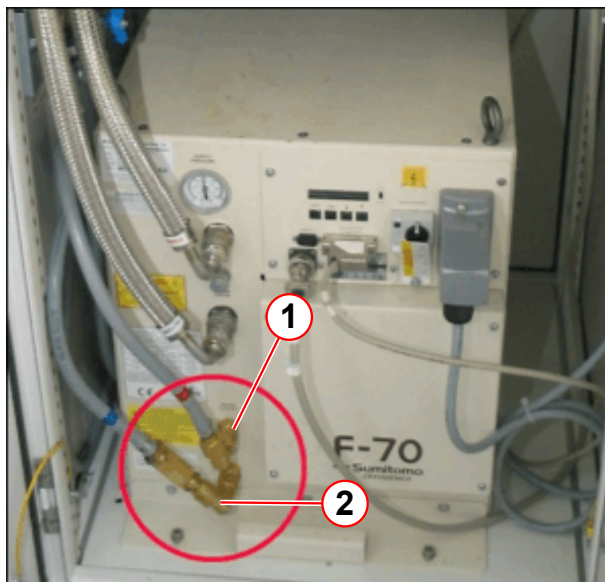
CAUTION

Chemical hazard

If the chilled water contains ethylene glycol, it must be collected in a suitable container while draining.

- » Return the collected water back to the water circuit after the new unit is installed.
- » If necessary, dispose of in line with local regulations for hazardous material.

Fig. 142: F70 connect the water lines



(1) Water line (inlet)
(2) Water line (outlet)

- Disconnect the **water pipes** from the compressor unit.



There is the potential for irreversible damage to the unit if the internal water pipes freeze during shipping.

Before returning a faulty compressor unit, use a gas line on either the inlet and outlet pipes to expel any water.

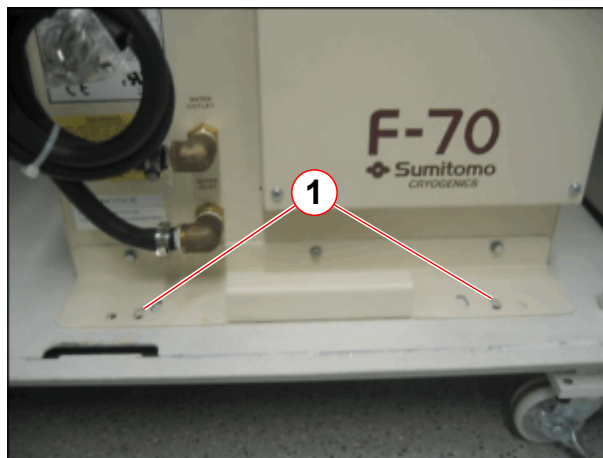
Fig. 143: F70 compressor - draining cooling water notice



Always **drain the coolant** from the cooling circuit when the compressor unit is shipped (or stored).

Refer to the disposal instructions for further information.

Fig. 144: F70 Compressor fixing plate



(1) M8 Hex hd screws (2 off)

RCA cabinet

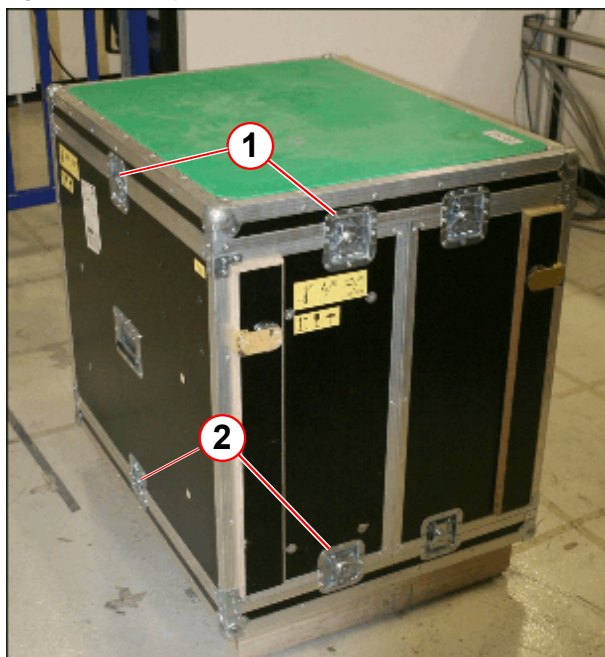
- **Remove** the 2 x M8 hex head screws securing the **fixing plate** to the cabinet.
- Leave the compressor unit in the cabinet until the replacement compressor unit is ready to install.

2.8.6 Unpacking and Pre-installation Checks

Before installation, check the compressor transport box for any transport damage or mishandling.

- Check the shock indicators when the transport box covers are removed.

Fig. 145: Transport box



- (1) Top retaining latch x 6
- (2) Bottom retaining latch x 6

- Unclip the 6 x latches securing the top cover.
- Lift the top cover away from the transport box.

Fig. 146: Cover lid of transport crate



- The transport box cover has x 4 casters fitted to the underside. The cover lid is used as a support ramp when removing the defective F70 compressor unit.
 - » The upper side includes a PTFE plate with gliding ability.

Fig. 147: Transport lid castors



When not being used stand the transport lid upright in a safe area against a wall.

- » Do not place the cover on the floor! Danger of rolling away.
- » Do not stand on the cover!



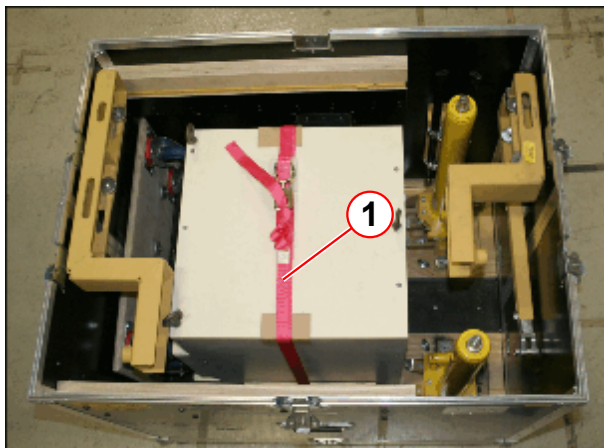
CAUTION

Storing the cover:

Risk of accident! Do not step on the cover!

- » When not in use, place cover upright in a safe area.

Fig. 148: Top view of compressor kit

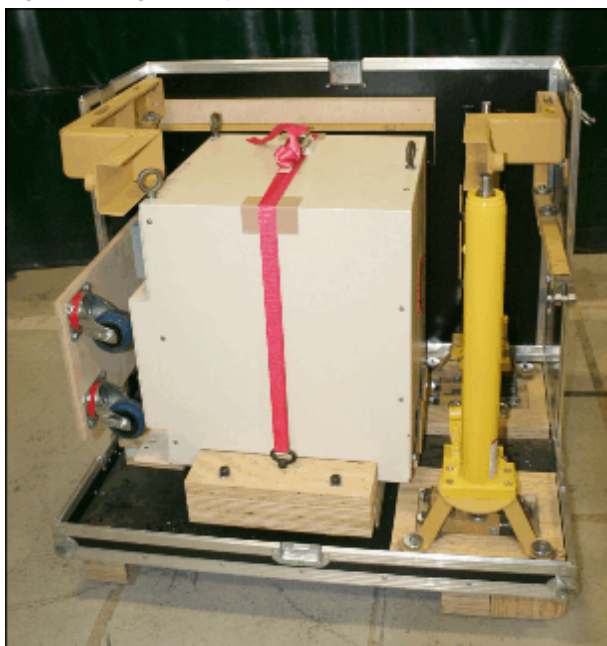


(1) Securing strap

Top view of the transport box.

- Check that the compressor unit has remained strapped securely to the transport base.

Fig. 149: Right side panel removed



All tooling components remain attached in position inside the transport box when removing the right side cover.

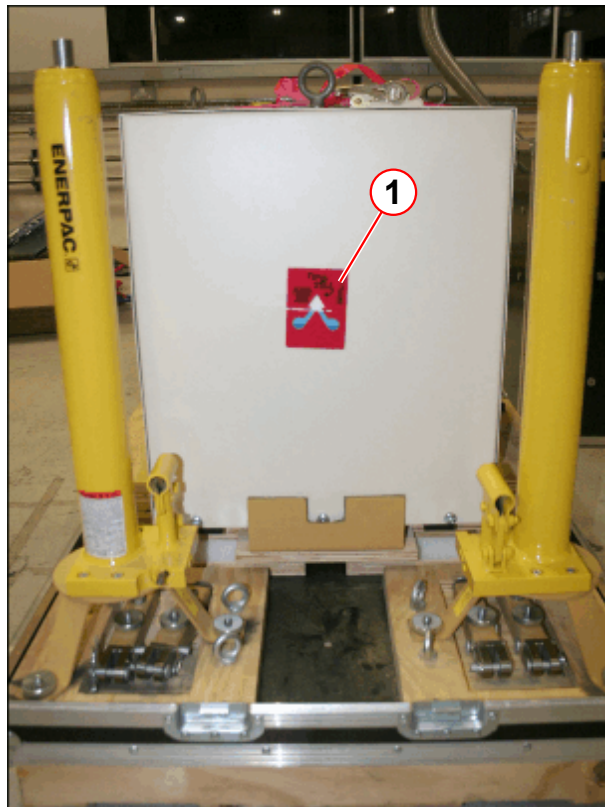
- Unclip the 3 latches and remove the right side cover of the transport box.
- Place the cover in a safe location to the side.



Do NOT tip the compressor unit more than 10 degrees from horizontal to avoid contamination.

DO NOT connect gas lines to the compressor unit if Tip-N-Tell indicator shows that mishandling has occurred.

Fig. 150: Crate sides removed



(1) Shock indicator

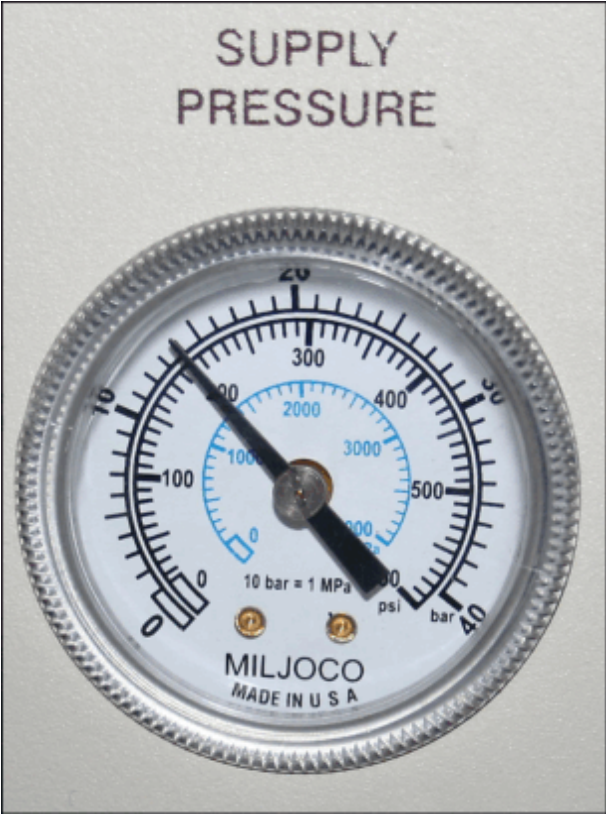
- Check the Tip-N-Tell indicator on the compressor unit.
- The sensor is blue (above the line) when the unit has been tipped or mishandled during transport.

Fig. 151: F70 Compressor Tip-N-Tell indicators



- If the sensor has been activated (the arrow point is blue), and the pressure is within specifications, allow the compressor unit to stand upright for **2 hours**. Refer to (→ Testing the Compressor Unit after Mishandling / Page 121)
- If the sensor has **not** been activated, continue with the replacement procedure.

Fig. 152: F70 Static pressure reading



- Check that the **static pressure** is **positive**.
 - If the gauge indicates **zero pressure** the compressor unit **cannot** be used.
 - Arrange a replacement compressor unit.

Tab. 5 F70 static pressure on arrival

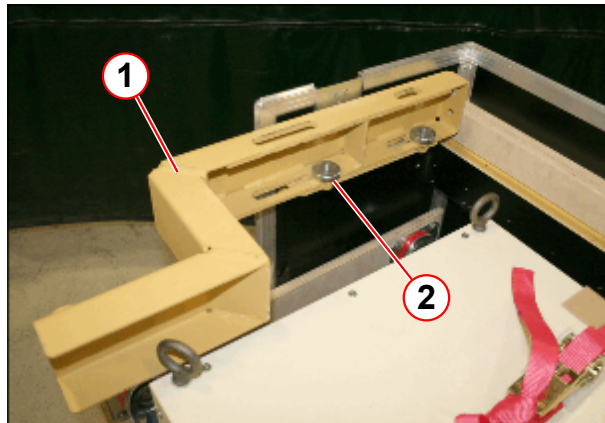
Pressure	Ambient temperature
207 - 212 psi (G)	20°C (68°F)
1430 - 1460 kPa	
13.5 - 14.0 bar	

Continue with the replacement procedure if:

- The sensor has **not** been activated
- The static compressor pressure is positive.

Remove the tooling components from the transport kit and assemble.

Fig. 153: Long Cross beam



- (1) Cross beam
(2) Knurled head fastener

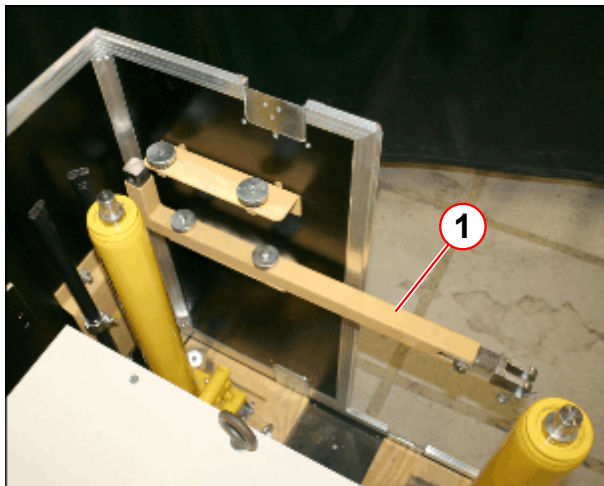
- Unscrew the 2 x knurled head fasteners securing the long 'cross beam'.
- Refit the fasteners to the case.

Fig. 154: Short cross beam



- Unscrew the 2 knurled head fasteners and remove the short cross beam.
- Refit the fasteners to the case.

Fig. 155: Vertical lifting beam

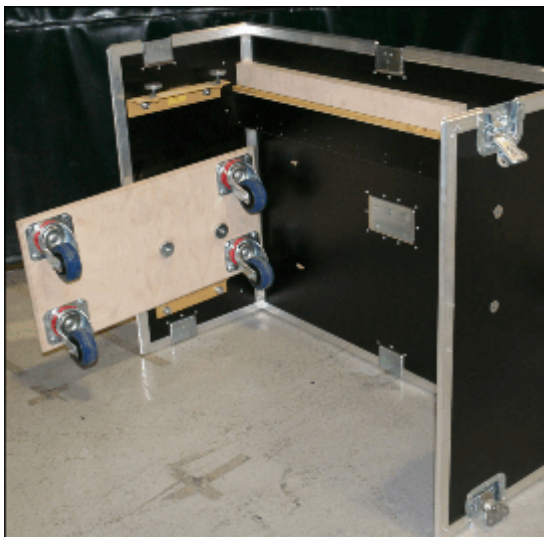


- (1) Vertical lifting beam tool

If there are only 2 x lifting eyes available for lifting the compressor unit:

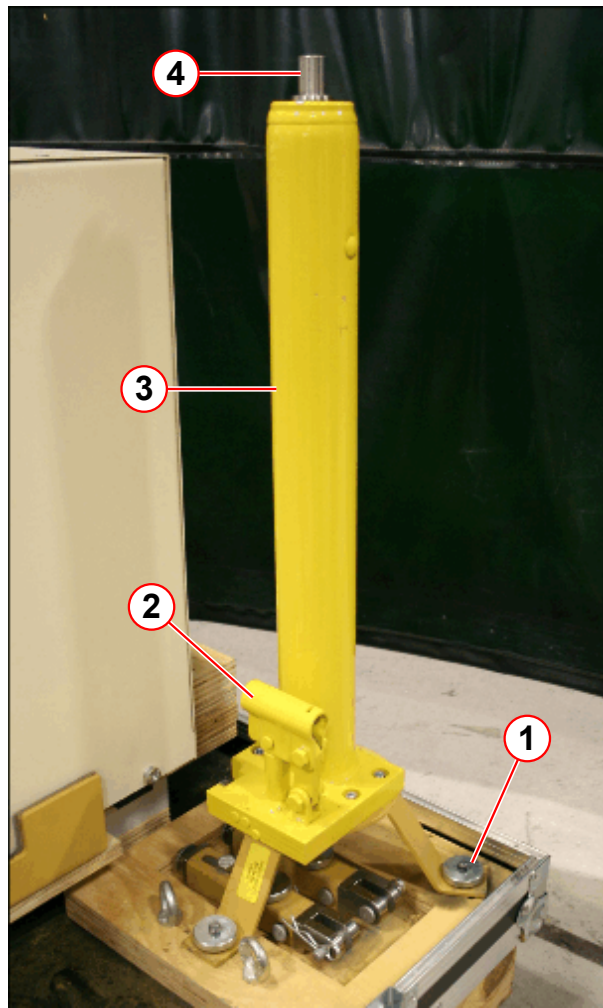
- Unscrew the 2 knurled head fasteners and remove the vertical support arm.
- Replace the fasteners.

Fig. 156: Board with castors



- Remove the knurled head fasteners and detach the moveable board with castors from the transport box side.
 - » Do not leave the board unattended.
 - » When not in use, place it upright in a safe area.

Fig. 157: Lifting Jack x 2



- Remove the knurled head fasteners x 3 from each of the 2 jacks.
- Remove the 2 lifting jacks from the base of the transport box.

- (1) Knurled fastner x 3
- (2) Hydraulic pump handle attachment
- (3) Cylinder body
- (4) Hydraulic lifter

**CAUTION****Hydraulic steel jack**

Non-observance of these warning notices may lead to damage or error of the steel jacks. The results are property damage and serious injuries.

- » Immediately support the compressor unit.
- » Working on a compressor unit when supported by the 2 x steel jacks is not permitted
- » Use only a hard, even floor area with sufficient load bearing capacity for the lifting device.
- » Prevent excess weight. The load bearing capacity cannot be exceeded!
- » Do not open the oil filler rubber plug under any circumstances. Exception: the jack has been fully lowered.
- » The operator has to monitor all movements of the load and the configuration.

Fig. 158: Support arms x 4



- Below the 2 jacks there are 4 x horizontal support arms.

The quantity required will correspond to the number of lifting eyes fitted to the compressor unit.

Fig. 159: Removing straps

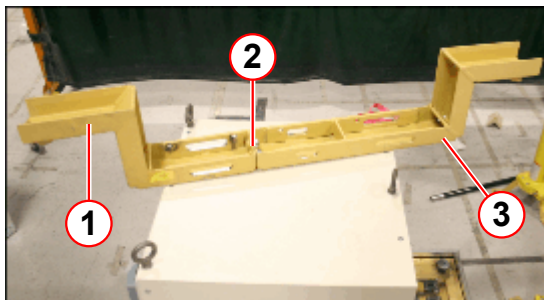


- Carefully open the restraining strap securing the compressor unit.
- Remove the edge protection pieces and the strap.



During assembly of the lifting frame place some cloth / cardboard on top to avoid paint damage to the paint.

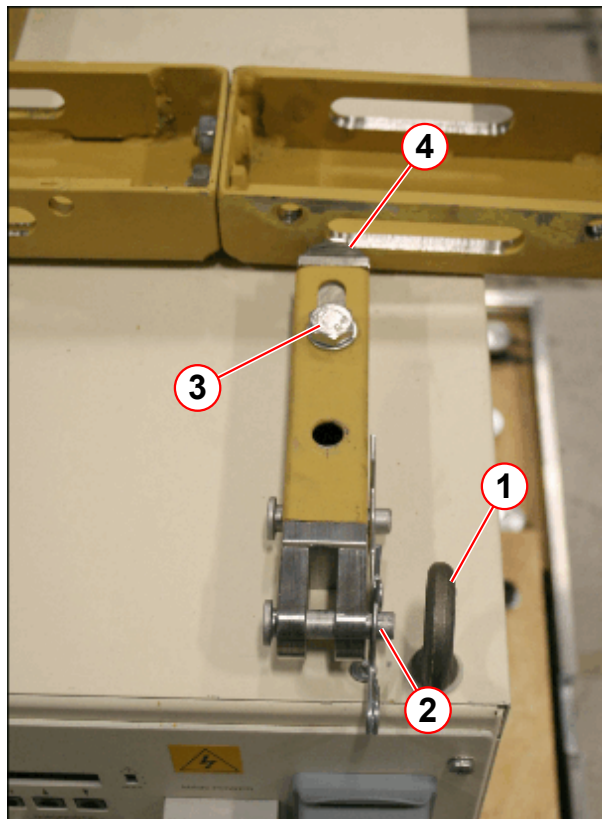
Fig. 160: Assemble lifting beams



- (1) Short beam
- (2) Bolts x 2
- (3) Long beam

- Screw both cross beams together with M10 nuts and washers.
- Torque screws to 23 Nm (17 Ft/lbs).

Fig. 161: Fitting the Support arms (x3)



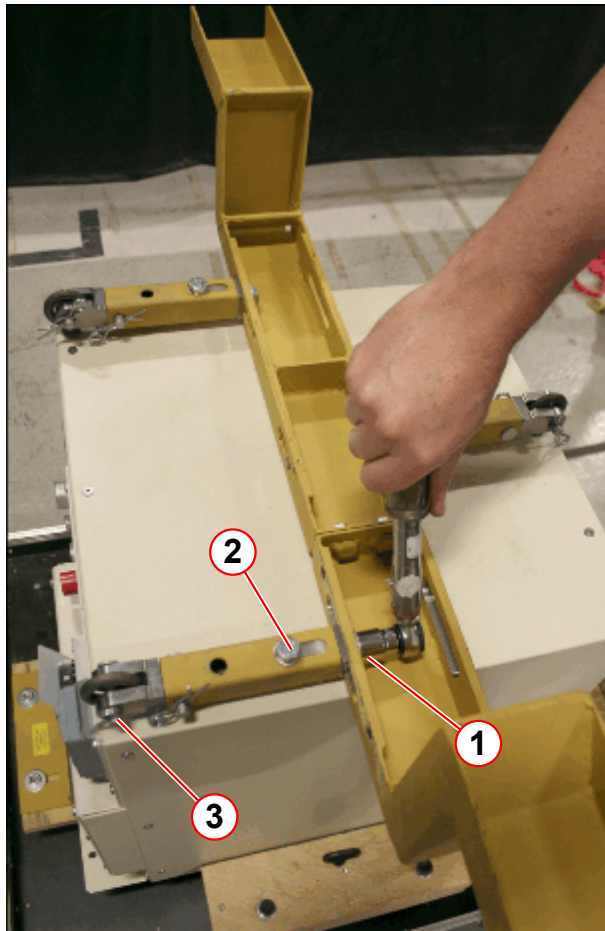
- (1) Lifting eye
- (2) Bolt and Split pin
- (3) Adjustment screw M10
- (4) Cross beam securing screw M10

Before attaching the support arms check that the lifting eyes are orientated at 90 degrees to fit into the lifting arms.

Attach the support arms:

- If there are no lifting eyes fitted on compressor unit, fit the lifting eyes supplied with the tooling.
- Insert the bolt and secure with the split pin, (pos: 2).
- For fine movement undo the adjustment screw, (pos: 3).
- Remove the cross beam securing bolt, (pos: 4).

Fig. 162: Secure support arms



- (1) Support arm & Lifting beam screw
- (2) Adjustment screw
- (3) Split pins

- Attach the support arms to the cross beams M10 x 20 screws and washers.
- Torque screws to 23 Nm (17 Ft/lbs).
- Center the complete cross beam assembly and tighten all screws with the 17-mm wrench.
- Torque the M10 adjustment screws to 23 Nm (17 Ft/lbs).
- Check that the split pins are all attached.

The cross beam lifting assembly is ready for the hydraulic jacks to be attached.

Fig. 163: Alternative lifting method



(1) Lifting eye

Alternative Lifting method

- If only 2 x lifting eyes are available on the upper side of the compressor unit:
- Attach the additional vertical support arm with the supplied crane lifting eye to the back of the compressor unit.

NOTE: The horizontal support arm seen in the figure is omitted here.

Fig. 164: Front F70 fixing adjustable plate



- Loosen the 2 x screws at the front angle adjustable plate.
- Move the angle plate away from the F70 cabinet housing.

2.8.7 Installing the Compressor Unit



Prevent stress situations caused by shocks generated by abruptly opening or closing the actuating valve while lowering the piston.

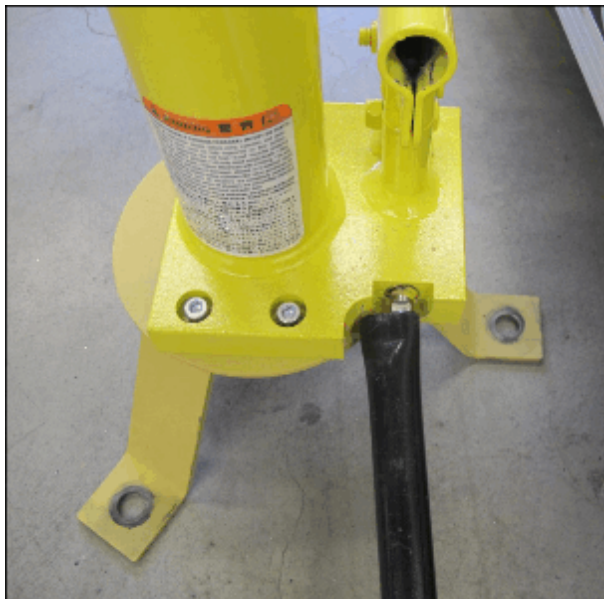
Shock loads can lead to hydraulic circuit overloads that damage the jack.

When the jack is not in use, the piston must be in the lowest position.



Before using, check the jack and the cross beam tooling for damage and oil leaks. If faulty, do not use the tooling.!

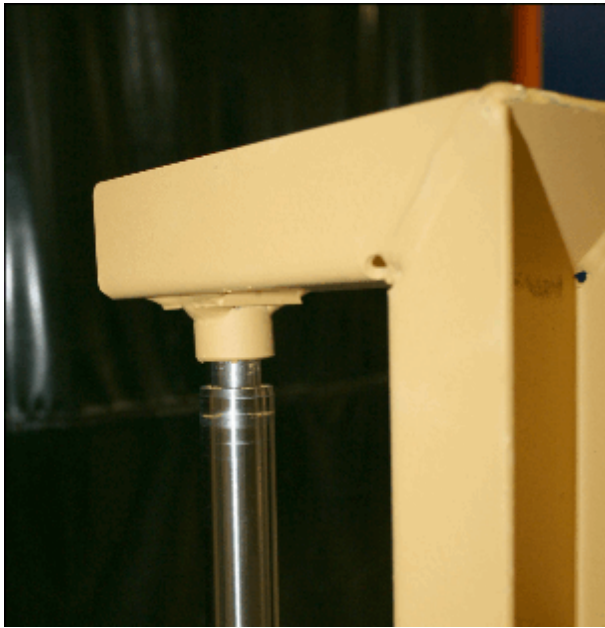
Fig. 165: Pressure relief valve



Lifting operation

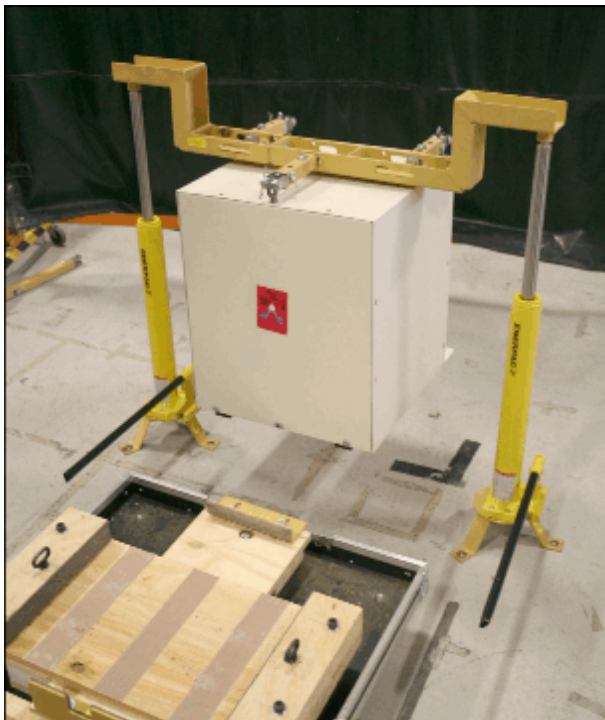
- Fit the rod onto the pressure relief valve screw.
- Turn it clockwise until the valve is firmly closed.
- Remove the rod and insert into the hydraulic pump bracket.
- Repeat this operation for the 2nd hydraulic jack.

Fig. 166: Locating the Hydraulic arms



- Slide hydraulic supports underneath the each lifting arm.
- Using the rod, pump each hydraulic arm up to the socket on the lifting beam.
- Check that the hydraulic arm is located into the socket on the lifting beam before taking the full weight of the compressor unit.

Fig. 167: Lifting the compressor



- Continue pumping the hydraulic jack rods **at the same time** to ensure a uniform lift of the compressor unit.
 - » Keep the compressor unit in a horizontal position. It must not be tipped at more than a 10 degree angle.
- Move the lower assembly of the transport box away from the compressor unit.

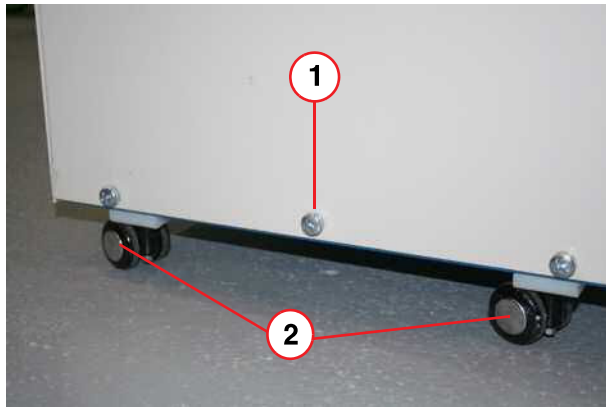


The new compressor unit is supplied fitted with 4 x caster wheels. When the compressor unit is held off the ground on the hydraulic jacks remove the caster wheels.

Compressor caster Removal

- The casters are to be removed for fitting the compressor into a SEP cabinet.
- For Standalone and IFP systems the caster wheels can remain fitted.

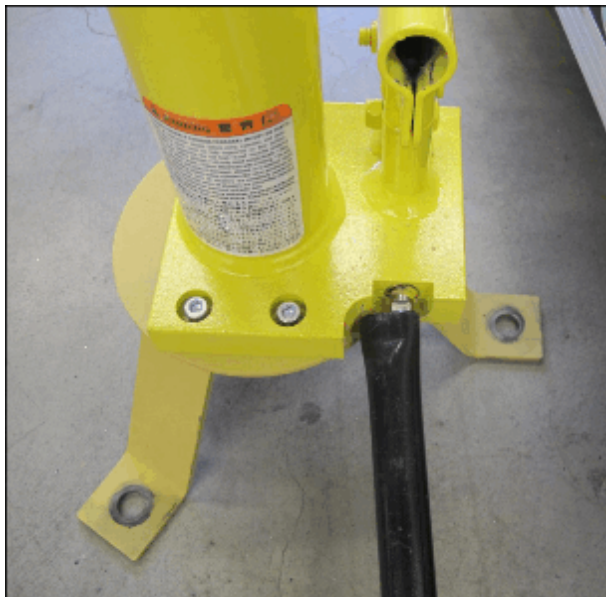
Fig. 168: F70 Back panel view



- (1) Center screw is removed for SEP cabinet installation
 (2) Casters x 4 (stand-alone units only)

- To remove the 4 casters from the compressor unit.
 - Use a 7 mm ratchet and socket and a 14 mm open ended wrench.
 - **Remove** the center screw (pos 1). (The screw hole aligns with the **guide pin** at the back of the SEP cabinet).

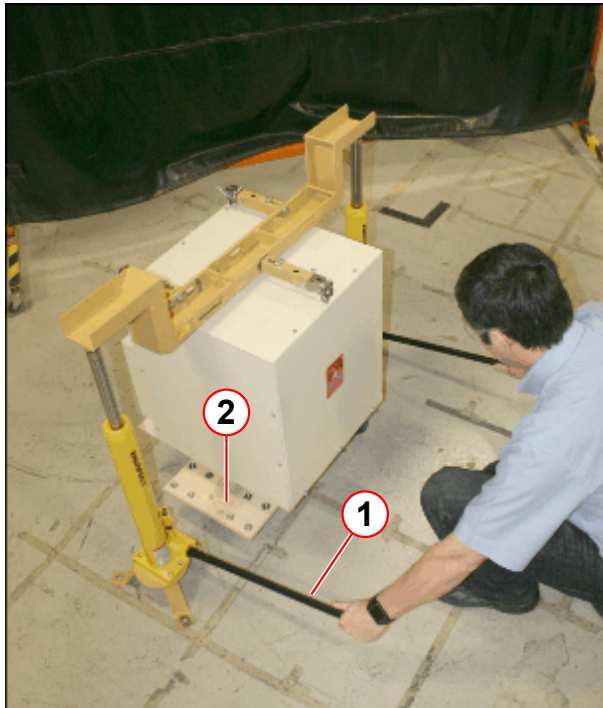
Fig. 169: Pressure relief screw



Lowering operation

- Remove the 2 x rods from the hydraulic pump bracket.
- Fit the rods over each pressure relief valve screws
- Slowly turn it counter-clockwise by not more than 1 full turn.
- Turn it slowly to be able to check the lowering speed.

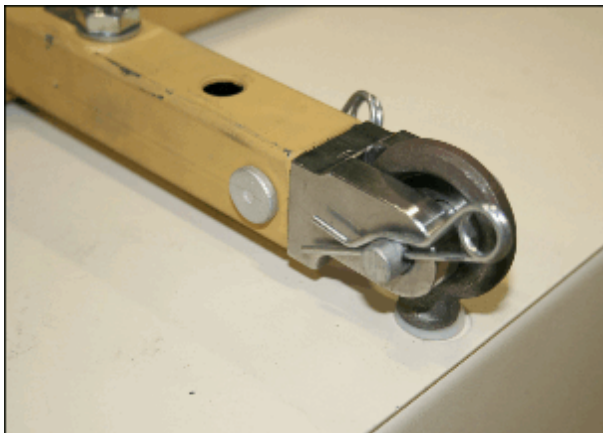
Fig. 170: Lowering compressor



- (1) Hydraulic jack rods x2
- (2) Moveable board

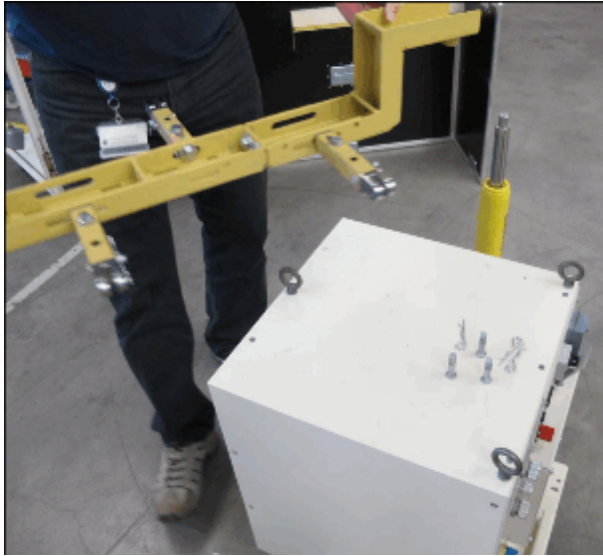
- Place the moveable board below the compressor unit.
- Turn both pressure relief valves slowly counter-clockwise to lower the compressor unit onto the moveable board.

Fig. 171: Support arm removal



- Remove the splints and bolts from the support arms fitted to the new F70 compressor unit.

Fig. 172: Removing lifting beam



- Remove the cross beam assembly.
- Carefully move the compressor unit to the final position near to the cabinet.



The moveable board casters are fixed. Movement will only be in a straight line. Check that the assembly tool is in the direct vicinity of the cabinet.



Uneven flooring leads to the hazard of tipping over!

Fig. 173: Removing the defective compressor



- Position the top cover board of the transport box in front of the cabinet.
- Carefully slide the defective compressor unit out of the cabinet onto the board.

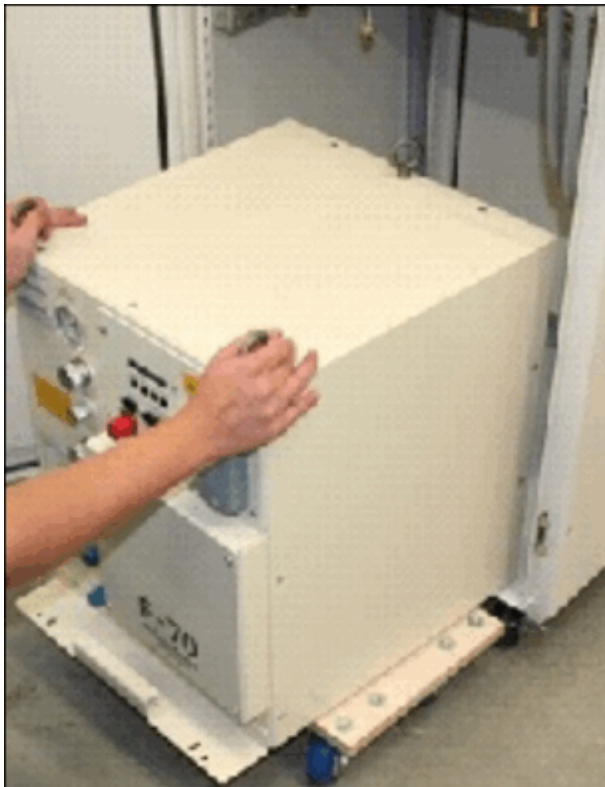
Fig. 174: Remove defective compressor (2)



When the defective compressor unit has been removed, move it away from the cabinet.

The cross beam and support arm assembly can be fitted later. The hydraulic jacks will be used to lift the defective compressor unit onto the transport box.

Fig. 175: Moving in the compressor

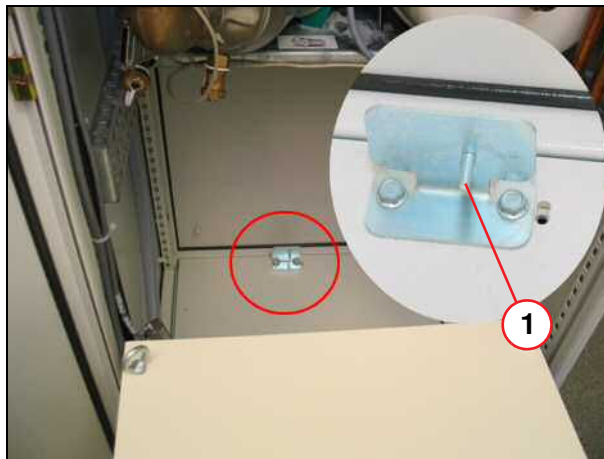


- When fitting the new compressor unit:

- » Check that the compressor unit is directly in line to the front of the cabinet.

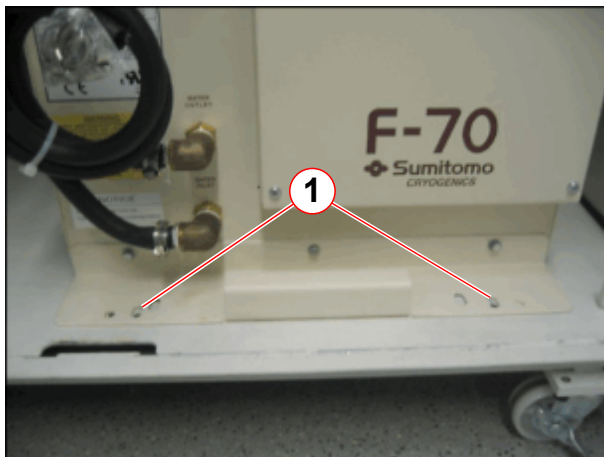
- Slide the new compressor unit into the cabinet.

Fig. 176: SEP - Compressor guide pin



- The (center) screw hole on the rear panel of the compressor unit should align with the **guide pin** at the back of the cabinet.

Fig. 177: F70 Compressor fixing plate



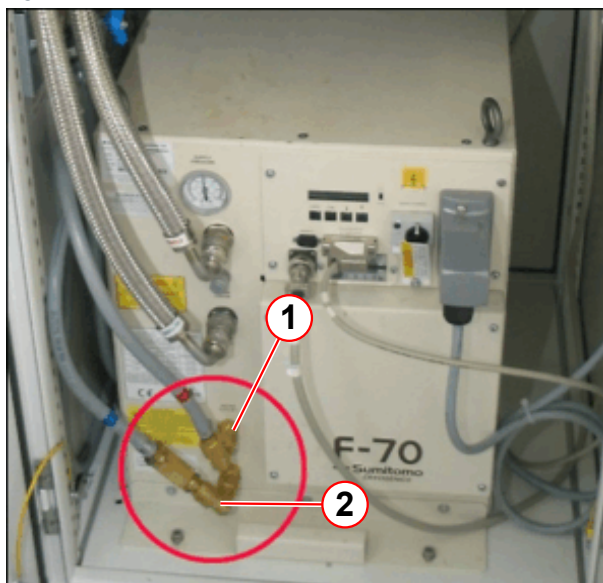
- Secure the **fixing plate** to the cabinet using the 2 x M8 hex head screws.

(1) M8 Hex hd screws (2 off)

2.8.8 Compressor Unit Configuration and Connections

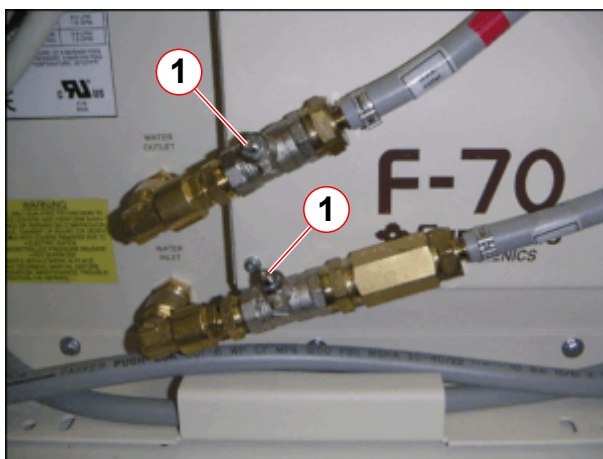
The new compressor unit must be operated for **10 minutes** to clean up any oil contamination incurred during transportation. Run the compressor unit **before** the **helium pressure lines** and **cold head** are connected. For this test, only the **water** and **power** connections are required.

Fig. 178: F70 connect the water lines



- (1) Water line (inlet)
- (2) Water line (outlet)

Fig. 179: F70 Compressor water inlet/outlet shutoff valves

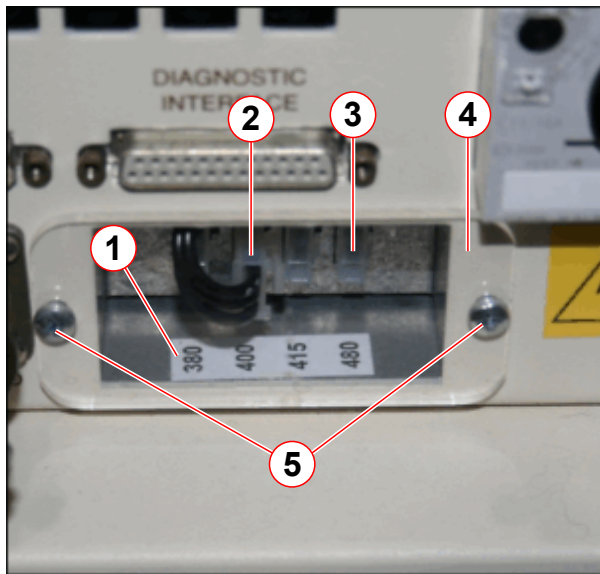


- (1) Shutoff valve

Connecting the water:

- Remove the protective caps from the water in and water out fixings.
- **Connect the water lines** to the compressor.
 - Ensure the **water in** is connected to the **water supply line**, and the water is turned **on**.
 - **Open the shut-off valves**, turn on the coolant supply, and check for leaks.

Fig. 180: F70 Compressor voltage selection access panel



- (1) Voltage label
- (2) Voltage selection connectors
- (3) Voltage tap
- (4) Clear plastic cover
- (5) Screws (2 off)

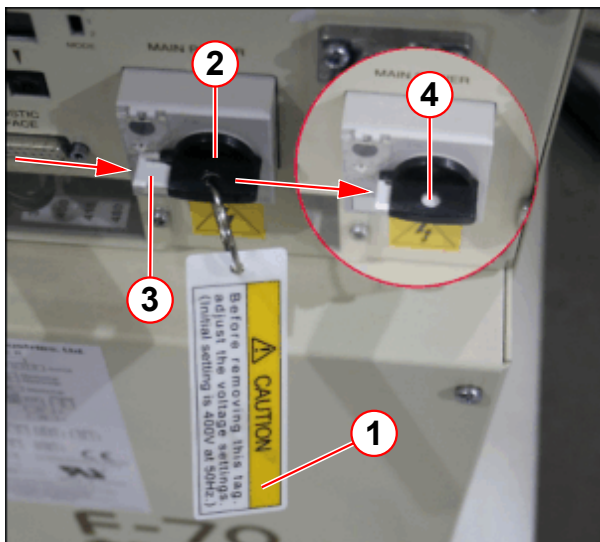
- Check that the **transformer voltage taps** are correctly selected to match the supply voltage.
- If necessary, remove the plastic cover (3), reposition the tap, and replace the cover.



The unit is configured in the factory to 400 V/3~50 Hz.

If the compressor unit is not wired correctly, connecting to a higher voltage may damage the control circuit.

Fig. 181: F70 Main power switch - locking device



- (1) Voltage caution label
- (2) Main power switch
- (3) White locking device
- (4) Device fully engaged (view through hole)

- **Remove** the voltage caution label (pos. 1) (fitted through the main circuit breaker).
- Engage the white switch locking device fully **before** attempting to turn on.
- **Push in** the white locking device fully.
 - » The device should be visible through the blocking hole (pos. 4).



Ensure that the white switch locking device is **FULLY** engaged. **Do not use too much force on the switch.** A new compressor unit will be required if damaged.

- Connect the mains voltage cable.

Fig. 182: F70 Main power supply connection

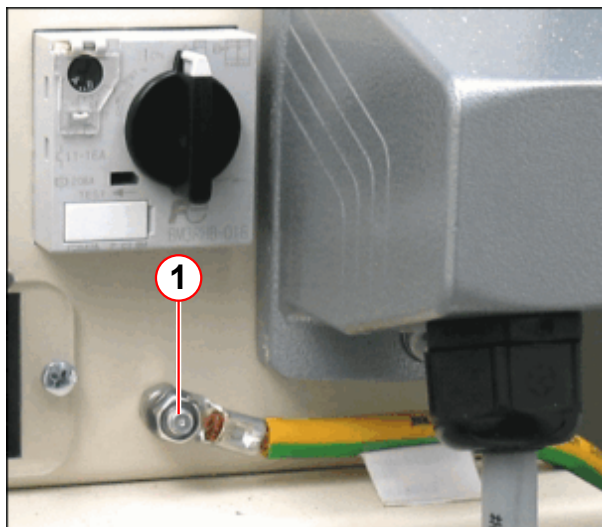


- (1) Pull cover to remove
- (2) Connect power supply cable
- (3) Fasten the cover by attaching the screws

2.8.9 Compressor Unit (Ground) Earth Connection

A protective ground connection test must be performed as a final step **before** switching on the compressor unit.

Fig. 183: Compressor ground wire connection



- (1) Ground wire connection

- Connect the ground wire to the helium compressor terminal.

Fig. 184: RCA



(1) Protective conductor

- Connect the protective conductor W0821 from the RCA to the CCA.

Test the protective earth connection

- Measure the protective earth connection resistance $\leq 100 \text{ m}\Omega$ with a ground test meter (Fluke 1503, or equivalent). Check between the protective earth connector on the compressor unit, and the protective earth wire connected to the top of the SEP (on the electronic box of the IFP).

2.8.9.1 Compressor Unit Run Mode



Compressor unit is supplied in configuration **mode 1**.

Before the compressor unit is switched on, the **mode selector switch** must be set to either mode **1** or **2**.

Tab. 6 Configuration mode switch

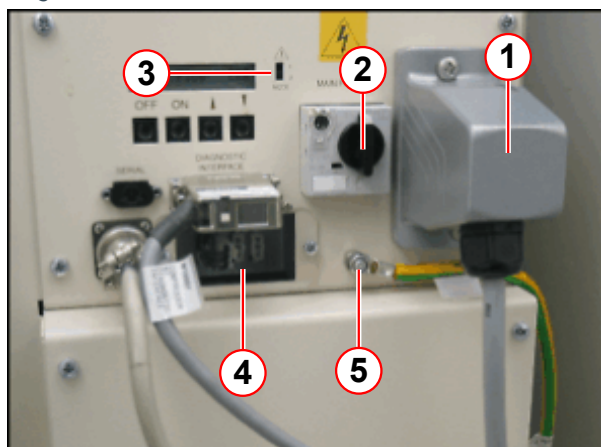
Configuration	Switch position	Magnet system type	ON/OFF
Mode 1	UP	10-K shield cooling	Enabled
Mode 1	UP	F70 Startup checks	Enabled
Mode 2	DOWN	4-K recondensing	Disabled



DO NOT change the configuration-mode switch position **AFTER** the main power has been switched **ON**. Loss of control and incorrect errors will result.

2.8.9.2 Testing the Compressor Unit on Startup

Fig. 185: F70 Compressor - connections for test running before connection



- (1) Power supply
- (2) Main power switch
- (3) Configuration mode switch
- (4) Voltage selector
- (5) Ground wire connection

- Ensure that the **main power** switch is **OFF** before adjusting the selector switch.
- Check that the configuration mode switch is set to **1**.

Use a flat-blade screwdriver to move the switch up and down.

- » The **ON/OFF** buttons only work in configuration mode **1**.
- » The compressor unit must **NOT** be run in stand-by in mode **2**.

In **mode 2** both **ON** and **OFF** buttons are disabled. Operation is controlled remotely via the DB25 diagnostic interface.

- Switch on the circuit breaker F4 (CCA cabinet).
- Switch on F3
- Turn the main power switch.
- Press the ON button.
- Check the temperature of the water supply (water in) and return (water out) lines:-
 - » The **return** (water out) pipe should be **warmer** than the **supply** (water in).
 - » **Stop** the compressor unit and reverse water pipes if the **return** line is **cooler**.
- Run the compressor unit for **10 minutes** and then press the **OFF** button.
- Turn **OFF** the main power switch.
- The test run is now complete.



DO NOT change the configuration-mode switch position **AFTER** the main power has been switched **ON**. The result is loss of control and incorrect indications.



The compressor unit is protected from running in reverse.



If the compressor unit starts but does not build pressure, **TURN OFF IMMEDIATELY**. Contact Siemens Magnet HSC.

2.8.10 Testing the Compressor Unit after Mishandling

Follow this procedure if:

1. The Tip-N-Tell indicator has been activated.
2. The compressor pressure is lower than specification.

Fig. 186: F70 Compressor Tip-N-Tell indicators



Pressure within specification:

- If the Tip-N-Tell has been activated but the pressure is in specification:
 - Stand compressor unit upright for **2 hours**.
 - Perform the test run.

Pressure outside specification:

- Notify the delivering carrier of the damage if:
 - The compressor unit has been delivered on its side or upside down.
 - There is physical damage to the unit.

NOTICE

If the compressor unit arrives at zero pressure - **DO NOT CONNECT THE HE PRESSURE LINES or COLD HEAD POWER CABLE** to the compressor unit.

Connecting the lines to the compressor unit at zero pressure may cause contamination of the whole refrigeration system.

- » Check the compressor pressure before attempting to connect the helium pressure lines.
- » Contact Siemens Magnet HSC for advice.

- **Remove** the compressor unit from the **transport box**.
 - **Retain** all packaging materials for reuse if returning the compressor unit for replacement or repair.

Install the replacement compressor unit into the SEP cabinet before performing the test.

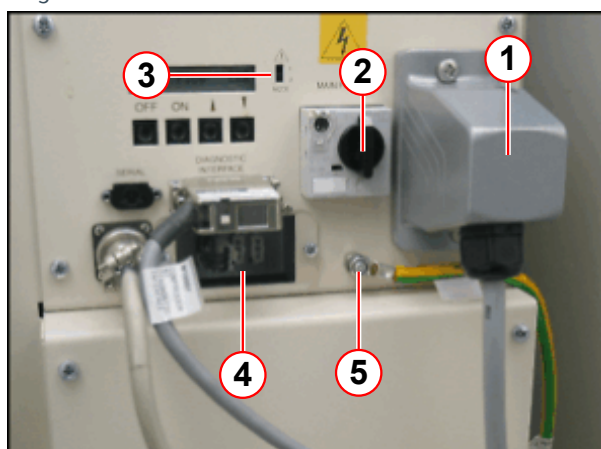
For the test, only compressor water and power connections are required.

Ensure all other lines and cable connections are disconnected from the compressor unit.



DO NOT change the configuration-mode switch position **AFTER** the main power has been switched **ON**. Loss of control and incorrect indications will result.

Fig. 187: F70 Compressor - connections for test running before connection



- (1) Power supply
- (2) Main power switch
- (3) Configuration mode switch
- (4) Voltage selector
- (5) Ground wire connection

- Check that the configuration mode selector switch is set to **1**.
Use a flat-blade screwdriver to move the switch up and down.
 - » The **ON/OFF** buttons only work in configuration mode **1**.
 - » The compressor unit must **NOT** be run in stand-by in mode **2**.
In configuration **2**, both **ON** and **OFF** buttons are **disabled**. Operation is controlled remotely via the **DB25 diagnostic interface**.
- Switch **ON** the circuit breaker F4 (CCA).
- Turn on the main power switch.
- Press the **ON** button to start the compressor unit. The LED displays **RUN**.

- Check the water supply and return line temperatures.
 - » The **return** line should be **warmer** than the **supply**.
- **Stop** the compressor unit and reverse the water pipes if the **return** line is **cooler**.
- Run the compressor unit for a **MINIMUM of 2 HOURS**.



If the compressor unit shuts down during the test run, contact Siemens Magnet HSC.

- If there are no problems during the test run, stop the compressor unit.
 - Press the **OFF** button.
 - Turn **OFF** the **main power** switch.

- The test run is complete.



The compressor unit is protected from running in reverse.



If the compressor unit starts but does not build pressure, **TURN OFF IMMEDIATELY**. Contact Siemens Magnet HSC.

2.8.11 Cold Head and Compressor Unit Start up



CAUTION

High-pressure helium gas

Failure to observe the following Safety Information may result in injury.

- » Protective clothing, gloves, and eye protection must be worn.

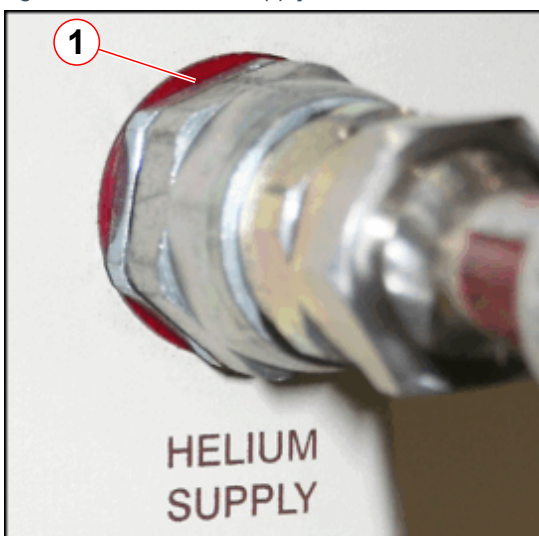
Helium gas lines

Ensure that the **Supply** (HP) and **Return** (LP) lines are connected correctly. The 4K helium pressure lines are colour coded as shown in the table below.

Tab. 7 Gas line connections

System	Gas line color	F70 He connection
4K supply (HP)	Yellow	Red
4K return (LP)	Green	Green

Fig. 188: F70 Helium supply connections - color coding



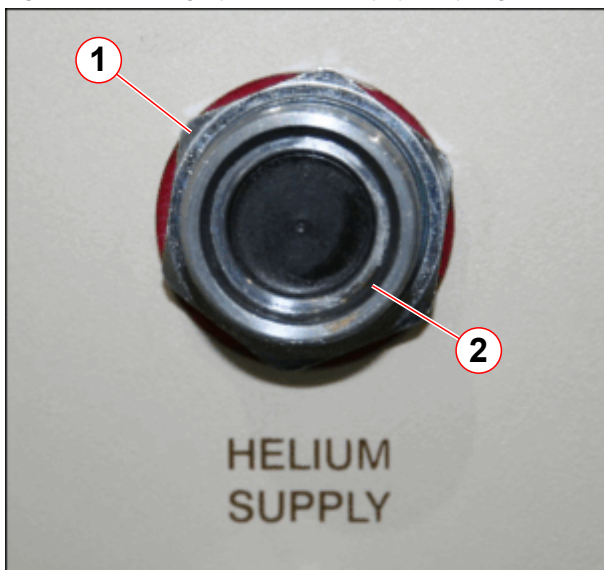
(1) Helium supply = RED

Fig. 189: F70 Helium supply connections - color coding



(1) Helium return = GREEN

Fig. 190: F70 High-pressure Aeroquip coupling

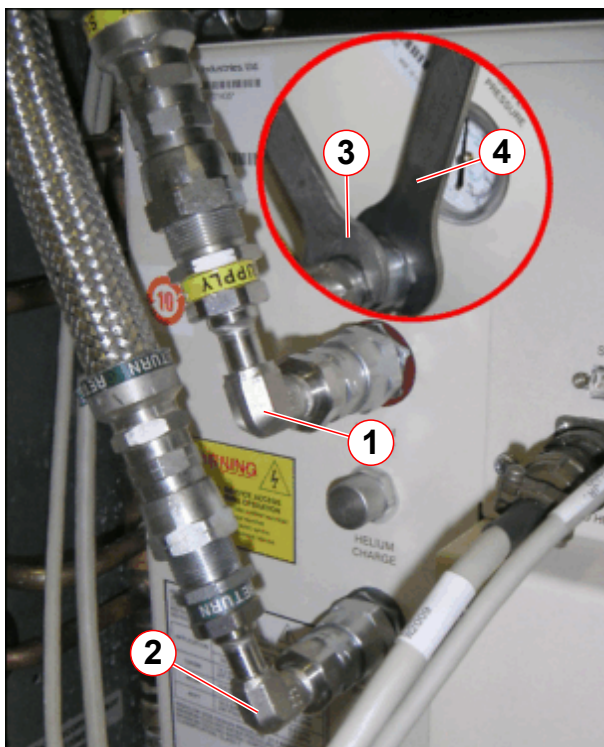


- (1) Locknut
- (2) Flat rubber gasket

- Remove the protective transport covers from the return and supply connections.
- Check the **rubber gasket** seal for dirt or dust and check that it is correctly fitted.

NOTE: An incorrect fit may cause a helium gas leak.

Fig. 191: F70 He compressor connections



- (1) SUPPLY (high pressure) 90-degree elbow
- (2) RETURN (low pressure) 90-degree elbow
- (3) Hold nut (1" spanner/wrench)
- (4) Turn nut (1 3/16" open end crowfoot spanner/wrench)

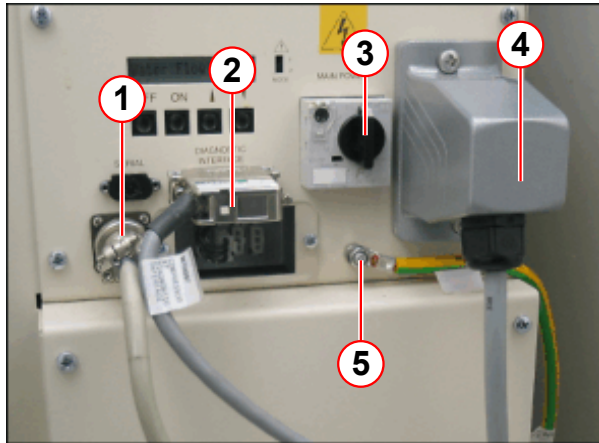
Each **helium pressure line** is connected to the compressor unit via a **90 degree elbow**.

- **Fit** the self-sealing helium pressure lines in the following order:
 - **Return** (low pressure).
 - **Supply** (high pressure).
- **Hold** nut (3) and **turn** nut (4).
- **Tighten down** all couplings as far as possible, and then turn them back by 1/4 turn to **relieve strain**.



When connecting the Helium pressure lines, **DO NOT OVER-TIGHTEN** the couplings.

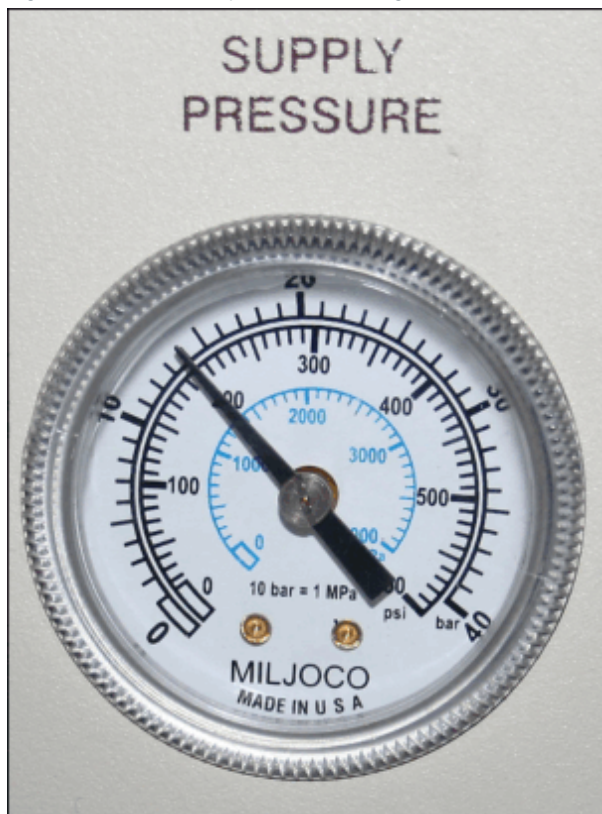
Fig. 192: F70 Electrical cable connections



- (1) Cold head cable
- (2) MSUP DB25 Diagnostic interface cable
- (3) Main power switch
- (4) Mains supply cable
- (5) Ground wire

- Connect the **cold head** cable (1) and **MSUP cable** (DB25 diagnostic interface) (2).

Fig. 193: F70 Static pressure reading



- Check the **static helium pressure** on the pressure gauge.
- Charge with helium gas if pressure is lower.

Equalization (static) pressure at 20⁰ C (68⁰ F) for 12 m to 20 m long gas lines:

Tab. 8 Helium gas pressures

Cold head	Static pressure	Dynamic pressure
10K - 50 Hz	14.0 - 14.5 bar (14.3 nominal) 203 - 210 psi (G)	18 - 23.2 bar 265 - 337 psi (G)
10K - 60Hz	13.5 - 14 bar (13.6 nominal) 196 - 203 psi (G)	18.0 - 23.0 bar 261 - 331 psi (G)

- Check that the **configuration switch** is set to **mode 1**.
- Check for **water flow** to the compressor unit.
- Switch **ON** the **main power** switch.
- Check the **alarm box** status and clear any faults.

2.8.12 Final Checks

- Wait **20 minutes** and check the operating (**Dynamic**) helium gas pressure on the pressure gauge.
 - **Supply: 21-22 bar** (304-319 psi (G)).



If the dynamic pressure is outside of this range, refill.

- **Check** the helium pressure line and connectors for **leaks**.
- **Check** for leakage at the water connections.
- Document the measurement in the **service report**. The (measured value, and measured equipment) **is required by IEC 62353**.

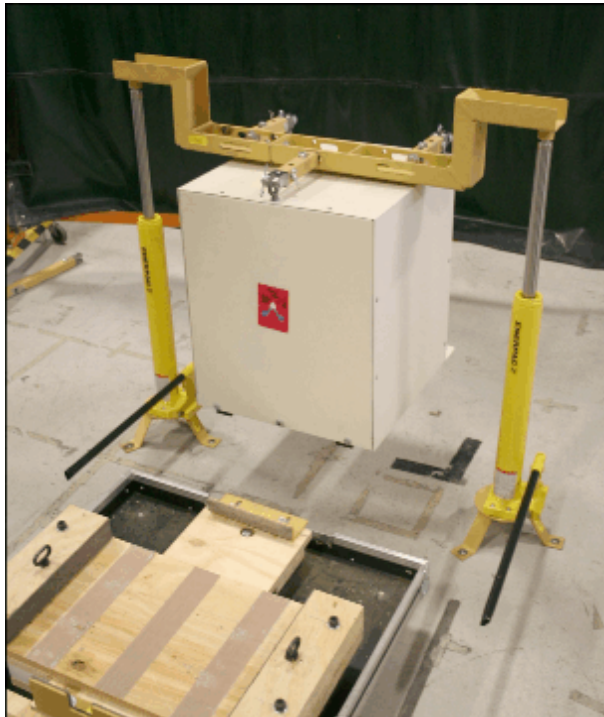
2.8.13 Packing the Compressor Unit

Fig. 194:



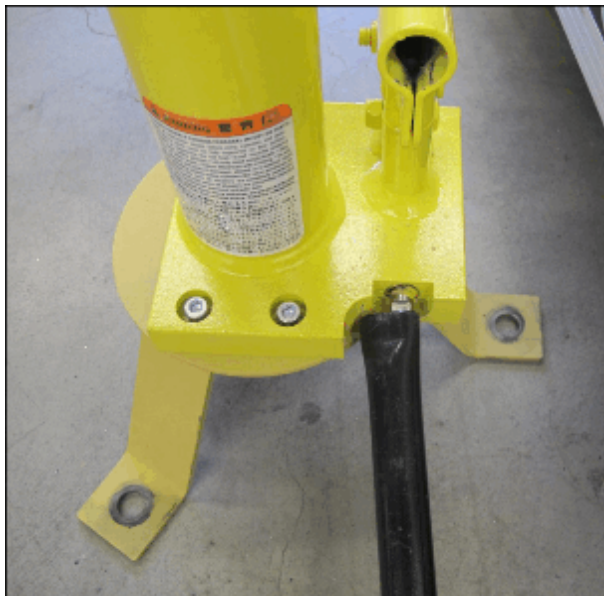
- Move the failed compressor unit on the moveable board next to the lifting jacks.
- Fit the lifting beam assembly.
- Pumping the 2 x hydraulic jacks at the same time lift the compressor unit.
- Remove the moveable board.

Fig. 195: Lifting the compressor



- Keep the compressor unit in a horizontal position
- Move the lower assembly of the transport box under the compressor unit.

Fig. 196: Pressure relief screw



Lowering operation

- Remove the 2 x rods from the hydraulic pump bracket.
- Fit the rods over each pressure relief valve screws
- Slowly turn it counter-clockwise by not more than 1 full turn.
- Turn it slowly to be able to check the lowering speed.

Fig. 197: Back F70 fixing plate



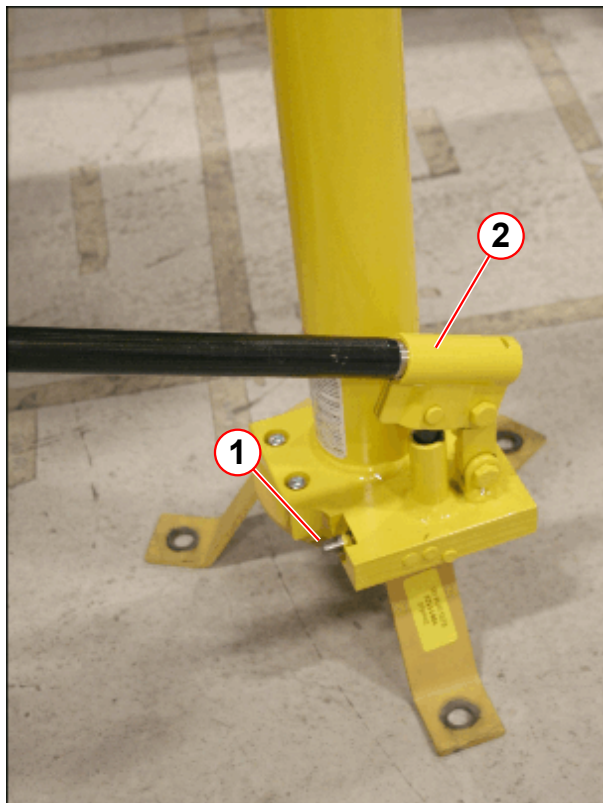
- Lower and place the defective F70 compressor unit onto the transport box in the location provided.
- Check that the F70 cabinet fits flush with the back angle fixing plate.

Fig. 198: Front F70 fixing adjustable plate



- Loosen the 2 x screws at the front angle adjustable plate.
- Move the angle flush and tight with the F70 cabinet housing and retighten the 2 x screws.

Fig. 199: Packing the hydraulic jacks



- (1) Pressure relief valve screw
(2) Hydraulic pump bracket

To removing air from the hydraulic jack

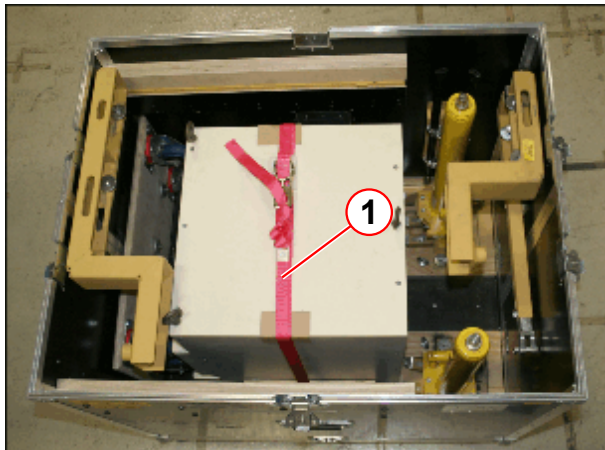
- Place the rod onto the pressure relief valve screw.
- Turn it counterclockwise by not more than 3 turns.
- Insert rod into the neck of the rod.
- Perform between 6 to 8 complete pump strokes.
- Repeat if necessary.

Fig. 200: Manual press piston



- If the piston has not fully lowered, manually press down.
- Repeat for the 2nd jack.

Fig. 201: Top view of compressor kit



(1) Securing strap

- For the return shipment, pack all the tooling parts back into the transport box in the correct locations and tighten all screws.

Fig. 202: Transport crate



- (1) Lifting handles (x 2)
- (2) top lid latch (x 6)
- (3) Lid with PTFE sheet covering

- Replace the top lid of the transport box.
- Return the compressor transport box to CSML.

**CAUTION**

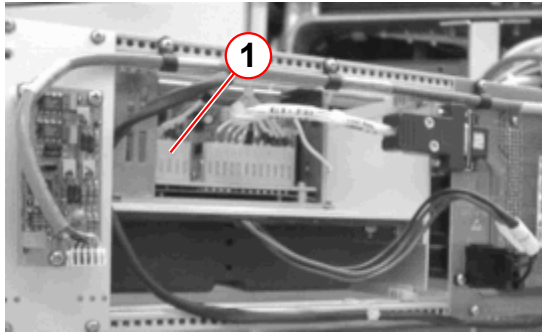
During working on the magnet supervision, as the emergency rundown unit will be inoperable.

If not observed, serious injury can occur.

- » Make sure, nobody is in the magnet room.

3.1 Replacing the battery

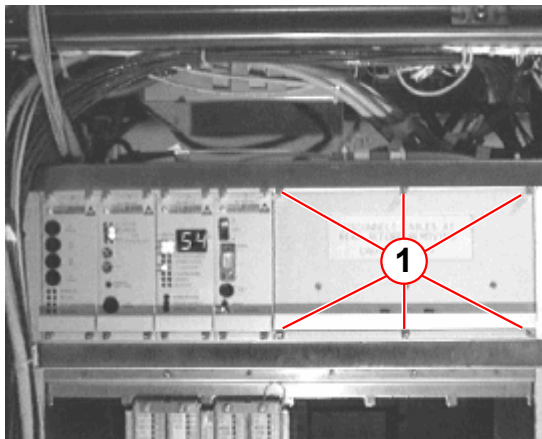
Fig. 203: Magnet supervision



(1) Power cable

- Switch off F24.
- Swivel out rack.
- Remove cable X3.
- Remove cable X7.
- Remove power cable.

Fig. 204: Magnet supervision



(1) Screw

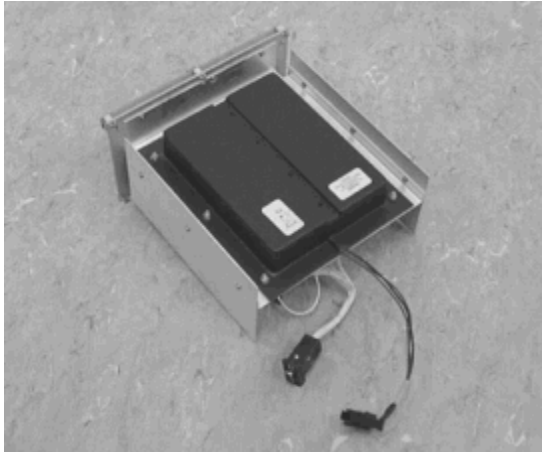
- Remove screws holding the slide-in-module.

Fig. 205: Magnet supervision



- Pull out slide-in-module.

Fig. 206: Battery



- Replace battery.

- Install in reverse order.

3.2 Replacing board D1; D2; D3



CAUTION

Emergency rundown unit will be inoperable during replacing board D1, D2,D3. MSUP has to be powered down.

If not observed, serious injury can occur.

- » Make sure, nobody is in the magnet room.

3.2.1 Procedure

- Disconnect heater cable from plug "B" on the turret of the magnet.
- Turn off power to MSUP by switching off F24 .
- Unplug X7 (Battery connection) from the rear of the control crate.
- Replace it with the new D1, D2 or D3.
- Reconnect the battery X7.
- Turn on F24 to power up the MSUP.
- Reconnect heater cable to plug "B" on the magnet turret.

3.2.2 Concluding step

Perform a (→ ERDU test / M1-010.833.01).

3.3 Replacing board D4



CAUTION

Emergency rundown unit will be inoperable during replacing board D4. MSUP has to be powered down.

If not observed, serious injury can occur.

» Make sure, nobody is in the magnet room.

3.3.1 Procedure

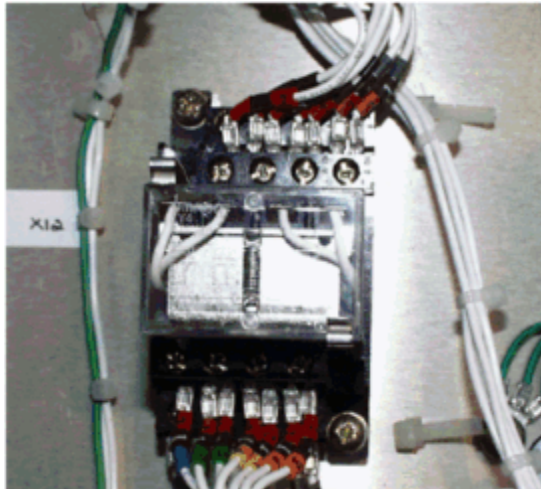
- Disconnect heater cable from plug "B" on the turret of the magnet.
- Turn off power to MSUP by switching off F24.
- Unplug X7 (Battery connection) from the rear of the control crate.
- Replace it with the new D4.
- Reconnect the battery X7.
- Turn on F24 to power up the MSUP.
- Reconnect heater cable to plug "B" on the magnet turret.
- Check the value of the helium probe resistance in the SESO MAGNET&COOLING.

3.3.2 Concluding step

Perform a (→ ERDU test / M1-010.833.01).

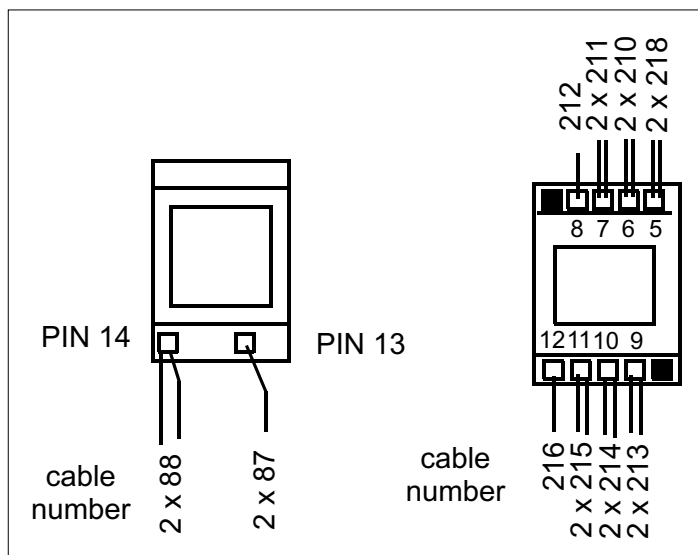
4.1 Replacing the Relay

Fig. 207: Relay



- Switch OFF F12.
- Disconnect the cables from the relay terminal.
- Undo the screw.
- Replace the relay.
- Connect the cables.
- Switch on F12.

Fig. 208: Relay terminal



4.2 Replacing the Power Modul

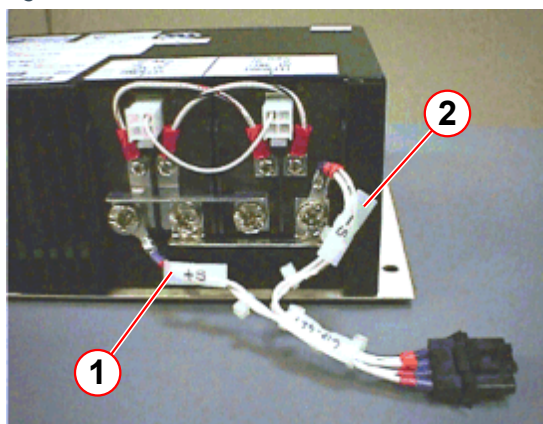
Fig. 209: Power modul



- (1) Plug A
- (2) Plug B

- Switch OFF F12.
- Disconnect plug A; B; C.
- Undo the 4 screws.
- Remove power modul.

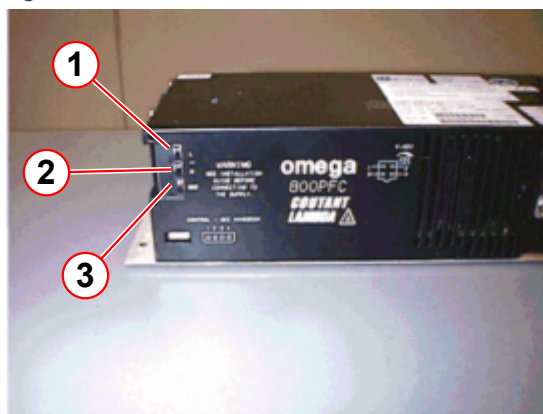
Fig. 210: Power modul



- (1) S+
- (2) S-

- Prepare the new PSU.

Fig. 211: PSU terminal



- (1) L: 2 x 87
- (2) N: 2 x 88
- (3) GND

- Connect cable L/N/GND.
- Install the PSU.
- Reconnect plug A;B; C.
- Switch ON F12.

Fig. 212: 10A Shim Power Supply



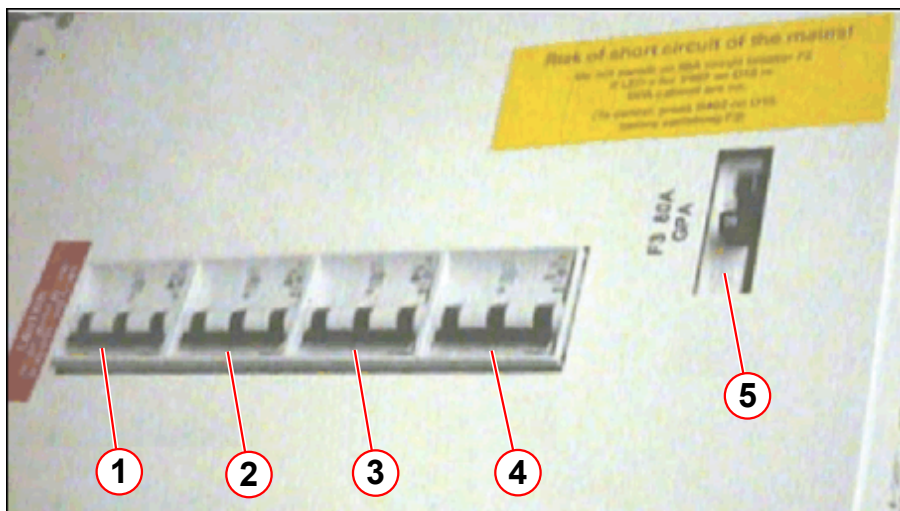
WARNING

Voltage continues to be present at the Shim Power Supply even after the MR system has been switched off.

If not observed, death or serious injury can occur.

- » The Circuit breaker F5 on the Mains box has to be set to OFF for service work.

Fig. 213: Mains Box



- (1) F4 RCA
- (2) F23 CCA
- (3) F5 MPSU SHIM
- (4) F25 RFPA
- (5) F4 GPA

5.1 Replacement of field replacable units



WARNING

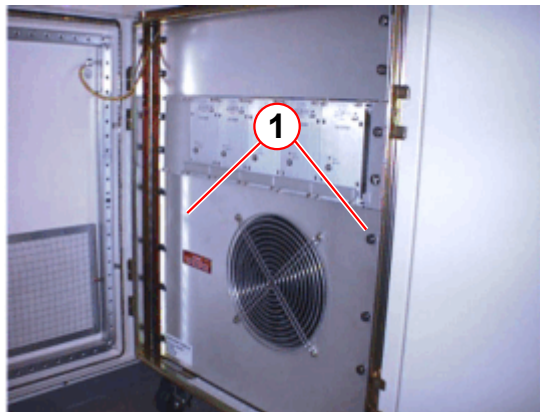
Voltage continues to be present at the Shim Power Supply even after the MR system has been switched off.

If not observed, death or serious injury can occur.

- » The Circuit breaker F5 on the Mains box has to be set to OFF for service work.

5.1.1 Replacement of the Shim Power Supply blower

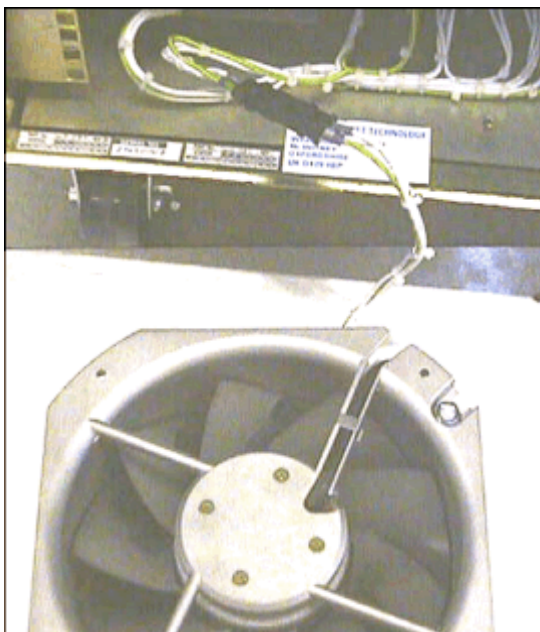
Fig. 214: Blower



(1) Screws

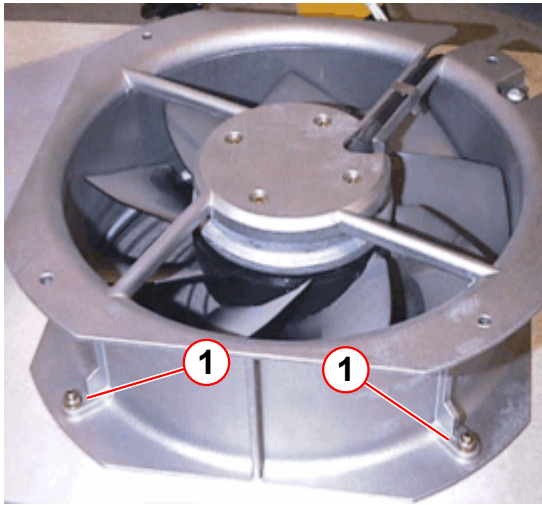
- Switch OFF F5.
- Undo the screws.

Fig. 215: Blower



- Disconnect the cable.

Fig. 216: Blower



(1) Screws

- Undo the screw.
- Replace the defective blower.
- Connect the cable.
- Tighten the front plate.
- Switch on F5.

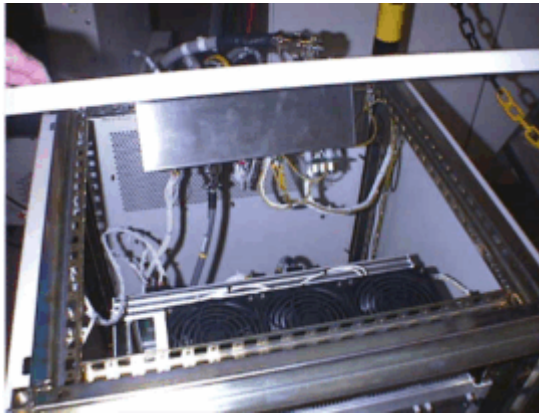
5.1.2 Replacement of the Shim Module blower

Fig. 217: Shim Module Blower



- Switch OFF F5.
- Remove cover above the shim module.
- Replace the defective blower.

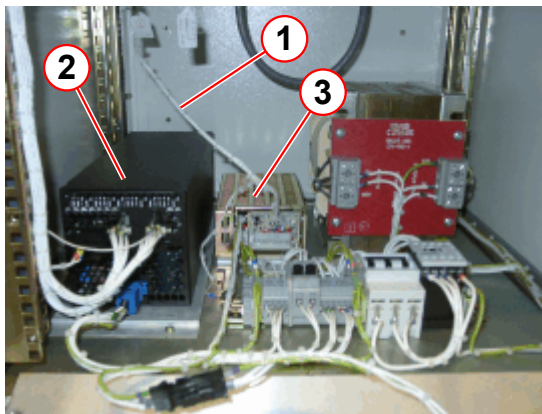
Fig. 218: Shim PS Top Cover



- For easy access open the top cover. Unscrew the 4 ring bolts.
- Replace the defective blower in reverse order.

5.1.3 Replacement of the DC power supply

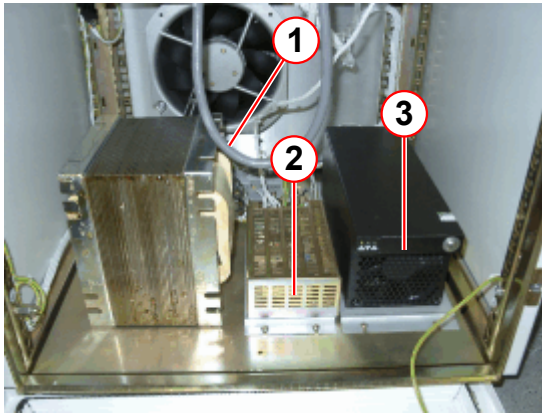
Fig. 219: DC Power Supply



- (1) DC power supply
- (2) PSU1
- (3) PSU2

- Switch OFF F5.
- Undo the screws of the front cover.
- Disconnect the cables.
- Undo the screws of the base plate.

Fig. 220: Rear view of SPS

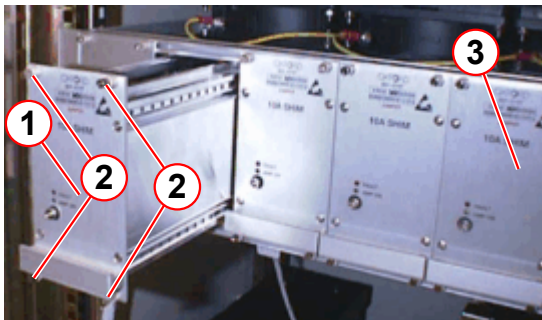


- (1) DC power supply
- (2) PSU2
- (3) PSU1

- Remove the rear cover of the cabinet.
- Undo the screws of the base plate.
- Replace the defective power supply in reverse order.

5.1.4 Replacement of the Shim Channel amplifier module

Fig. 221: Shim Channel amplifier



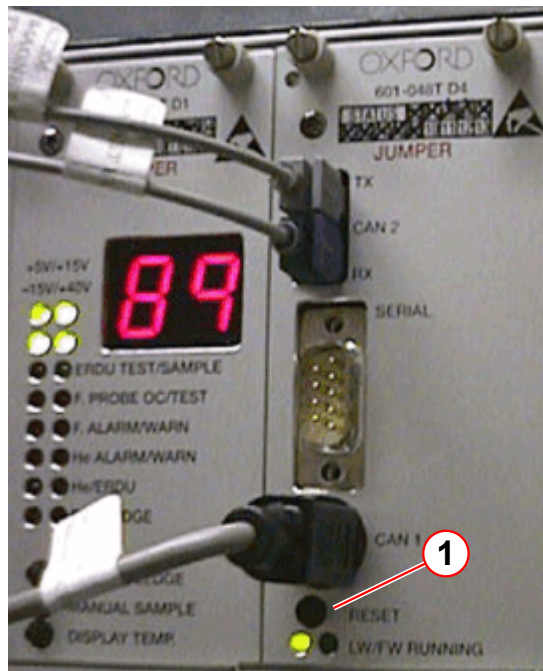
- (1) Channel 0
- (2) Screw
- (3) Channel 3

- Switch OFF F5.
- Loosen the screws holding the module.
- Pull out slide-in-module.
- Replace the module in reverse order.



Perform a RESET on the D4 of the MSUP after replacing a shim channel.

Fig. 222: MSUP D4

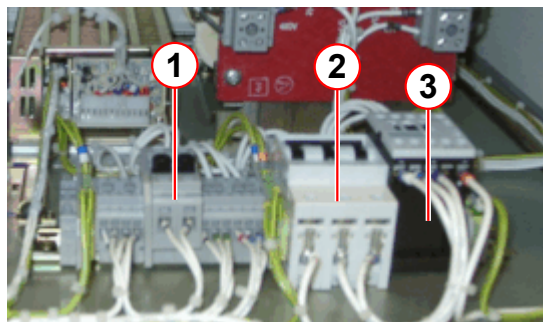


(1) RESET

- Perform a reset of D4.

5.1.5 Replacement of the replaceable unit K1, CB1

Fig. 223: Circuit breaker



(1) F1/F2
(2) CB1
(3) K1

- Switch OFF F5.
- Undo the screws of the front cover.
- Disconnect the cables.
- Remove CB1 or K1.
- Replace the K1 or CB1 in reverse order.

- » The common print number (MR-000.841.01) from Replacement of Parts is divided into separate Component print numbers.
- » A hint regarding the use of a protection mask during exchange of the displacer was added (→ Working with the heat gun / Page 32).

Version 02

- Hint: "Protective earth connection test" was added.

Version 03

- Updated 10K cold head displacer exchange (Charm MR_00454054) (→ Replacing the Displacer - APD system / Page 45)

Version 04

- Added safety precaution with lead (→ Prepare the new control disc / Page 29) and (→ Safety recommendations / Page 46).

Version 05

- Replacing the Displacer - APD system reworked (→ Replacing the Displacer - APD system / Page 45)

Version 06

- Replacing the F70 Compressor with Tooling kit, added. (→ Replacing the F70 Compressor with Tooling kit / Page 88)

Version 07

- Replacing the Leybold displacer, tooling Cryo Service and Cryo Service Set NMR replaced with 10119990 Displacer Exchange Kit.

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Print No.: MR-000.841.17.07.02 | Replaces: MR-000.841.17.06.02
Doc. Gen. Date: 10.16 | Language: English
Part No.: n.a.
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